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SYSTEMS AND METHODS FOR ENHANCED DETECTION AND QUANTIFICATION OF ANALYTES

Abstract

Devices, systems, and methods for detecting molecules of interest within a collected sample are described herein. In certain embodiments, self-contained sample analysis systems are disclosed, which include a reusable reader component, a disposable cartridge component, and a disposable sample collection component. The reader component may communicate with a remote computing device for the digital transmission of test protocols and test results. In various disclosed embodiments, the systems, components, and methods are configured to identify the presence, absence, and/or quantity of particular nucleic acids, proteins, or other analytes of interest, for example, in order to test for the presence of one or more pathogens or contaminants in a sample.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/587,403, filed Jan. 28, 2022, which is a continuation of U.S. application Ser. No. 16/480,677, filed Jul. 25, 2019, which is a 371 Application of International Application No. PCT/US2018/015111, filed Jan. 24, 2018 which claims priority to U.S. Provisional Application No. 62/450,518, filed Jan. 25, 2017, the entire contents of which are incorporated herein by reference.

TECHNOLOGICAL FIELD

[0002] The present technology relates generally to the field of molecule detection. In particular, the technology relates to microfluidic devices, systems, and methods for detecting the presence, absence and/or quantity of one or more particular analytes within a collected sample.

BACKGROUND

[0003] Conventional technologies for identifying the presence, absence and/or quantity of nucleic acids, proteins, and/or other molecules of interest within a sample often require expensive laboratory equipment and the expertise of highly-trained medical professionals. Consequently, such analyses are typically performed within laboratories or medical facilities. Such molecule detection can be important, for example, to detect the presence of pathogens, disease, contamination, overdoses, and poisonings within a person or other animal or within the environment.

Unfortunately, today, individuals may face long waits before the proper tests can be performed and before the results can be generated and analyzed. Due to the long waits and the inconvenience of traveling to a laboratory or medical facility, illnesses and contaminations often spread and may cause substantial harm before the presence of said illness or contamination is even identified.

SUMMARY

[0004] There is a significant need for improved molecule detection and quantification technologies. Described herein are devices that may detect molecules of interest in less time and with less technical expertise than the conventional devices used today. The devices may be utilized by consumers in non-clinical settings, for example, in schools, places of employment, and in the home. In addition, the devices may be used by consumers upon entering a pharmacy or healthcare facility, and may generate results quickly so that results are available by the time the consumer talks with a pharmacist or healthcare practitioner. The devices herein also may be configured to minimize biohazard risks.

[0005] One aspect of the disclosure is directed to a system for detecting molecules. In various embodiments, the system includes a cartridge device, a reader device removably coupled to the cartridge device, and a sample collection device.

[0006] Sample preparation reagents may be used in the system and may include a plurality of magnetic particles each having surface-bound affinity molecules, a plurality of detector agents which may each include a signaling agent, a plurality of amplification reagents, and/or a plurality of agents to facilitate access to a target analyte and binding between the target analyte and the surface-bound affinity molecules and the detector agents.

[0007] In accordance with one aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include an input tunnel, a reservoir, a reagent shuttle, and/or a sensor, each of which may be within

a housing of the cartridge. The input tunnel may extend from an aperture and may be configured to permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid, which may or may not be pre-filled within the reservoir. The reagent shuttle may be disposed between the reservoir and the aperture in a first position and the reagent shuttle may have a first end and a second end. The reagent shuttle may be configured to house a reagent ball comprising reagents (e.g., sample preparation reagents) between the first and second ends. The first end may be configured to seal the reservoir from the input tunnel in the first position. The reagent shuttle may be designed to move within the input tunnel when subjected to a force greater than a threshold force to a second position such that the reagent ball and the sample move into the reservoir. The reservoir may be continuously sealed from the input tunnel proximal to the reagent shuttle during movement from the first position to the second position. The sensor may be configured to analyze the fluid mixed with the reagent ball and the sample and further configured to generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample.

[0008] The second end of the shuttle may have an opening sized to wipe excess sample from a tip of the sample collection device such that, at most, a predetermined volume of the sample is mixed in the fluid within the reservoir. The reservoir may be sealed via the sample collection device partially inserted within the second end of the shuttle during movement from the first position to the second position. The cartridge may have one or more locking members configured to irreversibly lock the sample collection device within the input tunnel in the second position.

[0009] The shuttle may include one or more sample compartments configured to house, at most, a predetermined volume of the sample and one or more reagent ball compartments configured to house the reagent ball and, optionally, additional reagent balls. In some embodiments, the one or more sample compartments and the one or more reagent ball compartments are not exposed to the fluid within the reservoir in the first position. The one or more sample compartments and the one or more reagent ball compartments may be exposed to the fluid within the reservoir in the second position. The shuttle may have a compartment divider configured to divide at least one sample compartment from at least one reagent ball compartment. The compartment divider may have a slot configured to facilitate mixing when disposed within the sample preparation reservoir in the second position.

[0010] The reagents may include one or more of a plurality of solid particles, a plurality of affinity molecules, and/or a plurality of signaling agents. The reagents may include a plurality of magnetic particles configured to be magnetically held over a working electrode of the sensor. At least one magnetic particle of the plurality of magnetic particles may be configured to be indirectly bound to a signaling agent.

[0011] The cartridge may include an analysis channel, e.g., within the cartridge housing. At least a portion of the sensor may be disposed in the analysis channel and the fluid mixed with the reagent ball and the sample may travel to at least the portion of the sensor via the analysis channel.

[0012] The cartridge may include a contact switch, a sealing material configured to fluidically seal the fluid within the reservoir, and/or a seal piercer and each of those components may be within the cartridge housing. Insertion of the sample collection device within the input tunnel may cause: (i) the shuttle to move from the first position to the second position, (ii) the seal piercer to pierce the sealing material to vent the fluid in the reservoir, and/or (iii) activation of the contact switch.

[0013] In accordance with another aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include an input tunnel, a reservoir, a sealing material, and/or a seal piercer, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and the input tunnel may be configured to permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid, which may or may not be pre-filled. The sealing material may be configured to fluidically seal the fluid

within the reservoir. The seal piercer may be disposed partially within the input tunnel and the seal piercer may be configured to be contacted by the sample collection device within the input tunnel and to move, responsive to force applied by sample collection device, to cause the sealing material to be pierced to vent the fluid in the reservoir.

[0014] The seal piercer may be configured to move in a first direction and a second direction, different from the first direction, to pierce the sealing material. The first direction may be substantially parallel to movement of the sample collection device within the input tunnel and the second direction may be substantially perpendicular to the first direction. The seal piercer may include one or more piercers. The seal piercer may include a slider configured to move in a first direction and the one or more piercers may be configured to move in a second direction, different from the first direction, to pierce the sealing material.

[0015] The reservoir may be a sample preparation reservoir and may be configured to hold sample preparation reagents which may be within the fluid and/or introduced, e.g., via introduction of a reagent ball(s). The cartridge may also include a wash reservoir and/or a substrate reservoir. The seal piercer may be configured to cause the sealing material to be pierced to vent respective fluids in the sample preparation reservoir, the wash reservoir, and the substrate reservoir. The seal piercer may include an engager disposed within the input tunnel and the engager may be configured to engage an engagement zone of the sample collection device when the sample collection device is within the input tunnel.

[0016] The cartridge may include a contact switch which may be disposed on a circuit board within the housing. The contact switch may be configured to be activated upon insertion of the sample collection device within the input tunnel. Movement of the seal piercer may cause activation of the contact switch. The seal piercer may be configured to sequentially pierce the sealing material over the sample collection reservoir, the wash reservoir, and the substrate reservoir, in any order. The seal piercer may be configured to move out of one or more holes pierced in the sealing material after piercing to vent the fluid in the reservoir.

[0017] The cartridge may include a contact switch and a shuttle disposed between the reservoir and the aperture in a first position. The shuttle may have a first end and a second end and the first end may be configured to seal the reservoir from the input tunnel in the first position. The reagent shuttle may be configured to move within the input tunnel to a second position wherein the sample is moved into the reservoir. Insertion of the sample collection device within the input tunnel may cause: (i) the shuttle to move from the first position to the second position, (ii) the seal piercer to pierce the sealing material to vent the fluid in the reservoir, and/or (iii) activation of the contact switch. The cartridge may include one or more locking members configured to irreversibly lock the sample collection device within the input tunnel in the second position. The one or more locking members may irreversibly lock the sample collection device within the input tunnel during partial and/or full insertion of the sample collection device within the input tunnel.

[0018] Insertion of the sample collection device within the input tunnel may cause the seal piercer to pierce the sealing material to vent the fluid in the reservoir before the shuttle moves from the first position to the second position. Alternatively, or additionally, insertion of the sample collection device within the input tunnel may cause the seal piercer to pierce the sealing material to vent the fluid in the reservoir during movement of the shuttle from the first position to the second position.

[0019] In accordance with yet another aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include an input tunnel, a reservoir, an analysis channel, and/or a circuit board, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and the input tunnel may be configured to permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid and may be configured to receive the sample on the distal portion of the sample collection device. The analysis channel may be configured to receive, from the reservoir, the fluid having the sample

and reagents comprising a plurality of magnetic particles mixed therein. The circuit board may include a sensor having a working electrode and the sensor may be configured to be exposed to the mixed fluid in the analysis channel and to generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample. The working electrode may be masked with a plurality of striations configured to promote homogenous distribution of the plurality of magnetic particles over the working electrode and to promote resistance to movement of the plurality of magnetic particles off of the working electrode.

[0020] In accordance with yet another aspect, a kit is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The kit may include a sample collection device, a sample analysis cartridge, and/or a sample analysis reader. The sample collection device may have a distal portion adapted to be exposed to a sample. The sample analysis cartridge may include an input tunnel, a reservoir, an analysis channel, and/or a circuit board, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and may be configured to permit insertion of the sample collection device. The reservoir may be configured to hold a fluid and configured to receive the sample on the distal portion of the sample collection device. The analysis channel may be configured to receive, from the reservoir, the fluid having the sample and reagents comprising a plurality of magnetic particles mixed therein. The circuit board may include a sensor configured to be exposed to the mixed fluid in the analysis channel and to generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample. The sample analysis reader may be configured to receive the sample analysis cartridge. The sample analysis reader may have first and second magnetic generators configured to be disposed adjacent to a single working electrode of the sensor when the sample analysis cartridge is inserted in the sample analysis reader. The first and second magnetic generators may be configured to generate a magnetic field over the length of the single working electrode to promote homogenous distribution of the plurality of magnetic particles over the length of the single working electrode.

[0021] Receipt of the sample analysis cartridge by the sample analysis reader may cause electric coupling between the sample analysis cartridge and the sample analysis reader. The sample analysis cartridge may be configured to transmit the signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample to the sample analysis reader for processing. The sample analysis reader may be configured to transmit the processed signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample to a computer. The kit may include a computer readable medium with instructions that, when executed by a processor of the computer, cause a display of the computer to display information indicative of the presence, absence, and/or quantity of one or more target analytes.

[0022] A housing of the sample analysis cartridge may include a bottom surface having a magnetic generator depression. Receipt of the sample analysis cartridge in sample analysis reader may cause the first and second magnetic generators to move partially within the magnetic generator depression. The magnetic generator depression may be disposed adjacent the single working electrode.

[0023] In accordance with another aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include an input tunnel, a reservoir, a heater, an analysis channel, and/or a sensor, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and the input tunnel may be configured to permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid and configured to receive the sample on the distal portion of the sample collection device. The reservoir may include an outlet having a phase-changeable material therein to occlude an entire cross-section of the outlet. The heater may be configured to heat the phase-changeable material such that the phase-changeable material does not occlude the entire cross-section of the outlet. The analysis

channel may be configured to receive, from the reservoir through the outlet, the fluid having the sample and reagents comprising a plurality of magnetic particles mixed therein. The sensor may be configured to be exposed to the mixed fluid in the analysis channel and to generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample. The heater may be masked with a masking material configured to electrically isolate the heater from the sensor. The masking material may be a solder mask.

[0024] The cartridge may include a wash reservoir and a wash reservoir heater. The wash reservoir may be configured to hold a wash fluid and may include a wash reservoir outlet having a phase-changeable material therein to occlude an entire cross-section of the wash reservoir outlet. The wash reservoir heater may be configured to heat the phase-changeable material in the wash reservoir outlet such that the phase-changeable material does not occlude the entire cross-section of the wash reservoir outlet so that the wash fluid enters the analysis channel and travels to the sensor. The wash reservoir heater may be masked with a masking material configured to electrically isolate the wash reservoir heater from the sensor.

[0025] The cartridge may include a substrate reservoir and a substrate reservoir heater. The substrate reservoir may be configured to hold a substrate fluid and may include a substrate reservoir outlet having a phase-changeable material therein to occlude an entire cross-section of the substrate reservoir outlet. The substrate reservoir heater may be configured to heat the phase-changeable material in the substrate reservoir outlet such that the phase-changeable material does not occlude the entire cross-section of the substrate reservoir outlet so that the substrate fluid enters the analysis channel and travels to the sensor. The substrate reservoir heater may be masked with a masking material configured to electrically isolate the substrate reservoir heater from the sensor.

[0026] In accordance with yet another aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include a sample preparation reservoir, a substrate reservoir, an analysis channel, a fluidic isolator, and/or one or more heaters, each of which may be within a housing of the cartridge. The sample preparation reservoir may be configured to hold a fluid and may be configured to receive a sample from a sample collection device. The sample preparation reservoir may include a sample preparation reservoir outlet having phase-changeable material therein to seal the sample preparation reservoir outlet. The substrate reservoir may be configured to hold a fluid comprising a chemical substrate. The substrate reservoir may include a substrate reservoir outlet having phase-changeable material therein to seal the substrate reservoir outlet. Each of the sample preparation and substrate reservoirs may be, at least at times, in fluid communication with the analysis channel. The fluidic isolator may be phase-changeable material. At least one of the one or more heaters may be configured to heat phase-changeable material within the sample preparation reservoir outlet such that the phase-changeable material unseals the sample preparation reservoir outlet to permit the fluid having the sample mixed therein to flow into the analysis channel. At least one of the one or more heaters may be configured to heat phase-changeable material of the fluidic isolator after unsealing the sample preparation reservoir outlet such that phase-changeable material of the fluidic isolator flows into the analysis channel to fluidically isolate the sample preparation reservoir from the substrate reservoir. At least one of the one or more heaters may be configured to heat phase-changeable material within the substrate reservoir outlet, after the fluidic isolator fluidically isolates the sample preparation reservoir from the substrate reservoir, such that the phase-changeable material unseals the substrate reservoir outlet to permit the fluid comprising the chemical substrate to flow into the analysis channel, but not into the sample preparation reservoir.

[0027] The one or more heaters may include a sample preparation reservoir heater, a fluidic isolator heater, and/or a substrate reservoir heater. The sample preparation reservoir heater may be configured to heat the phase-changeable material within the sample preparation reservoir outlet. The fluidic isolator heater may be configured to heat the phase-changeable material of the fluidic isolator. The substrate reservoir heater may be configured to heat the phase-changeable material

within the substrate reservoir outlet. The sample preparation reservoir heater, the fluidic isolator heater, and the substrate reservoir heater may each be masked with a masking material configured to electrically isolate the respective heater from a sensor in the analysis channel.

[0028] The cartridge may include a wash reservoir configured to hold a wash fluid. The wash reservoir may include a wash reservoir outlet having phase-changeable material therein to seal the wash reservoir outlet. At least one of the one or more heaters may be configured to heat phase-changeable material within the wash reservoir outlet, after the fluidic isolator fluidically isolates the sample preparation reservoir from the substrate reservoir, but before the one or more heaters heats the phase-changeable material within the substrate reservoir outlet, such that the phase-changeable material unseals the wash reservoir outlet to permit the wash fluid to flow into the analysis channel to wash signaling agents unbound to magnetic particles from the sample preparation reservoir off a sensor in the analysis channel. The fluid having the chemical substrate may be configured to wash signaling agents unbound to magnetic particles from the sample preparation reservoir off a sensor in the analysis channel.

[0029] In accordance with another aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include an input tunnel, a reservoir, a shuttle, and/or a sensor, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and the input tunnel may be configured to permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample fluid. The reservoir may be configured to hold a fluid, which may or may not be pre-filled. The shuttle may be disposed between the reservoir and the aperture in a first position. The shuttle may have a first end and a second end and may define a sample compartment between the first and second ends. The sample compartment may be configured to receive the sample fluid compressed from the distal portion of the sample collection device. The shuttle may be configured to move within the input tunnel when subjected to a force greater than a threshold force to a second position such that the sample compartment having the sample fluid is moved into the reservoir. The sensor may be configured to be exposed to the fluid mixed with the sample fluid and the sensor may be further configured to generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample fluid.

[0030] The shuttle may further define a reagent ball compartment between the first and second ends. The reagent ball compartment may be configured to house one or more reagent balls comprising reagents. The reagent ball compartment may be outside the reservoir in the first position and in the reservoir in the second position. The sample compartment may be configured to receive, at most, a predetermined volume of the sample fluid compressed from the distal portion of the sample collection device. The cartridge may include an overflow compartment configured to receive a volume of the sample fluid from the sample compartment above the predetermined volume.

[0031] The reagent ball comprises reagents necessary to carry out amplification of the target analyte. The reagent ball can be of any appropriate and shape, non-limiting examples of such include a diameter of between about 1 mm to about 7 mm, or alternatively from about 2 mm to about 5 mm, or alternatively about 3 mm, or alternatively less than about 7 mm, or alternatively less than about 5 mm, or alternatively less than about 4 mm. The reagent ball can be of any appropriate shape, e.g., spherical, cylindrical, conical or ellipsoid.

[0032] The components of the reagent ball are preselected for the analyte and its method for amplification and subsequent detection and/or quantification. In one aspect, the components of the reagent ball comprise reagents for the detection or quantification of a hormone, other small molecule or a protein or fragment. In another aspect, the components of the reagent ball comprise reagents for the detection and/or quantification of a nucleic acid by a method that comprises amplifying the nucleic acid.

[0033] A kit may be provided including the sample analysis cartridge and the sample collection

device. The sample collection device may include a wicking portion at the distal portion. The wicking portion may be configured to wick and absorb the sample fluid. The wicking portion may be compressed to expel the sample fluid into the sample compartment. The sample compartment may be configured to receive, at most, a predetermined volume of the sample fluid compressed from the distal portion of the sample collection device. The sample analysis cartridge may further include an overflow compartment configured to receive a volume of the sample fluid from the sample compartment above the predetermined volume. The wicking portion of the sample collection device may be configured to wick and absorb sample fluid above the predetermined volume to permit a user to meter a quantity of sample fluid compressed into the sample compartment and the overflow compartment. At least some of the wicking portion may be slidably disposed within a shroud of the sample collection device.

[0034] The sample analysis cartridge may further include one or more locking members configured to irreversibly lock the sample collection device within the input tunnel when the sample collection device is fully inserted in the input tunnel. The sample collection device may further include a sample collection indicator configured to visually alert a collector based on a volume of sample fluid that has been collected. The sample collection indicator may be a colored thread embedded in the wicking portion that becomes increasingly visually exposed as the volume of sample collected increases.

[0035] In accordance with another aspect, compositions and method are provided for detecting and/or quantifying at least one of a presence, absence, or quantity of a target analyte within a sample in a cartridge. The method may include mixing, within a fluid in a reservoir of the cartridge, a plurality of affinity molecules, a plurality of de-binding agents, a plurality of signaling agents, a plurality of competitor molecules pre-bound to competitor binding molecules, each of the plurality of competitor molecules possessing a label, and the sample having a plurality of sample target analytes pre-bound to sample binding molecules; de-binding at least one competitor molecule from the pre-bound competitor binding molecule using at least one de-binding agent of the plurality of de-binding agents; de-binding at least one sample target analyte from the pre-bound sample binding molecule using at least one de-binding agent of the plurality of de-binding agents; binding the label of the de-bound competitor molecule to a signaling agent of the plurality of signaling agents; binding the de-bound competitor molecule to an affinity molecule of the plurality of affinity molecules; and/or generating a signal indicative of at least one of a presence, absence, or quantity of the sample target analyte within the cartridge. The reagent ball can be of many sizes, non-limiting examples of such include having an diameter of between about 1 mm to about 7 mm, or alternatively from about 2 mm to about 5 mm, or alternatively about 3 mm, or alternatively less than about 7 mm, or alternatively less than about 5 mm, or alternatively less than about 4 mm. While reagent ball is illustrated as a sphere, the disclosure is not limited thereto and many shapes may be used and multiple reagent balls each containing the same or different reagents also may be used.

[0036] In accordance with another aspect, compositions and methods are provided for amplifying and detecting and/or quantifying at least one of a presence, absence, or quantity of a target analyte, e.g., a target nucleic acid (deoxyribonucleic acid (DNA) or ribonucleic acid (RNA)) within a sample in a cartridge. The method comprises or alternatively consists essentially of preparing a plurality of amplicons comprising a plurality of capture elements, by mixing within a fluid in a reservoir of the cartridge: a plurality of enzymes to facilitate an amplification reaction, e.g., polymerases; reverse transcriptases; a plurality of magnetic beads having coupled thereto an affinity molecules; and a plurality of forward and reverse primers, that may or may not be labeled.

[0037] In a further aspect, provided herein are compositions and methods for the use, wherein the reagent ball further contains a plurality of forward primers selected to amplify the target nucleic acid and having coupled to each thereto a spacer element and a capture element; a plurality of reverse primers selected to amplify the target nucleic acid and having coupled to each thereto a

spacer element and a signaling agent or a signaling capture element; a plurality of nucleotides or analogs thereof (dNTPs) for the amplification reaction.

[0038] In a further aspect, the reagent ball further contains a reverse primer selected to bind a target nucleic acid, the reverse primer comprising a spacer element that is directly or indirectly conjugated to a reporter capture element. In a further aspect, the reagent ball further contains an effective amount of reverse transcriptase, effective to facilitate the amplification reaction.

[0039] In a yet further aspect, the reagent ball also contains a plurality of reporter affinity element conjugated to a reporter element.

[0040] In a further aspect, the reagent ball also contains single stranded binding proteins, e.g., from about 9 to about 18 amino acids, known to those of ordinary skill in the art such as but not limited to SSB from *E. coli* or GP 32 from phages.

[0041] In one aspect, the reagent ball further contains, a plurality of DNA template control nucleic acids. In another aspect, the reagent ball alternatively or also contains a plurality of RNA template control nucleic acids.

[0042] In a further aspect, the reagent ball may contain a plurality of reverse transcriptase-specific primers to facilitate reverse transcription of RNA to cDNA. In a yet further aspect, the reagent ball may also contain a plurality of reverse primers selected to serve as an internal control having coupled to each thereto a spacer element and a signaling agent; and a plurality of forward primers selected to serve as an internal control having coupled to each thereto a spacer element and a capture element.

[0043] In addition to the labeled primers, the reagent ball can contain an effective amount of a plurality of unlabelled primers designed to amplify at least the same target region but optionally flanking sequences to the target sequence. The presence of unlabeled primers can make amplification more efficient since labelled primers may be more sterically hindered as they bind to other elements e.g. solid particles or signaling agents.

[0044] The reagent ball can further contain an effective amount of one or lysis agents to free the target nucleic acid, templates and/or control(s) from a cell, microbe, or virus in the sample.

[0045] In a further aspect, the reagent ball contains an effective amount of a helicase to unwind dsDNA for loading primers or RecA or its analogues such as UvsX or RAD51. Further, the reagent ball may contain mutL, RecFOR enzymes, UvsY.

[0046] In one aspect the amplification reagents are selected for any one or more other amplification reactions, e.g., PCR methods or isothermal amplification. The reporter element and/or the capture element is located at 5' termini of the nucleic acid or along the nucleic acid sequence, i.e., internal to 5' end. The reporter and/or the capture element is covalently or non-covalently attached to the nucleic acid.

[0047] In one aspect the elements are provided in a sample reagent ball and mixed with the sample in the reservoir. Upon disintegration the reagents come in contact with the target nucleic acid and hybridization of the primers to the target nucleic acid and amplification of the target through a series of enzymatically driven melting and reconstruction of the DNA or in the case of target RNA, a complementary DNA (cDNA) molecule is first created from the RNA target nucleic acid and the double stranded cDNA containing the target sequence then serves as the template for further amplification. The reagent ball can be of many sizes, non-limiting examples of such include having an diameter of between about 1 mm to about 7 mm, or alternatively from about 2 mm to about 5 mm, or alternatively about 3 mm, or alternatively less than about 7 mm, or alternatively less than about 5 mm, or alternatively less than about 4 mm. While reagent ball is illustrated as a sphere, the disclosure is not limited thereto and many shapes may be used and multiple reagent balls each containing the same or different reagents also may be used. The spacer element separating the primer from a label comprises a polymer e.g., hexaethylene glycol or triethylene glycol.

Alternatively it can be a linear carbon polymer, e.g., hexane, pentane, containing from about 1 to about 18 carbon atoms or more.

[0048] As is apparent to the skilled artisan, combinations of the above embodiments, necessary to promote the specific amplification of a target nucleic acid, are intended within the scope of this disclosure.

[0049] The reagents and amounts thereof are pre-selected to facilitate the specific amplification of the target nucleic acids. For the purpose of illustration, non-limiting examples of reverse transcriptases are moloney murine leukemia virus (MMLV) or a derivative thereof or avian myeloblastosis virus (AMV) or a derivative thereof. The preferred reverse transcriptase will be dependent on the target and may not be the same between pellets for different targets, as it can be appreciated that the reagent ball can be used in a variety of different testing applications by modifying primers sequences and capture elements, primer concentrations, particle concentrations, affinity agents, lysis agents, polymerases, reverse transcriptases, and other enzymes to best match the optimal condition for each type of target (e.g. HIV quantification vs. influenza detection may have different reaction conditions).

[0050] Polymerases can include several different types, within the strand displacement polymerase category, options include for instance Bsu DNA polymerase or a fragment thereof such as Bsu DNA polymerase large fragment, Bst DNA polymerase, or a fragment thereof such as Bst DNA polymerase large fragment, phi29 DNA polymerase. It can be appreciated that for polymerases and reverse transcriptases it is often a desired property to have certain mutants such as polymerases lacking exonuclease activity or for the reverse transcriptase lacking RNase H activity.

[0051] Preferred reaction temperatures for an isothermal reaction can depend on the method and the reaction conditions and can include around 65 degrees Celsius as is often the case for LAMP and 55 degrees Celsius for Nicking Enzyme Amplification Reaction. A preferred but not limiting reaction temperature range is between about 37 and 42 degrees for the reverse transcriptase portion of an amplification reaction if the target nucleic acid is an RNA. Helicase Dependent Amplification, Strand Displacement Amplification, Recombinase Polymerase Amplification can all occur at about 37 degrees celsius or between 37 degrees Celsius and 42 degrees Celsius. FIG. 20 shows the temperature profile of an isothermal reaction occurring in a reservoir at around 40 degrees Celsius.

[0052] It may be desired to include single-stranded binding proteins (SSB) to facilitate several isothermal amplification techniques as these proteins help stabilize the unwinding and strand displacement polymerization of complement strands during amplification. Examples include but are not limited to RB 49 GP 32, RB 69 GP 32, T4 GP32, *E. coli*'s SSB protein, and others.

[0053] In some aspects, the reagent ball and the method further uses a helicase to unwind dsDNA for loading primers. Non-limiting examples include enzymes such as uvrD helicase from *E. coli*, T4 Gene 41 helicase, and many others. Recombinases that facilitate primer loading into dsDNA for enzymatic melting of duplex DNA for primer annealing can include RecA from *E. coli*, RAD51 human recombinase, DMC1 human meiotic recombinase, or analogues from phage such as T4 UvsX, RB 49 UvsX, RB 69 UvsX and many others. As is known to those of ordinary skill in the art, combinations of helicase and SSB are helpful for facilitating isothermal amplification. Accessory factors such as MutL can be added to facilitate helicase dependent amplification. Combinations of recombinases and SSB can be helpful in RPA and sometimes accessory factors such as RecFOR from *E. coli* and/or UvsY from various phages are employed as well to facilitate the reaction by helping the primary enzyme (RecA) or (uvrD) within recombinase polymerase amplification and helicase dependent amplification respectively. As appreciated by the skilled artisan, the reagent ball and/or reservoir can further contain any one or more of the above reagents as necessary to facilitate the specific amplification of the target nucleic acid.

[0054] Primer concentrations for isothermal amplification reactions such as SDA, HDA, and RPA can be between 0.01 and 10 micromolar, preferably closer to 0.5 micromolar. A LAMP Primer mix can be prepared with all 4 or 6 (with Loop) primers. A 10× Primer Mix could contain: 16 μM FIP, 16 μM BIP, 2 μM F3, 2 μM BE, 4 μM LoopF, 4 μM LoopB. dNTPs can be provided in

concentrations such as between 1 micromolar and 500 micromolar, preferably around 200 micromolar. SDA, HDA, RPA may require a high amount of ATP as some of the enzymes that allow for enzymatic melting and loading of primers into dsDNA require ATP to function and therefore as much as 100 micromolar to 4 millimolar of ATP can be used within a reaction.

[0055] Polymerase amounts can vary depending on the target but can be in the range of 1 unit to 1000 units per reaction. Reverse transcriptases can also be provided at such a range for a successful reaction. Magnesium is an essential co-factor for polymerase activity and can be provided in the reagent ball or in the reservoir at amounts well known in the art such as 5-50 millimolar, typically around 10 millimolar.

[0056] Methods and compositions for the non-covalent linkage of molecules that also can provide a label or signal for detection are known in the art. Non-limiting examples of such include avidin or streptavidin-biotin conjugation. Modifications to biotin are known in the art and intended within the scope of this disclosure. Non-limiting examples of such include biotin dT, biotin-TEG, dual biotin, PC biotin and desthioBiotin-TEG commercially available from Integrated DNA Technologies (see idtdna.com/pages/decoded/decoded-articles/core-concepts/decoded/2012/09/20/which-biotin-modification-to-use-, last accessed Jul. 16, 2016.) This disclosure also includes the use of additional conjugation chemistries for the linking of nucleic acids to proteins such as when a primer is conjugated directly to a signaling agent wherein in one aspect, the signaling agent is an enzyme such as HRP. Non-limiting examples of covalently joining a protein or polypeptide to another moiety include linking the moiety to a crosslinking reactive group, e.g., carbodiimides, imidoesters, and maleimides. See e.g., Bioconjugate Techniques, 3rd ed, Hermanson, G. T. (2013).

[0057] As is apparent to the skilled artisan, the components of the reagent ball are selected to facilitate the amplification of the target nucleic acid and/or targets, and/or controls by the appropriate method. In one aspect, the reagents are selected for rolling circle amplification (RCA), or loop-mediated isothermal amplification. In another aspect they are selected for amplification by the (LAMP) method. In another aspect they are selected for, strand displacement amplification (SDA). In another aspect they are selected for recombinase polymerase amplification (RPA). In another aspect they are selected for, helicase dependent amplification (HDA). In another aspect they are selected for, polymerase spiral reaction (PSR). In another aspect they are selected for nicking enzyme amplification reaction (NEAR). As indicated above, each particular reaction type has its own preferred combination of enzymes and components that allow efficient and selective amplification of the target and/or targets, and/or internal control nucleic acids and are known to the skilled artisan.

[0058] Each of the affinity molecules of the plurality of affinity molecules may be bound to a solid particle. The solid particle may be formed of magnetically responsive material and/or may be formed of non-magnetically responsive material. The non-magnetically responsive material may be gold nanoparticles.

[0059] The plurality of sample target analytes pre-bound to sample binding molecules may include 25-hydroxy Vitamin D2 or 25-hydroxy Vitamin D3 molecules pre-bound to vitamin D binding protein molecules. The plurality of competitor molecules pre-bound to sample binding molecules may include 25-hydroxy Vitamin D2 or 25-hydroxy Vitamin D3 molecules labeled with biotin and prebound to vitamin D binding protein molecules. At least one of the plurality of affinity molecules, the plurality of de-binding agents, the plurality of signaling agents, and the plurality of competitor molecules pre-bound to competitor binding molecules may be stored within a reagent ball.

[0060] As noted above, the primers (forward and reverse for the target nucleic acid and control templates) are designed based on the nucleotide sequence of the target nucleic acid to be amplified and detected if present. Methods to design optimal primers based on a target sequence are known in the art, see e.g., simgene.com/Primer3; quill.com; molbiol-tools.caPCR; and

ncbi.nlm.nih.gov/tools/primer-blast/, each last accessed on Jul. 15, 2016, and vary with the amplification method utilized, e.g., rolling circle amplification (RCA), loop-mediated isothermal amplification (LAMP), strand displacement amplification (SDA), recombinase polymerase amplification (RPA), helicase dependent amplification (HDA), polymerase spiral reaction (PSR), and nicking enzyme amplification reaction (NEAR).

[0061] In accordance with yet another aspect, a sample analysis cartridge is provided for detecting at least one of a presence, absence, or quantity of one or more analytes. The sample analysis cartridge may include a reagent ball, a reservoir, and/or a sensor, each of which may be within a housing of the cartridge. The reagent ball may include a plurality of competitor molecules pre-bound to competitor binding molecules, each of the plurality of competitor molecules may possess a label or bound to a signaling agent. In another aspect, the reagent ball comprises, or alternatively consists essentially of, the reagents necessary for amplification and detection of a target nucleic acid. The reservoir may be configured to hold a reservoir fluid, which may or may not be pre-filled in the reservoir. The reservoir may be configured to permit mixing of, within the reservoir fluid, a plurality of affinity molecules, a plurality of de-binding agents, a plurality of signaling agents, the plurality of competitor molecules pre-bound to competitor binding molecules, and a sample from a sample collection device, the sample having a plurality of sample target analytes pre-bound to sample binding molecules. A de-binding agent of the plurality of de-binding agents may be configured to de-bind a competitor molecule from the pre-bound competitor binding molecule or a sample target analyte from the pre-bound sample binding molecule. The label of the de-bound competitor molecule may be configured to bind to a signaling agent and the de-bound competitor molecule may be configured to bind to an affinity molecule of the plurality of affinity molecules. In another aspect, the reservoir is configured to hold a reservoir fluid, which may or may not be prefilled in the reservoir. The reservoir configured to permit mixing of, within the reservoir fluid, a plurality of enzymes to facilitate an amplification reaction, e.g., polymerases, reverse transcriptases; a plurality of magnetic beads having coupled thereto an affinity molecule; a plurality of forward primers selected to amplify the target nucleic acid and having coupled to each thereto a spacer element; a plurality of reverse primers selected to amplify the target nucleic acid and having coupled to each thereto a spacer element and a signaling agent or a signaling capture element; a plurality of nucleotides or analogs thereof (dNTPs) for the amplification reaction; a plurality of DNA template control nucleic acids; a plurality of reverse primers selected to serve as an internal control having coupled to each thereto a spacer element and a signaling agent; and a plurality of forward primers selected to serve as an internal control having coupled to each thereto a spacer element and a capture element. In one aspect the amplification reagents are selected for any one or more other PCR methods or isothermal amplification.

[0062] In a further aspect, the reagent ball and/or the reservoir contains an effective amount of a lysing agent to lyse a sample comprising a cell, e.g., a bacterial sample to release intracellular DNA, RNA and/or proteins that serve as analytes. Non-limiting example of lysing agents include NP-40, CHAPS, deoxycholate, Triton X-100, NP40, and Tween 20.

[0063] In a further aspect, the reagent ball and/or the reservoir contains an RNase inhibitor and/or DNase inhibitor and/or protease inhibitor in an amount to inhibit RNase, DNase or protease activity native or endogenous to the sample being added for analysis.

[0064] Spacer elements can be advantageous because in some cases the primer can participate better in the amplification reaction if the nucleic acid portion of the primer is more distant from the label where other sterically hindering events could be occurring or have already occurred such as being bound to a particle or being bound to a signaling agent, both of which may be bulky. The spacer element separating the primer from a label comprises a polymer e.g., hexaethylene glycol, triethlyene glycol, a C3 spacer phosphoramidite, a PC spacer, hexanediol and are commercially available from Integrated DNA Technologies (see idtdna.com/site/Catalog/Modifications/Category/6, last accessed Jul. 16, 2016).

[0065] The sensor may be configured to be exposed to the mixed reservoir fluid and the sensor may be further configured to generate a signal indicative of at least one of the presence, absence, or quantity of the sample target analyte within the sample. For example, particles from the mixed reservoir fluid comprising signaling agents may localize in an analysis channel over the sensor and the localized signaling agents may react with substrates from a substrate solution to generate electrical signals sensed by the sensor. The sensor may send the signal indicative of at least one of the presence, absence, or quantity of the sample target analyte within the sample using the electrical signals from the reaction.

[0066] The reservoir may be further configured to permit mixing of a plurality of solid particles within the reservoir fluid. Each solid particle may be pre-bound to an affinity molecule of the plurality of affinity molecules. The plurality of solid particles may be formed of magnetically responsive material and/or formed of non-magnetically responsive material. The non-magnetically responsive material may be gold nanoparticles. The plurality of sample target analytes pre-bound to sample binding molecules may include 25-Hydroxy vitamin D3 and/or 25-hydroxy vitamin D2 molecules pre-bound to binding protein molecules. The magnetically responsive material may be magnetically held over the sensor.

[0067] In accordance with yet another aspect, a sample analysis cartridge is provided. The sample analysis cartridge may include an input tunnel, a reservoir, a shuttle, and/or a collet, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and be configured to permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid. The shuttle may be disposed in the input tunnel between the reservoir and the aperture in a first position. The collet may be disposed in the input tunnel and coupled to the shuttle in the first position. The collet may decouple from the shuttle during insertion of the sample collection device in the input tunnel. The shuttle may move within the input tunnel from the first position to a second position after the collet is decoupled from the shuttle such that the shuttle is at least partially disposed within the reservoir in the second position.

[0068] The sample analysis cartridge may include a sensor configured to be exposed to the fluid mixed with the sample. The sensor may generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample. The shuttle may have a first end and a second end disposed proximal to the first end in the input tunnel. The second end of the shuttle may be configured to be disposed within a lumen of the collet in the first position. The first end of the shuttle may form a wall of the reservoir in the first position.

[0069] The collet may have one or more locking arms configured to couple the collet to the shuttle in the first position. The one or more locking arms may be configured to be deflected to decouple the one or more locking arms from the shuttle responsive to a force applied on the one or more locking arms by the sample collection device during insertion of the sample collection device in the input tunnel.

[0070] The sample analysis cartridge may include a sealing material configured to fluidically seal the fluid within the reservoir and a seal piercer disposed partially within the input tunnel. The seal piercer may be configured to be contacted by the sample collection device within the input tunnel and to move, responsive to force applied by the sample collection device, to pierce the sealing material to vent the fluid in the reservoir. The collet may have a slot and a portion of the seal piercer may extend through the slot into the input tunnel to permit contact between the seal piercer and the sample collection device.

[0071] The sample analysis cartridge may include a contact switch. The collet may have a deflector portion disposed adjacent the contact switch and configured to deflect to activate the contact switch responsive to a force applied on the deflector portion by the sample collection device during insertion of the sample collection device in the input tunnel. The deflector portion of the collet may include an arm configured to deflect downward to activate the contact switch. The contact switch

may be positioned in the input tunnel such that activation of the contact switch indicates full insertion of the sample collection device in the input tunnel.

[0072] The shuttle may be configured to house a reagent ball comprising reagents between first and second ends of the shuttle. Preferably, the reagent ball is not exposed to the fluid in the reservoir in the first position and is exposed to the fluid in the reservoir in the second position.

[0073] In accordance with yet another aspect, a sample analysis cartridge is provided. The sample analysis cartridge may include an input tunnel, a reservoir, a collet, and/or a contact switch, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and may permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid. The collet may be disposed in the input tunnel between the reservoir and the aperture. The collet may have a deflector portion and a lumen sized to receive the distal portion of the sample collection device therein. The contact switch may be disposed adjacent the deflector portion of the collet. The deflector portion may be configured to deflect to activate the contact switch responsive to a force applied on the deflector portion by the sample collection device during insertion of the sample collection device in the input tunnel.

[0074] The sample analysis cartridge may include a shuttle disposed within the input tunnel and configured to house a reagent ball(s) comprising reagents between first and second ends of the shuttle. The deflector portion of the collet may be an arm configured to deflect downward to activate the contact switch. The contact switch may be positioned such that activation of the contact switch indicates full insertion of the sample collection device in the input tunnel.

[0075] The sample analysis cartridge may include a sensor configured to be exposed to the fluid mixed with the sample. The sensor may generate a signal indicative of at least one of the presence, absence, or quantity of the one or more analytes within the sample.

[0076] In accordance with another aspect, a sample analysis cartridge is provided. The sample analysis cartridge may include an input tunnel, a reservoir, a shuttle, and/or a sonicator, each of which may be within a housing of the cartridge. The input tunnel may extend from an aperture and may permit insertion of a sample collection device having a distal portion adapted to be exposed to a sample. The reservoir may be configured to hold a fluid. The shuttle may define a first compartment and a second compartment. The first and second compartments may be configured to be disposed within the reservoir in a mixing position. The sonicator may be configured to emit acoustic waves to move the fluid in the reservoir in a wave pattern between the first and second compartments to mix the fluid in the reservoir.

[0077] The shuttle may include a compartment divider configured to divide the first compartment from the second compartment. The compartment divider may be a flange. Fluid flowing around the compartment divider may facilitate formation of the wave pattern. The compartment divider may have a slot configured to permit the fluid to flow through the compartment divider via the slot during mixing. The first compartment may be a reagent ball compartment configured to house a reagent ball including reagents and the second compartment may be a sample compartment configured to receive the sample from the sample collection device, e.g., expelled from the sample collection device and/or on the distal portion of the sample collection device. The reagent ball may include, as described above, reagents for the amplification of a target nucleic acid, e.g., polymerases, primers, and signaling agents. The first and second compartments are preferably not disposed within the reservoir in a pre-mixing position.

[0078] The sonicator may be a piezoelectric transducer such as a piezoceramic disc. The sonicator may form a wall of the reservoir, e.g., part of the bottom wall of the reservoir. The reservoir may be symmetric. Each of the walls of the reservoir may meet at an angle greater than a predetermined angle such as 60° to facilitate fluid emptying through an outlet of the reservoir. The sonicator may be positioned off-center of the reservoir to facilitate mixing of the fluid within the reservoir.

[0079] The sample analysis cartridge may include a printed circuit board coupled to the sonicator via one or more spring contacts. The sonicator may be electrically coupled to the printed circuit

board only via the one or more spring contacts. The sonicator may be activated responsive to a signal from a processor, e.g., the processor of the reader.

[0080] The sample analysis cartridge may include a temperature sensor configured to sense temperature indicative of temperature of the fluid in the reservoir. The temperature sensor may be disposed on a printed circuit board positioned adjacent the sonicator.

[0081] The sample analysis cartridge may include a contact switch configured to indicate insertion of the sample collection device in the input tunnel. The sonicator may be configured to emit the acoustic waves after actuation of the contact switch. For example, the reader may direct the sonicator to emit the acoustic waves after the reader receives an electrical signal indicating that the contact switch has been activated.

[0082] The acoustic waves emitted by the sonicator may be configured to isothermally amplify reactions of the fluid mixed in the reservoir.

[0083] In accordance with another aspect, a method for isothermal amplification of a target nucleic acid if present in a sample analysis cartridge is provided. The method comprises or alternatively consists of contacting in the reservoir a reagent ball as described above and containing a plurality of reagents pre-selected for the amplification and detection of the target nucleic acid with the sample to produce an amplicon-signaling agent complex coupled to the solid particle. The amplicon comprises a nucleic acid duplex comprising: a reverse primer complex comprising a nucleic acid comprising the target nucleic acid coupled to a spacer element that in turn is coupled to a signaling agent and a forward primer complex comprising a nucleic acid comprising the target sequence coupled at one end to a capture element. In a further aspect, the reverse primer complex further comprises a signaling affinity element conjugated to the signaling agent and the spacer element. The amplicon-signaling agent complex in turn is conjugated to an affinity element on the solid particle that in turn, can be bound or held to the sensor surface over a magnetic field. The amplicon-signaling agent complex in turn is conjugated to an affinity element on the solid particle that in turn, can be held to the sensor surface over a magnetic field. If the analyte is present, the sensor detects and/or quantifies the signaling agent-labeled amplicon.

[0084] A sonicator may emit acoustic waves toward the reservoir to promote amplification of the target nucleic acid in the reservoir. As noted above, the amplicon-signaling agent complex may be reacted with a substrate from a substrate reservoir. For example, the reaction may occur over a sensor in an analysis channel. A signal indicative of at least one of a presence, absence or quantity of amplified nucleic acid may be generated. The signal may be transmitted from the cartridge to another device such as a reader.

[0085] A reagent ball may be held in a shuttle. The shuttle may be disposed in an input tunnel of the cartridge. The reagent ball may comprise reagents for amplification of a target nucleic acid by an isothermal reaction as noted above. The reagents may comprise a polymerase, primers for amplification of the target nucleic acid and a signaling agent for the detection of the amplification of the target nucleic acid. One or more affinity molecules may be covalently or non-covalently bound to a solid particle for detection of the target nucleic acid.

[0086] In accordance with another aspect, a kit is provided. The kit may include a reservoir, a sonicator, a temperature sensor, and/or a processor. The reservoir may be configured to hold a fluid and to receive a sample collected by a sample collection device. The sonicator may be configured to emit acoustic waves to mix the fluid and the sample in the reservoir. The temperature sensor may be configured to generate a signal indicative of temperature of the fluid in the reservoir. The processor may be configured to activate the sonicator to emit the acoustic waves and to monitor the signal from the temperature sensor. The processor further may be configured to modify emission of the acoustic waves from the sonicator if the signal indicates a temperature of the fluid in the reservoir outside a threshold. A sample analysis cartridge may include the reservoir, the sonicator, and/or the temperature sensor, each of which may be within a housing of the cartridge, and a reader may include the processor. The reader may be configured to be electrically coupled to the sample

analysis cartridge.

[0087] The sample analysis cartridge may include a printed circuit board and the temperature sensor may be disposed on the printed circuit board positioned adjacent the sonicator. The sample analysis cartridge may include a contact switch configured to generate a signal to indicate insertion of the sample collection device in an input tunnel of the sample analysis cartridge. The processor of the reader may be configured to receive the signal from the contact switch and to activate the sonicator after receipt of the signal from the contact switch. The processor may modify emission of the acoustic waves from the sonicator by lowering a duty cycle of the sonicator if the signal indicates the temperature of the fluid in the reservoir is above the threshold. The processor may modify emission of the acoustic waves from the sonicator by increasing a duty cycle of the sonicator if the signal indicates the temperature of the fluid in the reservoir is below the threshold. The processor may modify emission of the acoustic waves from the sonicator by deactivating the sonicator if the signal indicates the temperature of the fluid in the reservoir is above the threshold.

[0088] The acoustic waves emitted by the sonicator may be configured to isothermally amplify reactions of the fluid mixed in the reservoir. The sample analysis cartridge may include a reagent ball disposed within the sample analysis cartridge. The sonicator may be configured to emit the acoustic waves to mix the fluid, the reagent ball, and the sample in the reservoir. The reagent ball may include polymerases, primers, and signaling agents. The sample analysis cartridge may include a shuttle configured to house the reagent ball.

[0089] In accordance with another aspect, a sensor for use in a microfluidic cartridge is provided. The sensor may include a positive control working electrode, a working electrode, and/or a negative control working electrode. The positive control working electrode may include affinity molecules pre-bound to the positive control working electrode. For example, the affinity molecules may be pre-bound to a surface of the positive control working electrode disposed within an analysis channel of the cartridge. The positive control working electrode may be configured to generate a first signal based on a reaction between signaling agents directly or indirectly bound to the affinity molecules and a chemical substrate. The signaling agents may be from a reagent ball(s). The chemical substrate may be from fluid stored in the substrate reservoir. The working electrode may be configured to generate a second signal based on a reaction between the signaling agents localized at the working electrode and the chemical substrate. The negative control working electrode may include a self-assembled monolayer. For example, the self-assembled monolayer may be at a surface of the negative control working electrode disposed within an analysis channel of the cartridge. The negative control working electrode may be configured to generate a third signal based on a reaction between the signaling agents localized at the negative working electrode and the chemical substrate. As should be understood, “first”, “second”, and “third” differentiate terms and do not necessarily mean order.

[0090] The second signal may be indicative of at least one of the presence, absence, or quantity of one or more analytes within a sample. The first signal may be indicative of reliability of a test. For example, the test may be determined to be reliable if the first signal indicates a quantity of the reaction within a predetermined range. The third signal is indicative of reliability of a test. For example, the test may be determined to be reliable if the third signal indicates a quantity of the reaction is below a threshold.

[0091] A cartridge may include the sensor. The cartridge may have an analysis channel and the positive control working electrode, the working electrode, and the negative control working electrode may be disposed in the analysis channel.

[0092] A kit including the cartridge is also provided. The kit may include a processor configured to process the second signal to generate information indicative of at least one of the presence, absence, or quantity of one or more analytes within a sample. The processor may process the first signal to determine if the first signal indicates a quantity of the reaction within a predetermined range. The processor may generate an alert if the quantity is outside the predetermined range. The

processor may process the third signal to determine if the third signal indicates a quantity of the reaction below a threshold. The processor may generate an alert if the quantity is above the threshold. The processor may be a component of a reader.

[0093] The working electrode may be masked with a plurality of striations configured to promote homogenous distribution of a plurality of magnetic particles directly or indirectly bound to the signaling agents localized over the working electrode and to promote resistance to movement of the plurality of magnetic particles off of the working electrode.

[0094] The working electrode may include a self-assembled monolayer. For example, the self-assembled monolayer may be at a surface of the working electrode disposed within an analysis channel of the cartridge.

[0095] The working electrode may include affinity molecules pre-bound to the working electrode. For example, the affinity molecules may be pre-bound to a surface of the working electrode disposed within an analysis channel of the cartridge.

[0096] In accordance with another aspect, a sensor for use in a microfluidic cartridge is provided. The sensor may include a positive control working electrode and a working electrode. The positive control working electrode may generate a first signal based on a reaction between signaling agents localized at the positive control working electrode and a substrate. The working electrode may generate a second signal based on a reaction between the signaling agents localized at the working electrode and the substrate. The positive control working electrode and/or the working electrode may have a self-assembled monolayer pre-bound to affinity molecules configured to directly or indirectly bind to the signaling agents. The self-assembled monolayer may be applied to a metallic sensor surface such as gold. The self-assembled monolayer may be backfilled with a backfiller that binds directly to the affinity molecules. In one example, the self-assembled monolayer includes hexaethylene glycol dithiol (HEGT) and the backfiller includes mercaptoundecanoic acid (MUDA). In one example, the positive control working electrode includes the self-assembled monolayer pre-bound to affinity molecules and the working electrode a self-assembled monolayer without pre-bound affinity molecules.

[0097] The sensor may include a negative control working electrode that generates a third signal based on a reaction between the signaling agents localized at the negative working electrode and the substrate. The negative control working electrode may have a self-assembled monolayer which may be pre-bound to affinity molecules. The second signal may be indicative of at least one of the presence, absence, or quantity of one or more analytes within a sample. The first signal is indicative of reliability of a test which may be determined to be reliable if the first signal indicates a quantity of the reaction within a predetermined range. The third signal may be indicative of reliability of a test which may be determined to be reliable if the third signal indicates a quantity of the reaction is below a threshold.

[0098] A cartridge having the sensor therein is also provided. The cartridge may include an analysis channel and the positive control working electrode and the working electrode may be disposed in the analysis channel. A kit including the cartridge is also provided. The kit may further include a processor configured to process the second signal to generate information indicative of at least one of the presence, absence, or quantity of one or more analytes within a sample.

[0099] In accordance with yet another aspect, systems and methods for multiplexing first and second types of target analytes in a sample to determine at least one of a presence, absence, or quantity of the first and second types of target analytes in the sample are provided. A method may include: introducing the sample comprising the first and second types of target analytes into fluid in a sample preparation reservoir of a sample analysis cartridge; directly or indirectly binding the first type of target analyte to a first signaling agent in the fluid in the sample preparation reservoir, the first signaling agent having a first detectable potential for the first type of target analyte; directly or indirectly binding the second type of target analyte to a second signaling agent in the fluid in the sample preparation reservoir, the second signaling agent having a second detectable potential,

different from the first detectable potential, for the second type of target analyte; releasing the fluid from the sample preparation reservoir into an analysis channel comprising one or more sensors such that the first type of target analyte directly or indirectly bound to the first signaling agent and the second type of target analyte directly or indirectly bound to the second signaling agent localize over the one or more sensors in the analysis channel; releasing a substrate solution from a substrate reservoir into the analysis channel, the substrate solution comprising a substrate; reacting the substrate with the first signaling agent localized over the one or more sensors; electrically stimulating the reacted substrate and the first signaling agent using the one or sensors to generate a first signal; reacting the substrate with the second signaling agent localized over the one or more sensors; electrically stimulating the reacted substrate and the second signaling agent using the one or sensors to generate a second signal; and processing the first and second signals to determine at least one of the presence, absence, or quantity of the first type of target analyte and the second type of target analyte.

[0100] The first type of target analyte directly or indirectly bound to the first signaling agent may be a first amplicon complex amplified in the sample preparation reservoir and the second type of target analyte directly or indirectly bound to the second signaling agent may be a second amplicon complex amplified in the sample preparation reservoir. The first and second amplicons may be amplified in the sample preparation reservoir with polymerase and primers. The first and second signaling agents may be redox dye and the substrate may be ferricyanide or phosphate buffering saline (PBS). The electrically stimulating may be conducted using cyclic voltammetry, square wave voltammetry, or amperometry. Any one or more of the electrodes of the one or more sensors may be used for the electrically stimulating. For example, the positive control working electrode, the reference electrode, the counter electrode, negative control working electrode, and/or the working electrode may be used for the electrically stimulating. The first type of target analyte directly or indirectly bound to the first signaling agent and the second type of target analyte directly or indirectly bound to the second signaling agent may be each directly or indirectly bound to a magnetic particle that permits localization over the one or more sensors in the analysis channel via magnetic attraction to one or more magnetic generators.

[0101] The substrate may be a first type of substrate and the substrate solution may further include a second type of substrate. The method may further include reacting the second type of substrate with the first signaling agent localized over the one or more sensors; electrically stimulating the reacted second type of substrate and the first signaling agent using the one or sensors to generate a third signal; reacting the second type of substrate with the second signaling agent localized over the one or more sensors; electrically stimulating the reacted second type of substrate and the second signaling agent using the one or sensors to generate a fourth signal; and processing the third and fourth signals to determine at least one of the presence, absence, or quantity of a third type of target analyte and a fourth type of target analyte.

[0102] In accordance with yet another aspect, systems and methods for multiplexing first and second types of target analytes in a sample to determine at least one of a presence, absence, or quantity of the first and second types of target analytes in the sample are provided. A method may include: introducing the sample comprising the first and second types of target analytes into fluid in a sample preparation reservoir of a sample analysis cartridge; directly or indirectly binding the first type of target analyte to a signaling agent in the fluid in the sample preparation reservoir; directly or indirectly binding the second type of target analyte to a signaling agent in the fluid in the sample preparation reservoir; releasing the fluid from the sample preparation reservoir into an analysis channel comprising one or more sensors such that the first type of target analyte directly or indirectly bound to the signaling agent and the second type of target analyte directly or indirectly bound to the signaling agent localize over the one or more sensors in the analysis channel; releasing a substrate solution from a substrate reservoir into the analysis channel, the substrate solution comprising a first type of substrate having a first detectable potential for the first type of target

analyte and a second type of substrate having a second detectable potential, different from the first detectable potential, for the second type of target analyte; reacting the first type of substrate with the signaling agent localized over the one or more sensors; electrically stimulating the reacted first type of substrate and the signaling agent using the one or sensors to generate a first signal; reacting the second type of substrate with the signaling agent localized over the one or more sensors; electrically stimulating the reacted second type of substrate and the signaling agent using the one or sensors to generate a second signal; and processing the first and second signals to determine at least one of the presence, absence, or quantity of the first type of target analyte and the second type of target analyte.

[0103] The first type of target analyte directly or indirectly bound to the signaling agent may be a first amplicon complex amplified in the sample preparation reservoir and the second type of target analyte directly or indirectly bound to the signaling agent may be a second amplicon complex amplified in the sample preparation reservoir. The first and second amplicons may be amplified in the sample preparation reservoir with polymerase and primers. The signaling agents may be horseradish peroxidase (HRP) and/or redox dye and the first type of the substrate may be Tetramethylbenzidine (TMB) and the second type of substrate may be ferricyanide. The electrically stimulating may be conducted using cyclic voltammetry, square wave voltammetry, or amperometry. The first type of target analyte directly or indirectly bound to the signaling agent and the second type of target analyte directly or indirectly bound to the signaling agent may be each directly or indirectly bound to a magnetic particle that permits localization over the one or more sensors in the analysis channel via magnetic attraction to one or more magnetic generators.

[0104] The signaling agent may be a first signaling agent and the fluid in the sample preparation reservoir may further include a second signaling agent. The method may further include reacting the first type of substrate with the second signaling agent localized over the one or more sensors; electrically stimulating the reacted first type of substrate and the second signaling agent using the one or sensors to generate a third signal; reacting the second type of substrate with the second signaling agent localized over the one or more sensors; electrically stimulating the reacted second type of substrate and the second signaling agent using the one or sensors to generate a fourth signal; and processing the third and fourth signals to determine at least one of the presence, absence, or quantity of a third type of target analyte and a fourth type of target analyte.

[0105] Such methods and devices may be used, for example, to determine: from which illness, among many, a person is suffering; to which drug or poison, among many, a person is adversely reacting; or which chemical, among many, has contaminated the water. Other examples include quantifying the concentrations of various agents, that include without limitation vitamins, hormones, proteins, or other analytes of interest within one's body, waterborne and foodborne pathogens, microbial growth and/or contamination of medical equipment, and other potential disease-causing contaminants from pets and livestock. Examples of contaminants include, but are not limited to viral, bacterial, and fungal pathogens, bloodstream infection (BSI), pneumonia (e.g., ventilator-associated pneumonia [VAP]), urinary tract infection (UTI), and surgical site infection (SSI), *Staphylococcus aureus*, Methicillin resistant *Staphylococcus aureus*, *Candida albicans*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Stenotrophomonas maltophilia*, *Clostridium difficile*, Tuberculosis, Gastroenteritis, Vancomycin-resistant *Enterococcus*, Legionnaires' disease, Puerperal fever, MRSA, and *E. coli*. Examples of foodborne pathogens include, but are not limited to *shigella*, *salmonella*, *vibrio*, *Yersinia*, *Listeria*, *Escherichia coli*, and *Campylobacter*. The application of this technology is not limited to pathogens or analytes that are important to the health of human patients but also includes the health and maintenance of pets and livestock, e.g., veterinary uses.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0106] Exemplary embodiments are described below with reference to the accompanying drawings, wherein like numerals denote like elements. In the drawings:

[0107] FIGS. **1A-1B** provide schematic depictions of an exemplary analyte detection system for analyzing the presence, absence, and/or quantity of one or more target analytes within a collected sample and for viewing analysis results, wherein FIG. **1A** shows components uncoupled and FIG. **1B** shows components coupled for analysis and charging.

[0108] FIG. **1C** provides a schematic depiction of another exemplary analyte detection system for analyzing the presence, absence, and/or quantity of one or more target analytes within a collected sample and for viewing analysis results, wherein a charger is not provided.

[0109] FIGS. **2A** and **2B** illustrate perspective views of an exemplary sample collection device for use in the detection system.

[0110] FIG. **3** illustrates a perspective view of an exemplary cartridge device for use in the detection system.

[0111] FIGS. **4A** and **4B** illustrate perspective views of the cartridge device with the sample collection device locked therein for analysis of the collected sample, wherein FIG. **4A** shows the top surface of the cartridge device and FIG. **4B** shows the bottom surface of the cartridge device.

[0112] FIG. **5** illustrates an exploded view of the exemplary cartridge device showing internal components that may be within the cartridge housing.

[0113] FIGS. **6A**, **6B**, and **6C** illustrate an exemplary circuit board and an exemplary layer that may be used within the housing of the cartridge device, wherein FIG. **6A** depicts the circuit board, FIG. **6B** depicts the layer, and FIG. **6C** depicts the layer disposed on and coupled to the circuit board.

[0114] FIGS. **7A**, **7B**, and **7C** illustrate another exemplary circuit board and another exemplary layer that may be used within the housing of the cartridge device, wherein FIG. **7A** depicts the circuit board, FIG. **7B** depicts the layer, and FIG. **7C** depicts the layer disposed on and coupled to the circuit board and an absorbent pad.

[0115] FIGS. **7D** through **7F** illustrate alternative exemplary sensors that may be used within the housing of the cartridge device.

[0116] FIG. **7G** are images generated by Atomic Force Microscopy (AFM) of exemplary sensor surfaces that may be used on the one or more sensors within the housing of the cartridge device.

[0117] FIG. **7H** is a graph showing measured current in nA before and after self-assembled monolayer treatment for Electrode **1** and Electrode **2**.

[0118] FIG. **7I** is a graph showing measured current in nA versus CRP concentration (mg/L) for a new sensor with a self-assembled monolayer and a sensor with a self-assembled monolayer aged the equivalent of 18 months in an aging chamber.

[0119] FIG. **8A** illustrates an exemplary internal component coupled to an exemplary circuit board via a layer positioned therebetween, all of which may be disposed within the housing of the cartridge device.

[0120] FIGS. **8B** and **8C** show a close-up views of certain components of the circuit board and a valve for use in the housing of the cartridge device.

[0121] FIG. **8D** shows a close-up views of alternative components of the circuit board and a valve for use in the housing of the cartridge device.

[0122] FIGS. **9A** and **9B** illustrate exemplary shuttles that may be disposed in the housing of the cartridge device, wherein the shuttles are each shown housing a reagent ball.

[0123] FIG. **10A** is a cross-sectional perspective view showing a sample collection device partially inserted within an input tunnel of a cartridge device.

[0124] FIG. **10B** is a cross-sectional perspective view showing a tip of the sample collection device entering a shuttle disposed within the input tunnel of the cartridge device.

[0125] FIG. **10C** is a top view showing an exemplary orientation of piercing elements over

reservoirs within the housing of the cartridge device.

[0126] FIG. **10D** is a cross-sectional perspective view showing engagement between the sample collection device and a slider of a seal piercer within the input tunnel of the cartridge device, wherein part of the cartridge device is removed.

[0127] FIG. **10E** is a cross-sectional side view showing engagement between the sample collection device and the seal piercer and sealing between the sample collection device and the shuttle, wherein the sample preparation reservoir remains sealed by the shuttle.

[0128] FIG. **10F** is a top view showing engagement between the sample collection device and the seal piercer in a pre-venting position.

[0129] FIG. **10G** is a cross-sectional perspective view showing the piercing element and the slider of the seal piercer in a pre-venting position, wherein the sealing material over the sample preparation reservoir within the cartridge device has not yet been pierced.

[0130] FIG. **10H** is a cross-sectional side view illustrating transition from the pre-mixing and pre-venting positions towards the mixing and venting positions within the input tunnel of the cartridge device.

[0131] FIG. **10I** is a top view showing movement of the sample collection device causing movement of the seal piercer to the venting position.

[0132] FIG. **10J** is a cross-sectional perspective view showing the piercing element piercing the sealing material over the sample preparation reservoir within the cartridge device.

[0133] FIG. **10K** is a cross-sectional side view showing the sample collection device in the venting and mixing positions, wherein the sealing material over the sample preparation reservoir is vented and the collected sample and the reagent ball are mixed within the fluid in the sample preparation reservoir which has been re-sealed by the shuttle.

[0134] FIG. **10L** is a cross-sectional perspective view showing the sample collection device in the venting and mixing positions within the cartridge device.

[0135] FIG. **10M** is a cross-sectional perspective view showing the sample collection device in the venting and mixing positions within the sample preparation reservoir of the internal component of the cartridge device.

[0136] FIGS. **10N**, **10O**, and **10P** are cross-sectional side views showing enhanced mixing of the fluid in the sample preparation reservoir with the collected sample and the reagent ball via a sonicator element.

[0137] FIGS. **11A** through **11E** illustrate an alternative seal piercer wherein insertion of a sample collection device within the input tunnel of the cartridge device also activates a switch to represent proper sample collection device insertion.

[0138] FIGS. **12A** through **12E** illustrate cross-sectional side views of an alternative sample collection device and an alternative cartridge device for collecting and analyzing a fluid sample.

[0139] FIG. **13A** illustrates an exploded view of another exemplary cartridge device showing internal components that may be within the cartridge housing.

[0140] FIGS. **13B** through **13SS** illustrate various views of exemplary sample collection devices that may be used in the detection system.

[0141] FIGS. **14A** through **14D** illustrate exemplary collets that may be disposed in the housing of the cartridge device, wherein FIGS. **14A** and **14C** show perspective views and FIGS. **14B** and **14D** show cross-sectional views of the collets in FIGS. **14A** and **14C**, respectively.

[0142] FIG. **15A** illustrates a cross-sectional view through the center of the input tunnel of an exemplary cartridge in the pre-mixing, pre-venting, storage position.

[0143] FIG. **15B** illustrates a cross-sectional view of the exemplary cartridge and an exemplary sample collection device fully inserted in the input tunnel in the mixing, venting, analysis position.

[0144] FIGS. **15C** and **15D** illustrate perspective views of the exemplary cartridge having the exemplary sample collection device inserted therein in the pre-venting position (FIG. **15C**) and in the venting position (FIG. **15D**).

[0145] FIGS. **16A** through **16E** illustrate cross-sectional side views showing insertion of the sample collection device in the cartridge.

[0146] FIGS. **16F** and **16G** illustrate cross-sectional top views showing further distal insertion of the sample collection device in the cartridge.

[0147] FIGS. **16H** through **16J** illustrate cross-sectional side views showing further distal insertion of the sample collection device in the cartridge, wherein the sample collection device is fully inserted in the input tunnel in the mixing, venting, analysis position in FIG. **16J**.

[0148] FIGS. **17A** through **17D** illustrate various views of an exemplary sonicator electrically coupled to a circuit board via spring contacts for use within an exemplary cartridge housing.

[0149] FIGS. **18A** and **18B** are cross-sectional side and top views, respectively, of another exemplary cartridge device.

[0150] FIG. **19** illustrates an exemplary process for monitoring temperature during enhanced mixing via the sonicator.

[0151] FIG. **20** is a graph showing measured temperature over time during enhanced mixing via the sonicator.

[0152] FIG. **21A** illustrates a perspective view of an exemplary reader device for use in the detection system.

[0153] FIG. **21B** illustrates an exploded view of the exemplary reader device of FIG. **21A** showing internal components that may be within the reader housing.

[0154] FIG. **22A** shows a cross-sectional perspective view of the exemplary reader device.

[0155] FIG. **22B** is a cross-sectional side view showing a cartridge device (having a sample collection device partially inserted therein) partially inserted within the exemplary reader device.

[0156] FIGS. **22C** and **22D** are cross-sectional perspective and side views, respectively, showing the cartridge device inserted within the exemplary reader device in the analysis position.

[0157] FIGS. **23A** and **23B** are graphs showing magnetic field strengths over the length of a single working electrode for a single magnet (FIG. **23A**) versus a dual magnet (FIG. **23B**) design.

[0158] FIG. **24** provides a flowchart of one embodiment of a method for detecting the presence, absence, and/or quantity of one or more target analytes in a sample.

[0159] FIG. **25A** illustrates a perspective view of an exemplary charger that may be used in the detection system.

[0160] FIG. **25B** illustrates an exploded view of the exemplary charger of FIG. **25A** showing internal components that may be within the charger housing.

[0161] FIGS. **26A** and **26B** provide schematic depictions of molecules and reactions found within one embodiment of the presently disclosed analyte detection system.

[0162] FIGS. **26C** and **26D** provide schematic depictions of molecules and reactions found within another embodiment of the presently disclosed analyte detection system.

[0163] FIGS. **27A** and **27B** provide schematic depictions of molecules and reactions found within yet another embodiment of the presently disclosed analyte detection system.

[0164] FIG. **28A** is a schematic depiction of molecules within a sample on a sample collection device.

[0165] FIG. **28B** is a schematic depiction of molecules within two reagent balls for reacting with the molecules within the collected sample.

[0166] FIG. **28C** is a schematic depiction of molecules within a single reagent ball for reacting with the molecules within the collected sample.

[0167] FIG. **28D** is a schematic depiction of molecules showing the collected sample being introduced to a sample preparation reservoir.

[0168] FIG. **28E** is a schematic depiction of molecules showing mixing of the molecules of the collected sample with sample preparation reagent molecules within the fluid of the sample preparation reservoir.

[0169] FIGS. **28F**, **28G**, and **28H** are schematic depictions of molecules showing reactions between

the molecules of the collected sample and the sample preparation reagent molecules within the fluid of the sample preparation reservoir.

[0170] FIG. **29A** is a graph showing electrochemical sensor readings versus concentration of a target analyte using pre-bound competitor binding molecules and FIG. **29B** shows a graph comparing electrochemical sensor readings versus concentration when a competitor binding molecule is not pre-bound.

[0171] FIG. **30A** is another schematic depiction of molecules within a sample on a sample collection device.

[0172] FIG. **30B** is another schematic depiction of molecules within two reagent balls for reacting with the molecules within the collected sample.

[0173] FIG. **30C** is another schematic depiction of molecules within a single reagent ball for reacting with the molecules within the collected sample.

[0174] FIG. **30D** is another schematic depiction of molecules showing the collected sample being introduced to a sample preparation reservoir.

[0175] FIG. **30E** is another schematic depiction of molecules showing mixing of the molecules of the collected sample with sample preparation reagent molecules within the fluid of the sample preparation reservoir.

[0176] FIGS. **30F**, **30G**, and **30H** are schematic depictions of molecules showing reactions between the molecules of the collected sample and the sample preparation reagent molecules within the fluid of the sample preparation reservoir.

[0177] FIGS. **31A** through **31J** show an exemplary process for detecting the presence, absence, and/or quantity of a target analyte(s) within a sample in a cartridge.

[0178] FIGS. **32A** through **32N** show an exemplary process for detecting the presence, absence, and/or quantity of a target analyte(s) within a sample in a cartridge using isothermal amplification.

[0179] FIGS. **33A** through **33H** show an exemplary multiplexing process for detecting the presence, absence, and/or quantity of multiple target analytes within a sample in a cartridge.

[0180] FIG. **34** provides a schematic depiction of the exemplary analyte detection system FIGS. **1A-1B** communicatively coupled to one or more servers via a network.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

[0181] In the following detailed description, reference is made to the accompanying drawings, which form part of the present disclosure. The embodiments described in the drawings and description are intended to be exemplary and not limiting. As used herein, the term “exemplary” means “serving as an example or illustration” and should not necessarily be construed as preferred or advantageous over other embodiments. Other embodiments may be utilized and modifications may be made without departing from the spirit or the scope of the subject matter presented herein. Aspects of the disclosure, as described and illustrated herein, can be arranged, combined, and designed in a variety of different configurations, all of which are explicitly contemplated and form part of this disclosure.

[0182] Various devices, systems, kits, and methods disclosed herein are intended to isolate, tag, and detect a target analyte within a sample taken from a specimen. In certain embodiments, chemical reactions are employed to enable such detection.

[0183] Various embodiments of systems described herein are designed to create a self-contained environment in which any of the chemical reactions occur in an automated manner entirely or substantially without human intervention, for example, as described in commonly assigned U.S. Pat. No. 9,034,168 to Khattak, U.S. Pat. No. 9,052,275 to Khattak, U.S. Pat. No. 9,086,417 to Khattak, U.S. Pat. No. 9,207,244 to Khattak, U.S. Pat. No. 9,207,245 to Khattak, U.S. Pat. No. 9,360,491 to Sever, U.S. Pat. No. 9,522,397 to Khattak, U.S. Pat. No. 9,623,409 to Khattak, U.S. Pat. No. 9,636,676 to Sever, U.S. Pat. No. 9,718,058 to Khattak, U.S. Pat. No. 9,724,691 to Khattak, U.S. Pat. No. 9,789,483 to Khattak, U.S. Pat. No. 9,808,804 to Khattak, WO 2016/040642 to Khattak, and WO 2017/015172 to Khattak, the entire contents of each of which are incorporated

herein by reference. In some designs described herein, one or more chemical reactions proceed without any need for an operator to add or remove reagents from the system. In certain embodiments, the systems are closed such that biohazard risks, such as the risk of spilling sample collected from a specimen, are minimized. In various embodiments, such systems include at least, a sample collection device, a cartridge device, and a reader device. Some exemplary embodiments of such devices are described in detail below.

[0184] FIGS. **1A** and **1B** illustrate an exemplary analyte detection system constructed in accordance with the principles of the present disclosure. Detection system **100** may include sample collection device **200**, cartridge device **300**, reader device **400**, charger **500**, and/or software-based detection interface system **600**. Detection system **100** may be used to detect the presence, absence, and/or quantity of one or more target analytes.

[0185] Sample collection device **200** is configured to be exposed to a sample for analysis. For example, sample collection device **200** may be exposed to a biological sample, such as, but not limited to, blood, plasma, urine, saliva, mucous, cellular material and/or other biological material for determining the presence, absence, and/or quantity of one or more target analytes within the sample. In addition or alternatively, the sample collection device is exposed to a solid or other surface that is suspected of harboring a target analyte, e.g., a food-borne pathogen and the surface is a cooking or food preparation surface.

[0186] Cartridge device **300** is configured to analyze the sample collected with sample collection device **200**. Cartridge device **300** may include input tunnel **301** that extends from aperture **302** into the cartridge housing. Input tunnel **301** is configured to permit insertion of sample collection device **200** as shown in FIG. **1B** such that the collected sample may be analyzed within cartridge device **300**. Based on the analysis, cartridge device **300** is configured to generate electric signals indicative of the presence, absence, and/or quantity of one or more target analytes within the sample.

[0187] Reader **400** is configured for electric coupling with cartridge device **300** to permit transmission of the electric signals indicative of the presence, absence, and/or quantity of one or more target analytes within the sample generated by cartridge device **300**. Cartridge device **300** may be electrically coupled to reader **400** by inserting cartridge device **300** within reader opening **401** of reader **400** as shown in FIG. **1B** such that respective electrical connectors of cartridge device **300** and reader **400** contact one another. Reader **400** may comprise a computer readable medium with instructions that, when executed by a processor of reader **400**, cause electrical components of cartridge **300** to perform steps for analyzing the sample on sample collection device **200**. Preferably, the instructions are not executed until cartridge device **300** is electrically coupled to reader **400** and sample collection device **200** is suitably disposed within cartridge device **300**, for example, as shown in FIG. **1B** or **4A**.

[0188] In one embodiment, sample collection device **200** and cartridge device **300** are each disposable and designed for one time use while reader **400** is designed for multi-use and for receiving many different cartridge devices throughout the life of reader **400** such that many samples are analyzed by reader **400** for determining the presence, absence, and/or quantity of one or more target analytes within the respective samples. Such a configuration is expected to promote sanitary use of the system, as the components exposed to the sample are disposable, while reducing costs as the components with more expensive electronics, e.g., reader **400**, may be used repeatedly.

[0189] Charger **500** is configured to charge one or more batteries within reader **400**, e.g., via respective inductive coils disposed within the housings of charger **500** and reader **400**. Charger **500** may be plugged into a conventional socket, e.g., via a cord or a cord with an AC to DC power converter, for charging components within charger **500** to permit charging of reader **400**.

[0190] As will be readily apparent to one skilled in the art, the detection system need not require a charger. For example, referring to FIG. **1C**, detection system **100'** is constructed similarly to detection system **100** of FIGS. **1A** and **1B**, wherein like components are identified by like-primed reference numbers. Thus, for example, cartridge device **300''** in FIG. **1C** corresponds to cartridge

device **300** of FIGS. **1A** and **1B**, etc. As will be observed by comparing FIGS. **1C** and **1B**, detection system **100'** does not include charger **500**. In such an embodiment, reader **400'** may be plugged into a conventional socket, e.g., via a cord or a cord with an AC to DC power converter, for powering components of reader **400'** and/or reader **400'** may include a suitable battery such as a replaceable battery or rechargeable battery and reader **400'** may include circuitry for charging the rechargeable battery, and a detachable power cord.

[0191] In FIGS. **1A** and **1B**, software-based detection interface system **600** is installed and runs on computing device **601** to permit a user to review analyte detection test results, e.g., on display **602** of computing device **601**. Computing device **601** may be, for example, a smartphone, smartwatch, tablet, wearable device, a laptop or other computer. As shown in FIGS. **1A** and **1B**, reader **400** may communicate with computing device **601** wirelessly to transmit data indicative of the presence, absence, and/or quantity of one or more target analytes based on the electrical signals generated within cartridge device **300**. In addition or alternatively, a removable wired connection, such as a cable connection, may be provided between reader **400** and computing device **601**. Software-based detection interface system **600** may comprise a computer readable medium with instructions that, when executed by a processor of computing device **601**, cause display **602** to display information indicative of the presence, absence, and/or quantity of one or more target analytes.

Sample Collection Devices and Cartridges

[0192] The sample collection device of various embodiments is configured to collect a sample from a specimen. Sample collection devices may be configured to collect cells and other biological material from any desired region or location, for example, an inner cheek, the throat, a nasal passageway, an ear, from urine, from blood, from plasma, from saliva, or from another body part. One exemplary sample collection device includes a unit that wicks a small droplet of blood or urine into a small capillary channel. In other embodiments, the sample collection device may be configured to collect biological material, particulates, or other chemicals from the environment, such as, for example, from the air or the water, or from a physical surface or other structure.

[0193] The sample collection device of various embodiments is sized and shaped to collect a sufficiently large sample from an appropriate location of a specimen such that it is possible, using the other devices described below, to detect the presence, absence, and/or quantity of one or more target analytes in and/or on the specimen. For example, for some target analytes, such as ones associated with a virus causing cold or flu-like symptoms, the sample collection device may be a nose-insertion swab; the swab is sized and shaped to collect a sufficient amount of sample from a nasal passageway of an individual to enable detection of target analytes associated with the virus causing cold or flu-like symptoms, if present in the individual. For other target analytes, such as, for example, ones associated with strep throat, the sample collection device may be a throat swab shaped to scrape sufficient cells from an individual's throat or mouth. As another example, the sample collection device appropriate for collecting a target analyte associated with HIV may comprise a blood lancet. In another example, a sample collection device configured to collect urine may be appropriate for collecting target analytes for various tests, including, for example, tests for tracking testosterone levels, drug levels, vitamin levels, and/or fertility. A sample collection device for collecting fluid, such as urine, blood, plasma, or saliva, may include features for compressing a wicking portion of the device to expel sample absorbed on the wicking portion for analyzing the expelled sample. In yet a further aspect, the sample collection device is shaped to collect a sample from a solid surface, e.g., on a medical device, on medical equipment or from the surface of a food preparation surface such as a cutting board or flat surface.

[0194] Referring to FIGS. **2A** and **2B**, sample collection device **200** is illustrated. Sample collection device **200** is configured to collect a small quantity of a sample to be analyzed and configured for full or partial insertion within cartridge device **300** after sample collection. Sample collection device **200** may include distal portion **201**, proximal portion **202**, and shaft **203** extending therebetween. Distal portion **201** may include tip **204** having tube **205** therein. Sample

collection device **200** also may include handle **206**, proximal sealing zone **207**, distal sealing zone **208**, and/or engagement zone **209**.

[0195] Distal portion **201**, including tip **204**, is configured to be exposed to a sample such that, at most, a predetermined volume of the sample is disposed in tube **205** for analysis. Collection of a predetermined volume of the sample is expected to promote accuracy of analyte analysis as a substantially known quantity of the sample will be analyzed. Tip **204** may be transparent to permit a collector to verify that sample is disposed in tube **205**. Tip **204** may have a rounded end as illustrated although various shapes may be used including any blunt or substantially blunt tip shape. Tip **204** may be configured to collect a sample from any desired region or location, for example, an inner cheek, the throat, the mouth, a nasal passageway, an ear, from urine, from blood, from plasma, from saliva, or from another body part.

[0196] Proximal portion **202** may include handle **206** sized and shaped to be held by a collector's hand. Handle **206** may include gripping protrusions as illustrated. Handle **206** may further lock sample collection device **200** within the input tunnel of cartridge device **300**. Sample collection device **200** also may include proximal sealing zone **207** configured for sealing the input tunnel of cartridge device **300** when sample collection device **200** is inserted in the input tunnel. Proximal sealing zone **207** may include a protrusion extending around shaft **203** and sized greater than the input tunnel opening so as to seal off the input tunnel. As such, the protrusion may further lock sample collection device within the input tunnel of cartridge device **300**. Handle **206** may be breakable or otherwise removable from the remainder of sample collection device **200** following insertion of the remainder of sample collection device **200** into cartridge device **300**.

[0197] Shaft **203** is elongated to facilitate easy and sanitary collection, with a collector's hand removed from the site of collection. For example, shaft **203** may be elongated such that tip **204** may be exposed to a sample within an inner cheek, the throat, the mouth, a nasal passageway, an ear, from urine, from blood, from plasma, from saliva, etc. to collect fluid, cells, and other biological material, while handle **206** is not exposed to the sample. Shaft **203**, tip **204**, and handle **206** may be formed of the same material or of different materials. Shaft **203**, tip **204**, and handle **206** may be formed of a plastic. Sample collection device **200** may be pre-packaged within sterile packaging and is preferably configured for one-time use.

[0198] Sample collection device **200** may have distal sealing zone **208** for facilitating the formation of a liquid-tight seal between sample collection device **200** and cartridge device **300** after insertion of sample collection device **200** into cartridge device **300**. For example, distal sealing zone **208** may be sized and shaped to seal the collected sample on tip **204** and fluid within a sample preparation reservoir of cartridge device **300** within cartridge device **300**. Distal sealing zone **208** may be of greater radial size than tip **204**. For example, distal sealing zone **208** may include a shoulder extending further from the longitudinal axis of sample collection device **200** than tip **204** such that shoulder abuts against a portion of cartridge device **300**, e.g., a shuttle, to form the liquid-tight seal. In this manner, the sample may be sealed within cartridge device **300** to reduce leakage and exposure of the sample outside the cartridge. In addition, the shoulder may be used to move a seal piercer to vent one or more reservoirs within cartridge device **300** before, during, or after formation of the liquid-tight seal.

[0199] Sample collection device **200** may include engagement zone **209** configured for engagement with one or more components of cartridge device **300**. For example, engagement zone **209** may be configured to be coupled, permanently or temporarily, to a seal piercer of the cartridge device to move the seal piercer within the cartridge device responsive to movement of sample collection device **200**. Engagement zone **209** also may facilitate fixed engagement between sample collection device **200** and the cartridge device such that sample collection device **200** is mated irreversibly and immovably with the cartridge when sample collection device **200** is inserted a predetermined distance in the input tunnel of the cartridge. Engagement zone **209** may be a groove around shaft **203** as illustrated or may be multiple grooves extending a shorter distance from the longitudinal

axis than the regular shaft surface and/or may be one or more protrusions extending a greater distance from the longitudinal axis of sample collection device **200**.

[0200] In various embodiments, a cartridge is formed of a housing, which defines an enclosed space and has various features that enable the cartridge to do one or more of the following: receive a sample with target analytes from a sample collection device, store the sample with sample preparation reagents, provide a space for mixing and binding of the target analytes with sample preparation reagents, provide an analysis zone wherein bound target analytes localize over sensors for detection, provide a fluid medium for transporting the bound target analytes to the analysis zone, store and provide a substrate that can undergo a detectable reaction when introduced to the bound target analytes, provide a fluid medium for transporting the substrate to the bound target analytes in the analysis zone, and provide a waste collection zone where waste is stored.

[0201] In various embodiments, the cartridge is a substantially closed system wherein the reactions needed to detect the presence, absence, and/or quantity of one or more target analytes occur within the cartridge. The cartridge of such embodiments is said to be “substantially closed” because the only inputs needed into the cartridge system are one or more of the following: a sample from a specimen, energy to facilitate mixing and bound, and a magnetic force to facilitate localization of bound target analytes within an analysis zone; the only outputs from the cartridge are electrical signals. In various embodiments, the cartridge is target-analyte-specific with the included sample preparation reagents selected to detect one or more specific target analytes. Different cartridge types include different reagents intended to identify different target analytes. For example, different cartridge types may include inflammation, influenza, testosterone, fertility, HIV, and Vitamin D which each include application-specific reagents intended to identify different target analytes.

[0202] Referring now to FIG. **3**, an exemplary cartridge is illustrated. Cartridge device **300** may include input tunnel **301** that extends from aperture **302** on front surface **303** into cartridge housing **304**. Cartridge housing **304** may have a substantially rectangular prism shape as illustrated, although the present disclosure is not limited thereto. Cartridge housing **304** has front surface **303**, top surface **305**, right side surface **306**, left side surface **307** (shown in FIG. **4A**), bottom surface **308** (shown in FIG. **4B**), and back surface **309** (shown in FIG. **4B**). Cartridge housing **304** may be formed of a single component or multiple components. For example, cartridge housing **304** may include first cover component **310** configured to be laterally coupled to second cover component **311** such that the internal components of cartridge device **300** are housed therein.

[0203] FIGS. **4A** and **4B** illustrate sample collection device **200** inserted within input tunnel **301** of cartridge device **300** in a mixing position. In the mixing position, proximal sealing zone **207** of sample collection device **200** may seal input tunnel **301** at aperture **302** to reduce or eliminate leakage from input tunnel **301**. Input tunnel **301** may be integrally formed with housing **304** of cartridge device **300** or may be detachably coupled to cartridge housing **304**. As shown in FIG. **4B**, cartridge device **300** may include electrical connector **312** configured for electrical coupling with the reader, e.g., via an electrical connector of the reader. Accordingly, signals indicative of the presence, absence, and/or quantity of one or more target analytes may be transmitted from cartridge device **300** via electrical connector **312** to the reader. Electrical connector **312** may be positioned on bottom surface **308** and back surface **309** to facilitate coupling with the electrical connector within the opening of the reader.

[0204] Bottom surface **308** of cartridge device **300** may include first ramp portion **313**, second ramp portion **314**, and magnetic generator depression **315**. First ramp portion **313** is configured to gradually depress one or more magnetic generators of the reader during insertion of cartridge device **300** within the opening of the reader. First ramp portion **313** may begin in the depression of cartridge housing **304** where electrical connector **312** is positioned and ramp down to bottom surface **308**. As cartridge device **300** is inserted past first ramp portion **313**, the magnetic generators remain in a depressed position until contacting the second ramp portion **314** which ramps up into magnetic generator depression **315**. Second ramp portion **314** is configured to gradually guide the

one or more magnetic generators of the reader into magnetic generator depression **315**. Magnetic generator depression **315** is disposed beneath one or more working electrodes of cartridge device **300** such that the one or more magnetic generators of the reader move up into magnetic generator depression **315** and are disposed adjacent the one or more working electrodes when cartridge device **300** is fully inserted in the reader. Second ramp portion **314** also facilitates removal of cartridge device **300** from the reader by gradually depressing the one or more magnetic generators of the reader during cartridge removal.

[0205] Referring now to FIG. 5, an exploded view of cartridge device **300** is shown. Cartridge device **300** may include internal component **316**—which may include sample preparation reservoir **317**, wash reservoir **318**, and substrate reservoir **319**—sealing material **320**, seal piercer **321**—which may include slider **322** and piercer **323**—shuttle **324**, desiccant **325**, input tunnel component **326**, sonicator element **327**, absorbent pad **328**, layer **329**, analysis channel **330**, and circuit board **331** electrically coupled to memory **332**. The internal components may be disposed within housing **304**, e.g., between first and second cover components **310** and **311**. Alternatively, one or more internal components may be disposed within one housing while other internal components may be disposed within another housing(s). In the case of multiple housings, such separate housing may be configured to couple to one another.

[0206] Internal component **316** is configured to define one or more reservoirs, illustratively sample preparation reservoir **317**, wash reservoir **318**, and substrate reservoir **319**. Internal component **316** may further define a portion of analysis channel **330** such as by creating the upper boundary of analysis channel **330** when cartridge device **300** is assembled. Internal component **316** may house other internal components such as absorbent pad **328**. Further, internal component **316** may be formed of a suitable material, such as plastic, and may have a base sized in a generally rectangular shape to sit over circuit board **331**.

[0207] Sample preparation reservoir **317** is configured to hold a fluid, preferably a liquid having sample preparation reagents. For example, the fluid may be water, saline solution, water/saline solution mixed with one or more of magnetic particles, affinity molecules, connection molecules, signaling agents, competitor binding molecules, competitor molecules, labels, and/or signaling agents, as described in further detail below. Sample preparation reservoir **317** is positioned adjacent to the distal end of input tunnel **301** such that input tunnel **301** leads to sample preparation reservoir **317**. As described further below, sample preparation reservoir **317** may be partially formed with sonicator element **327**, e.g., as part or all of the bottom surface, which facilitates mixing of the fluid and additional particles in the fluid. In addition, sample preparation reservoir **317** may be partially formed with an end of shuttle **324** during the pre-mixing state and partially formed with another portion of shuttle **324** during the mixing state when one or more reagent balls and the sample are disposed in sample preparation reservoir **317**. In this manner, sample preparation reservoir **317** remains fluidically sealed in the pre-mixing state by shuttle **324** and in the mixing state by shuttle **324** and continuously fluidically sealed throughout movement from the pre-mixing state to the mixing state such that fluid does not leak proximally past shuttle **324**. Sample preparation reservoir **317** is positioned such that upon insertion of sample collection device **200** into input tunnel **301**, distal portion **201** having the sample, e.g., at tip **204** and/or tube **205**, enters sample preparation reservoir **317**. When sample collection device **200** enters sample preparation reservoir **317**, sample preparation reservoir **317** becomes further filled with sample particles, including one or more target analytes, if present in the sample. The fluid may be gently mixed, e.g., via sonicator element **327**, with the one or more reagent balls and the sample to suspend and hybridize particles within sample preparation reservoir **317**. The target analytes in the sample may hybridize and/or bind, at least, to the magnetic particles and/or to the affinity molecules present among the sample preparation reagents forming magnetic particle-bound complexes and/or affinity molecule-target complexes. Sample preparation reservoir **317** is configured to permit release, e.g., via an outlet, of the fluid having the sample and sample preparation reagents mixed therein into analysis channel **330** for

analyzing the presence, absence, and/or quantity of one or more target analytes within the sample. The outlet of sample preparation reservoir **317** may be sealed with a heat actuated valve. When the valve opens, fluid from sample preparation reservoir **317** acts as a transport medium causing the magnetic particle-bound complexes and/or affinity molecule-target complexes and other particles to flow from sample preparation reservoir **317** into the analysis channel **330**. Advantageously, the fluid serving as the mixing medium and storage medium within sample preparation reservoir **317** also acts as the flow medium to transport the contents of sample preparation reservoir **317** to an analysis zone within analysis channel **330** without the need for a pump.

[0208] Wash reservoir **318** is configured to hold a fluid, preferably a liquid configured as a wash solution. Wash reservoir **318** is further configured to permit release, e.g., via an outlet, of the wash solution into analysis channel **330** to move particles in the mixed fluid previously released from sample preparation reservoir **317** that are not bound to a magnetic particle or a pre-bound surface affinity molecule off a working electrode and/or off a positive control working electrode in the analysis channel. The outlet of wash reservoir **318** may be sealed with a heat actuated valve. When the valve opens, the wash solution flows from wash reservoir **318** into analysis channel **330**, thereby removing all or substantially all unbound detector agents and/or unbound competitive binding agents from analysis channel **330**. In one aspect, most or all free-floating, unbound molecules from sample preparation reservoir **317** are washed from analysis channel **330** to reduce the likelihood of having any non-specific binding of significance and/or non-specific signal generated by free floating signaling agents, e.g., HRP, of significance occur within an analysis zone of analysis channel **330**.

[0209] Substrate reservoir **319** is configured to hold a fluid, preferably a substrate solution comprising a substrate such as a chemical substrate. The fluid of substrate reservoir **319** may include a substrate that undergoes a reaction in the presence of a signaling agent from sample preparation reservoir **317**. For example, the substrate of substrate reservoir **319** may undergo an oxidation or reduction reaction in the presence of an oxidizing enzyme or a redox label from sample preparation reservoir **317**. The fluid may be a substrate solution including acceptor molecules, such as hydrogen peroxide, and the substrate which may be an enzyme substrate such as Tetramethylbenzidine (TMB) and/or o-phenylenediamine dihydrochloride (OPD) molecules. As an example, the substrate may be a commercially available enzyme-linked immunosorbent assay (ELISA) substrate. Preferably, the substrate is oxidizable and/or reducible. The acceptor molecules may be configured to receive electrons stripped from the substrate by the signaling agent (thereby oxidizing the substrate) during the reaction between the substrate and the signaling agent. For example, when the acceptor molecules are hydrogen peroxide, an oxidase reaction between the substrate, e.g., TMB, OPD, and the signaling agent, e.g., HRP, SBP, causes electrons to be stripped from the substrate and donated to the acceptor molecules (e.g., hydrogen peroxide) during the oxidase reaction such that the acceptor molecules convert to another molecule (e.g., water). In some embodiments, ferricyanide or phosphate buffered saline (PBS) is used as the substrate (and reacted with a signaling agent, that may be an oxidation/redox dye such as Methylene Blue, ferrocene, Nile blue ([9-(diethylamino)benzo[a]phenoxazin-5-ylidene]azanium; sulfate, available from Sigma-Aldrich and other commercial vendors), and/or anthraquinone, from sample preparation reservoir **317**). Substrate reservoir **319** is further configured to permit release, e.g., via an outlet, of the fluid with the substrate into analysis channel **330**. The outlet of substrate reservoir **319** may be sealed with a heat actuated valve. When the valve opens, fluid from substrate reservoir **319** acts as a transport medium causing the chemical substrate to flow from substrate reservoir **319** into analysis channel **330**.

[0210] One skilled in the art will appreciate that while three reservoirs are depicted, in various embodiments, the plurality of reservoirs may include two reservoirs or four or more reservoirs and may adopt alternative spatial configurations. For example, wash reservoir **318** and substrate reservoir **319** could be combined into a reservoir configured to hold a fluid that acts as a wash

solution and having chemical substrates. In addition, while the reservoirs are preferably pre-filled with the respective fluids described above, the disclosure is not limited thereto and one or more reservoirs may be empty in the non-use state and filled with the respective fluid during the mixing state.

[0211] Sealing material **320** is configured to fluidly seal the fluid in one or more reservoirs. For example, sealing material **320** may fluidly seal the respective fluids in sample preparation reservoir **317**, wash reservoir **318**, and substrate reservoir **319**. Sealing material **320** may be a single piece of material configured to cover all reservoirs, as illustrated, or may be separate pieces each configured to cover one or more reservoirs within cartridge device **300**. Sealing material **320** may be any material that can fluidly seal fluid, such as a foil. Preferably, sealing material **320** is a liquid-impermeable membrane.

[0212] Seal piercer **321** is configured to pierce sealing material **320** to vent the fluid in sample preparation reservoir **317**, wash reservoir **318**, and/or substrate reservoir **319**. Seal piercer **321** may be configured to be contacted by distal portion **201**, e.g., at a shoulder or engagement zone **209**, of sample collection device **200** within input tunnel **301** and to move within housing **304**, responsive to force applied by sample collection device **200**, to cause sealing material **320** to be pierced to vent the fluid in sample preparation reservoir **317**, wash reservoir **318**, and/or substrate reservoir **319**. Seal piercer **321** is disposed within housing **304** and may be partially disposed within input tunnel **301**. In one embodiment, seal piercer **321** is configured to move in a first direction, e.g., laterally, responsive to insertion of sample collection device **200** in input tunnel **301** and in a second direction, e.g., vertically, to pierce into sealing material **320**.

[0213] Seal piercer **321** may be a single piece or may include multiple pieces. Illustratively, seal piercer **321** includes slider **322** and piercer **323**. Slider **322** is disposed within housing **304** and may be partially disposed within input tunnel **301**. For example, slider **322** may have an engager adapted to be disposed within input tunnel **301**. The engager may be configured to be temporarily or permanently coupled to sample collection device **200**, e.g., at a shoulder or at engagement zone **209**, to permit movement of slider **322** responsive to insertion of sample collection device **200** into input tunnel **301**. The engager may be sized to fit within a groove of engagement zone **209**, e.g., U-shaped as illustrated, or to receive a protrusion of engagement zone **209**. Slider **322** may be configured to move within housing **304**, responsive to force applied by sample collection device **200** resulting from a collector pushing sample collection device distally into input tunnel **301**, to cause sealing material **320** to be pierced by piercer **323** to vent the fluid in sample preparation reservoir **317**, wash reservoir **318**, and/or substrate reservoir **319**. Piercer **323** may be one or more piercing elements with ends sufficiently sharp to cut open sealing material **320**. As described in detail below, piercer **323** may include three different piercers, each disposed within housing **304** above one of sample preparation reservoir **317**, wash reservoir **318**, or substrate reservoir **319**. As slider **322** moves within input tunnel **301** as caused by insertion of sample collection device **200**, slider **322** contacts piercer **323** and moves piercer **323** in a direction to pierce sealing material **320**. In one embodiment, slider **322** is configured to move in a first direction, e.g., laterally/substantially parallel to movement of sample collection device **200**, responsive to insertion of sample collection device **200** in input tunnel **301** to cause piercer **323** to move in a second direction, e.g., vertically, to pierce into sealing material **320**.

[0214] Shuttle **324** is configured to be disposed within housing **304**, preferably between sample preparation reservoir **317** and aperture **302**. For example, shuttle **324** may be disposed within input tunnel **301** when the cartridge is in a non-use state such that a distal end of shuttle **324** forms a wall of sample preparation reservoir **317** to seal fluid therein. Shuttle **324** may define one or more compartments configured to receive the collected sample from sample collection device **200** when inserted in input tunnel **301**. Shuttle **324** also may define one or more additional compartments configured to house one or more reagents balls. Shuttle **324** may be configured to move within housing **304** when subjected to a threshold force, e.g., caused by contacting sample collection

device **200** to shuttle **324**, to a second position such that the one or more sample compartments having the sample and/or the one or more reagent ball compartments having the one or more reagent balls therein are disposed in the fluid within sample preparation reservoir **317**. The proximal end of shuttle **324** may, in conjunction within sample collection device **200**, re-form the wall of sample preparation reservoir **317** to seal fluid therein when the sample and/or the one or more reagent ball are in sample preparation reservoir **317**. In this manner, sample preparation reservoir **317** remains fluidically sealed in the non-use state by shuttle **324** and in the mixing state by shuttle **324**. In addition, unlike a breakable membrane housing reagents, shuttle **324** may remain intact as the sample is moved into sample preparation reservoir **317** for analysis.

[0215] Desiccant **325** may be disposed within housing **304** and in fluidic communication with one or more reagent balls housed in shuttle **324**. Desiccant **325** is configured to absorb moisture that enters into housing **304** to reduce moisture exposure to the one or more reagent balls in the non-use state. Desiccant **325** may be a pad and may be at least partially disposed within input tunnel **301**. Desiccant **325** may have a lumen sized to permit the sample collection device to be inserted therethrough.

[0216] Input tunnel component **326** forms a portion of input tunnel **301** and is sized and shaped to secure shuttle **324** within input tunnel **301**. For example, input tunnel component **326** may have a U-shape to house a generally cylindrical shuttle.

[0217] Sonicator element **327** is disposed within housing **304** and preferably adjacent to, or integral with, sample preparation reservoir **317** to permit mixing of the fluid therein. Sonicator element **327** is configured to transmit controlled amounts of energy into sample preparation reservoir and may include piezoelectric components. Sonicator element **327** may be disposed on or form a bottom wall of sample preparation reservoir **317**. Sonicator element **327** may be electrically isolated, e.g., via use of a relay, from other electrical components within housing **304** such as the components on circuit board **331**, including the sensor and the heaters. Sonication energy may be controlled to achieve mixing and binding of components within sample preparation reservoir **317** while limiting damage caused to fragile DNA probes or other molecules such as antibodies and enzymes.

Sonicator element **327** may include a pressure-sensitive piezoelectric disk. Sonicator element **327** also may include a high water content blister disposed between the sample preparation reservoir **317** and the piezoelectric disk. Such a high water content blister may be affixed under sample preparation reservoir **317** in the cartridge production process. The high water content blister may facilitate delivery of sonic energy from sonicator element **327** to sample preparation reservoir **317** with minimal attenuation. The blister may be replaced with another appropriately conducting sonication medium and the component serving as a sonication medium may be dry on the outside, with no liquid residue present.

[0218] Absorbent pad **328** is disposed within housing **304** at the downstream-most end of analysis channel **330**. Absorbent pad **328** wicks fluid from analysis channel **330**, thereby encouraging fluid to flow downstream to absorbent pad **328**. Absorbent pad **328** may act as a waste receptacle, collecting all waste fluids and waste particles after they have flowed through analysis channel **330**. The size and degree of absorbency of absorbent pad **328** may be selected to meter the flow of fluids and particles within the analysis channel **330**. For example, the volume of fluid that absorbent pad **328** can wick must be great enough to drain all fluid from sample preparation reservoir **317** and wash reservoir **318** and draw the fluid carrying the chemical substrate from the substrate reservoir **319**. Such a condition may serve as the lower limit of absorbency.

[0219] Layer **329** is disposed between internal component **316** and circuit board **331** and forms part of analysis channel **330**. Layer **329** may be an adhesive layer configured to couple internal component **316** to circuit board **331**. For example, layer **329** may be a double-sided adhesive tape which may be hydrophilic to support the capillary flow of fluid.

[0220] Analysis channel **330** may be defined by a wall(s) of internal component **316**, a wall(s) of layer **329**, and/or a wall(s) of circuit board component **331**. For example, the top wall of analysis

channel **330** may be defined by internal component **315**, the side walls of analysis channel **330** may be defined by layer **329**, and the bottom wall of analysis channel **330** may be defined by circuit board **331**. Additionally, each reservoir **317**, **318**, **319** includes an outlet which connects the reservoir to analysis channel **330**. In this manner, fluid within each of the reservoirs can flow through their respective outlets and into analysis channel **330**. Analysis channel **330** may extend from the reservoirs to absorbent pad **328**. Preferably, one or more sensors on circuit board **331** are at least partially positioned within analysis channel **330**.

[0221] Circuit board **331** is disposed within housing **304** and may be coupled to internal component **316**, e.g., via layer **329**. Circuit board **331** includes electrical components, for example, one or more of: resistors, electrical leads, vias, and sensors needed for detection of target analytes. Although described separately, it is to be appreciated that electrical components of circuit board **331** need not be separate structural elements. One or more electrical components and/or circuits may perform some of or all the roles of the various components described herein.

[0222] Memory **332** is disposed within housing **304** and electrically coupled to circuit board **331**. Memory **332** may be any type of memory suitable for storing data related to cartridge device **300** such as an EPROM, EEPROM, flash memory, or the like. Memory **332** may store data such as information on cartridge type (e.g., inflammation, influenza, testosterone, fertility, Vitamin D), cartridge identification information (e.g., serial number), and/or calibration information. When cartridge device **300** is electrically coupled to reader device **400**, reader device **400** may receive data transmitted from memory **332** to facilitate determination of the presence, absence, and/or quantity of one more target analytes using such data. In one embodiment, one or more cartridge devices of a select group of cartridges (e.g., common lot of production cartridges) may be tested using a known quantity of target analytes to determine electrical properties associated with one or more target analytes sensed by the sensor of the tested devices. Calibration information based on the test results may be stored in memory **332** in the select group of cartridges to precisely and consistently determine the presence, absence, and/or quantity of one more target analytes using the electrical signals generated by the sensor of the cartridge and the calibration information. Memory **332** also may store test result reliability information such as a predetermined range(s) of parameter(s), e.g., voltage, current, that may be compared to electrical signals generated by a positive control working electrode to determine whether the parameter(s) are within the predetermined range(s), as described below.

[0223] Referring now to FIGS. **6A**, **6B**, and **6C**, exemplary circuit board and layer are illustrated, wherein FIG. **6A** depicts circuit board **331**, FIG. **6B** depicts layer **329**, and FIG. **6C** depicts layer **329** disposed on and coupled to circuit board **331**.

[0224] As shown in FIG. **6A**, circuit board **331** may include heating elements **333**, **334**, **335**, **336**, and/or **337**, and sensor **338** which may include reference electrode **339**, working electrode **340**, counter electrode **341**, background working electrode **342**, and/or reference electrode **343**. Working electrode **340** may be masked with one or more striations **344** and background working electrode **342** may be masked with one or more striations **345**. Circuit board **331** may further include contacts **346** and **347** for electrically coupling circuit board **331** to sonicator element **327** via wires, although sonicator element **327** also may be electrically coupled to circuit board **331** with a spring contact as described below.

[0225] Heating elements **333**, **334**, **335**, **336**, and **337** are configured to generate heat within housing **304**, e.g., based on electric signals transmitted from reader **400** at times specified in a protocol stored within the memory of reader **400**. Each of heating elements **333**, **334**, **335**, **336**, and **337** may form part of circuit board **331**. For example, heating elements **333**, **334**, **335**, **336**, and **337** may be a resistive heating element appearing as a serpentine trace located on the bottom side of circuit board **331**, surrounding a via. In other embodiments, the heating element is located external to the cartridge, for example, on the reader. In various embodiments in which a resistive heating element is used, in order to generate heat, current is allowed to flow through the resistive heating

element, for example, through actuation of a transistor. Current passing through the resistive heating element generates heat through Joule heating. The heat is conducted to the via due to physical contact between the resistive heating element and the via. Heating elements **333**, **334**, **335**, **336**, and **337** may be masked, e.g., with a solder mask, to maintain heat transfer while promoting electrical isolation from sensor **338** to minimize interference with electrical signals sensed by sensor **338**.

[0226] Heating element **333** may be positioned adjacent to an outlet of sample preparation reservoir **317**. The outlet may have a phase-changeable material therein to occlude the entire cross-section of the outlet, thereby fluidically sealing the outlet. Heating element **333** may be configured to heat the phase-changeable material within the outlet of sample preparation reservoir **317** such that the phase-changeable material unseals the outlet of sample preparation reservoir **317** to permit the fluid having the sample mixed therein held in sample preparation reservoir **317** to flow into analysis channel **330**. Heating element **333** may be caused to heat the phase-changeable material at a time specified by a protocol stored in the memory of reader **400**, e.g., after reader **400** detects cartridge device **300** electrically coupled thereto and after reader **400** detects proper insertion of sample collection device **200** into cartridge device **300**.

[0227] Heating element **334** may be positioned adjacent to an outlet of wash reservoir **318**. The outlet may have a phase-changeable material therein to occlude the entire cross-section of the outlet, thereby fluidically sealing the outlet. Heating element **334** may be configured to heat the phase-changeable material within the outlet of wash reservoir **318** such that the phase-changeable material unseals the outlet of wash reservoir **318** to permit the wash solution held in wash reservoir **318** to flow into analysis channel **330**. Heating element **334** may be caused to heat the phase-changeable material at a time specified by the protocol stored in the memory of reader **400**, e.g., a predetermined time after reader **400** causes heating element **333** to be heated and/or a predetermined time after reader **400** causes heating element **336** to be heated.

[0228] Heating element **335** may be positioned adjacent to an outlet of substrate reservoir **319**. The outlet may have a phase-changeable material therein to occlude the entire cross-section of the outlet, thereby fluidically sealing the outlet. Heating element **335** may be configured to heat the phase-changeable material within the outlet of substrate reservoir **319** such that the phase-changeable material unseals the outlet of substrate reservoir **319** to permit the fluid with the substrates held in substrate reservoir **319** to flow into analysis channel **330**. Heating element **335** may be caused to heat the phase-changeable material at a time specified by the protocol stored in the memory of reader **400**, e.g., a predetermined time after reader **400** causes heating element **334** to be heated.

[0229] Heating element **336** may be positioned adjacent to a fluidic isolator which may comprise a phase-changeable material. Heating element **336** may be configured to heat the phase-changeable material of the fluidic isolator after the outlet of sample preparation reservoir **317** is unsealed such that the phase-changeable material of the fluidic isolator flows into analysis channel **330** to fluidically isolate sample preparation reservoir **317** from substrate reservoir **319**. Heating element **336** may be caused to heat the phase-changeable material at a time specified by the protocol stored in the memory of reader **400**, e.g., a predetermined time after reader **400** causes heating element **333** to be heated.

[0230] Heating element **337** may be positioned adjacent to a pocket of gas, e.g., air, within analysis channel **330**. Heating element **337** may be configured to heat the pocket of air to cause the air to expand and put pressure on the phase-changeable material, thereby facilitating movement of the phase-changeable material in analysis channel **330**. Heating element **336** may be caused to heat the phase-changeable material at a time specified by the protocol stored in the memory of reader **400**, e.g., a predetermined time after reader **400** causes heating element **333** to be heated. Placement of heating element **337** in the downstream direction of heating elements **333**, **334**, and **335** is expected to reduce bubble formation in analysis channel **330**.

[0231] Electrical leads (shown in FIG. 8) of circuit board 331 may be provided to establish electrical connections and continuity with a reader device. The electrical leads may be electrically coupled to heating elements 333, 334, 335, 336, 337, sensor 338 including each of reference electrode 339, working electrode 340, counter electrode 341, background working electrode 342, and reference electrode 343, contacts 346, 347, and to memory 332. In this manner, such components may receive electrical current when activated by the reader device. Advantageously, while the electrical leads are exposed at the electrical connector portion on the bottom surface of circuit board 331 (shown in FIG. 4B), the electrical leads electrically coupling the connectors to the components may be traceless on the top surface of circuit board 331 as shown in FIG. 6A. The traceless configuration of circuit board 331 between the electrical connector portion and these components creates a smooth top surface of circuit board 331 to reduce bonding interferences, thereby promoting secure adhesion, with layer 329. The bonding interferences may cause leakage when fluid enters analysis channel 330 due to malformation of layer 329 caused by such interferences.

[0232] Heating elements 333, 334, 335, 336, 337 may be formed of a conductor and each may include a via. A via is a standard product on printed circuit boards and is typically used to enable signal traces on one layer of a circuit board to continue electrically with another layer. The vias provide electrical continuity through multiple layers. Such vias are excellent conductors of heat; they are able to transfer heat to a very precise location without affecting the surrounding areas, because the surrounding material that comprises most circuit boards is an excellent insulator of heat. Thus, in various embodiments, a plurality of vias are provided in circuit board 331 as heating elements, and each via is disposed under, over, or adjacent to a phase-changeable, heat-actuated valve disposed in a reservoir outlet to create a valve actuating element. The precision of heat transfer associated with the vias allows for minimal crosstalk between valves located close to each other; thus, the timing of valve actuation can be carefully controlled for each valve. The valves may be formed of a phase-changeable material such as wax, for example, a hydrophilic wax, and the vias act as conductors of heat to melt wax at precise points of time, as controlled by a reader device. Upon phase transition, e.g., melting, of a wax valve disposed in the outlet of a reservoir, the outlet is no longer occluded and the reservoir has an opening through which its fluid contents can drain into the analysis channel. The holes in the vias may be filled with a filling material, e.g., solder, and the vias may be masked, e.g., with a solder mask, to maintain heat transfer while promoting electrical isolation from sensor 338 to minimize interference with electrical signals sensed by sensor 338.

[0233] In order to ensure full melting of the wax with precise timing, in various embodiments, the wax valves are carefully constructed within the outlets of the reservoirs. For example, in some embodiments, it is preferable for the wax valves to have the minimum height necessary to occlude the outlet of the reservoir; the minimal height minimizes the distance heat must travel to melt the wax. One example method for realizing a wax barrier having such characteristics involves applying melted wax to a pre-heated via. Advantageously, when the via is pre-heated, it takes longer for the wax valve to solidify relative to a room-temperature via; thus the wax has more time to flatten and expand outward before hardening. "Pancaking" of the wax is desirable to minimize the height, which will maximize the chance of proper melting actuation of the valve. Additionally, the heating of the via facilitates a greater level of contact area between the wax and the via such that a greater proportion of the wax experiences the heat, also maximizing the chance of proper valve actuation. The method of heating the via prior to deposition of wax is further enhanced with the following method: the opening of the reservoir is aligned over the via such that when the melted wax is applied to the pre-heated via, the opening at the bottom of the reservoir is spatially close to the via such that when the wax hardens, the wax adheres simultaneously to multiple inner walls of the reservoir and the via itself. This is advantageous for enhancing the manufacturing yield of intact valves that fully occlude the opening to the analysis channel such that no inadvertent flow of fluid

from the reservoir occurs.

[0234] Sensor **338** may be configured to be exposed to the fluid in analysis channel **330** and to generate a signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. Sensor **338** may detect electrical signals resulting from chemical reactions over sensor **338**. For example, the mixed fluid from sample preparation reservoir **317** may be introduced into analysis channel **330** and signaling agents in the mixed fluid may localize over sensor **338** (e.g., responsive to magnetic fields holding magnetic particles (if present) directly or indirectly bound to the signaling agents). The chemical reactions may occur when fluid from substrate reservoir **319** reacts with the particles from mixed fluid from sample preparation reservoir **317** localized over sensor **338**. For example, a substrate solution having a substrate may be introduced from substrate reservoir **319** and sensor **338** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS) and the signaling agents (e.g., HRP, SBP, redox dye) localized over sensor **338**. The reactions may cause electrons to be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules from the substrate solution) thereby generating electrical signals detectable by sensor **338**. Such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signal may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300** and reader device **400**.

[0235] Sensor **338** may include reference electrode **339**, working electrode **340**, counter electrode **341**, background working electrode **342**, and/or reference electrode **343**. Sensor **338** is disposed in analysis channel **330** and the area of analysis channel **330** above sensor **338** may be referred to as the “analysis zone.” Sensor **338** is strategically located such that, when circuit board **331** is included within the assembled cartridge **300** with a surface of circuit board **331** forming one wall of analysis channel **330**, sensor **338** is at least partially disposed within analysis channel **330**. While one sensor is illustrated, a plurality of sensors may be provided, each spaced relative to the others, and preferably all aligned within analysis channel **330**. In addition, working electrode **340** and background working electrode **342** may be disposed upstream and downstream of one another, or vice versa, and sensor **338** may include additional working electrodes beyond working electrode **340** and background working electrode **342**.

[0236] Sensor **338** may be an electrochemical sensor that forms an electrochemical cell within analysis channel **330**. Reference electrode **339** may be configured to create a voltage differential between itself and working electrode **340**. Counter electrode **341** may provide electrons (e.g., from the substrate stripped by the signaling agents) which gather on working electrode **340** when the electrical environment created by reference electrode **339** and working electrode **340** results in a positive charge over working electrode **340**. As is explained above, an oxidation reaction may occur at sensor **338** if an oxidizing enzyme (e.g., a signaling agent described herein such as HRP, SBP, redox dye which may be introduced into analysis channel **330** from sample preparation reservoir **317**) bound indirectly to a particle (e.g., a magnetic particle which may be introduced into analysis channel **330** from sample preparation reservoir **317**) is present at sensor **338** and an appropriate chemical substrate (e.g., TMB, OPD, ferricyanide, PBS) is introduced into analysis channel **330** (e.g., from substrate reservoir **319**). In such embodiments, working electrode **340** releases electrons to replenish electrons stripped from the substrate by the oxidizing enzyme in a quantity proportional to the amount of oxidizing enzyme present. The release of electrons from working electrode **340** (e.g., from the substrate reacting with the signaling agent over working electrode **340**) is a current which may be detectable as a signal within a circuit connected to sensor **338**. Sensor **338** can thereby indirectly detect the presence, absence, and/or quantity of oxidizing enzymes localized in the analysis zone. A processor, for example, within the reader device described below, can then correlate the presence, absence, and/or quantity of one or more target analytes to the presence, absence, and/or quantity of oxidizing enzymes. The functions of such a processor are described in more detail below. One or more magnetic fields may be used to facilitate

localization of the enzymes or other signaling agents within the analysis zone. Advantageously, in such embodiments, no affinity molecules need to be pre-bound to sensor **338** to achieve localization, which would otherwise significantly slow the analyte quantification process due to the limits of diffusion-based hybridization kinetics. Details of the magnetic fields are also provided below.

[0237] Electrochemical scanning at the sensor **338**, such as with cyclic voltammetry, facilitated by the substrate from the substrate reservoir can regenerate the signaling agent from the sample preparation reservoir localized over sensor **338** such that the substrate continuously accepts more electrons. For example, a substrate such as ferricyanide can promote regeneration of the redox dyes (e.g., methylene blue, ferrocene, nile blue, anthraquinone) during scanning such that they can continuously accept more electrons. A substrate such as ferricyanide can also act as a mediator for a label such as methylene blue, wherein the ferricyanide can diffuse to the surface of the working electrode after reacting with the methylene blue and thus act as an electron transfer mediator in proportion to how much methylene blue label is localized over the working electrode. Moreover, the concept applies to other potential mediators wherein a reducible/oxidizable molecule within a solution can be the mediator for a specific redox dye. For example, if molecule A is known to accept electrons from redox dye X and molecule B accepts electrons preferentially from redox dye Y, then measuring the electrical response at the redox potential of molecule A indicates the quantity of redox dye X and measuring the electrical response at the redox potential of molecule B indicates the quantity of redox dye Y. In this way, multiple redox dyes can be measured by measuring their specific mediators and thereby allowing for the detection and/or quantification of multiple analytes, as these redox dyes are the signaling agents specifically associated with a specific target analyte.

[0238] Sensor **338** may include gold surfaces made through an ENIG process. In other embodiments, gold or gold-plated sensors are used that have not been made through an ENIG process. A person skilled in the art can appreciate that there are many plating processes for catalytic and autocatalytic deposition of gold utilized to create electrically active pads within the printed circuit board industry.

[0239] The various electrodes of sensor **338** (e.g., working electrode, background working electrode, reference electrode, and/or positive control working electrode) may have a surface chemistry formed of a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol for added stability. Additionally, a back-filing agent with a carboxylic acid head group such as mercaptoundecanoic acid can be used with the thiolated ethylene glycol dithiol to improve the self-assembled monolayer. Alternatively or additionally, the various electrodes of sensor **338** may have surface-bound affinity molecules pre-bound thereto. Sensor **338** may have select electrodes having a surface chemistry formed of a self-assembled monolayer while other electrodes have surface-bound affinity molecules pre-bound thereto. Additionally or alternatively, the various electrodes of sensor **338** may overlay surface-bound affinity molecules pre-bound to surface chemistry of a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol.

[0240] Working electrode **340** may have a surface chemistry formed of a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol for added stability. The hydrophilic nature of the head groups of such surface chemistry facilitates flow and protein resistance. Additionally or alternatively, the surface of one or more of the electrodes may be backfilled with mercaptoundecanoic acid, mercaptohexanol, or other suitable backfiller. The surface of one or more of the electrodes within sensor **338** may be formed through sequential addition and incubation of the ethylene glycol dithiol and the backfiller at unelevated temperatures.

[0241] Background working electrode **342** may be ambient electrochemical noise sensors spaced within analysis channel **330** away from the site of magnetic particle localization. Background working electrode **342** may be used to quantify background noise downstream or upstream of working electrode **340**, depending on the selected order of working electrode **340** and background

working electrode **342** within analysis channel **330**. Such noise may be due to, for example, the presence of non-specifically bound enzyme. During processing of the detection results, a processor at reader **400** may apply an algorithm to remove the background working electrode signal(s) (from background working electrode **342**) from the detection sensor signal (from working electrode **340**) to account for and/or eliminate system noise and to thereby allow for proper quantification or detection of the one or more target analytes. The signal from background working electrode **342** may be used for error detection and diagnosis of an improperly functioning cartridge, e.g., as evidenced by the signal from background working electrode **342** having electrical value(s) outside a predetermined range(s).

[0242] Reference electrode **343** may be configured to create a voltage differential between itself and background working electrode **342**. Counter electrode **341** also may provide electrons which gather on background working electrode **342** when the electrical environment created by reference electrode **343** and background working electrode **342** results in a positive charge over background working electrode **342**.

[0243] In some embodiments, the detection is carried out using a standard electrochemical circuit that utilizes a bias potential generated at background working electrode **342** for the oxidation/reduction reaction to proceed. The potential is held at the reduction potential of the chemical substrate (low enough that there is little nonspecific reduction of reducible species in the solution) so that the flow of electrons to the oxidized molecules can be quantified using an operational amplifier based current-to-voltage (op amp) circuit topology in reader device **400** electrically connected to working electrode **340**.

[0244] A common substrate molecule, tetramethylbenzidine, may be used for HRP. When present, HRP oxidizes TMB molecules, and these molecules are in turn reduced by working electrode **340**. Since this event occurs in proportion to the amount of HRP present which in turn is proportional to the amount of target analyte present, a change in the current-to-voltage op amp measurement results. Using an analog-to-digital converter, the actual signal can be delivered to a processor for processing. As described in more detail below, in various embodiments, the processor and signal processing components are provided within the reader device.

[0245] Working electrode **340** may be masked, e.g., solder masked, with a plurality of striations **344** configured to promote homogenous distribution and retention of the plurality of magnetic particles released from sample preparation reservoir **317** over working electrode **340**. Accuracy of analyte detection may be adversely impacted by premature wash away of magnetic particles during analysis due, for example, force on the particles in the analysis channel flow direction caused by release of the wash solution from wash reservoir **318** and/or the fluid having the chemical substrates from substrate reservoir **319** being greater than the magnetic force toward the magnetic generators of reader **400** disposed beneath working electrode **340** during analysis. Such striations **344** promote resistance to movement of the plurality of magnetic particles off working electrode **340**. In addition, background working electrode **342** may be masked, e.g., solder masked, with a plurality of striations **345**, although such striations are not necessary on background working electrode **342** and are merely exemplary of an alternate embodiment.

[0246] In alternative embodiments, sensor **338** may be configured to analyze the fluid in analysis channel **330** and to generate a signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample, wherein the signal is visible. For example, the housing of the cartridge device may include a window that permits a user to view, e.g., with a camera, fluorescence and quantify that fluorescence to determine the presence, absence, and/or quantity of one or more analytes within the sample.

[0247] Referring now to FIG. **6C**, layer **329** is disposed on the top surface of circuit board **331** such that sensor **338** is disposed at least partially in analysis channel **330**. Heating elements **333**, **334**, **335**, **336**, and **337** may be covered, partially or fully, with masks **348**, **349**, **350**, **351**, and **352**, respectively. Masks **348**, **349**, **350**, **351**, and **352** may be solder masks. Mask **348** is configured to

maintain heat transfer from heating element **333** to the phase-changeable material within the outlet of sample preparation reservoir **317** at an energy level sufficient to cause a phase change of the material while promoting electrical isolation between heating element **333** and sensor **338** to minimize interference by heating element **333** with the electrical signals sensed by sensor **338**. Mask **349** is configured to maintain heat transfer from heating element **334** to the phase-changeable material within the outlet of wash reservoir **318** at an energy level sufficient to cause a phase change of the material while promoting electrical isolation between heating element **334** and sensor **338** to minimize interference by heating element **334** with the electrical signals sensed by sensor **338**. Mask **350** is configured to maintain heat transfer from heating element **335** to the phase-changeable material within the outlet of substrate reservoir **319** at an energy level sufficient to cause a phase change of the material while promoting electrical isolation between heating element **335** and sensor **338** to minimize interference by heating element **335** with the electrical signals sensed by sensor **338**. Mask **351** is configured to maintain heat transfer from heating element **336** to the phase-changeable material of the fluidic isolator at an energy level sufficient to cause a phase change of the material while promoting electrical isolation between heating element **336** and sensor **338** to minimize interference by heating element **336** with the electrical signals sensed by sensor **338**. Mask **352** is configured to maintain heat transfer from heating element **337** to the pocket of gas in analysis channel **330** above heating element **337** at an energy level sufficient to cause movement downstream in analysis channel **330** of phase-changeable material while promoting electrical isolation between heating element **337** and sensor **338** to minimize interference by heating element **337** with the electrical signals sensed by sensor **338**.

[0248] Referring now to FIGS. 7A, 7B, and 7C, circuit board **331'** and layer **329'** are constructed similarly to circuit board **331** and layer **329** of FIGS. 6A, 6B, and 6C, wherein like components are identified by like-primed reference numbers, except that heating elements **333'**, **334'**, **335'**, **336'**, and **337'** are positioned in a different configuration on circuit board **331'** and analysis channel **330'** is re-shaped accordingly. In addition, FIG. 7C depicts absorbent pad **328** coupled to layer **329'** at the downstream end of analysis channel **330'**.

[0249] Referring now to FIGS. 7D and 7E, alternative exemplary sensors that may be used in the cartridges described herein are provided. Sensor **338''** may include reference electrode **339''**, working electrode **340''**, counter electrode **341''**, negative control working electrode **342''** (also referred to herein as a background working electrode), and/or positive control working electrode **376**. Sensor **338''** may detect electrical signals generated by chemical reactions at sensor **338''** as described above for sensor **338**. Sensor **338''** is disposed in the analysis channel in the same manner as sensor **338** described above. While one sensor is illustrated, a plurality of sensors may be provided, each spaced relative to the others, and preferably all aligned within the analysis channel. Preferably, fluid flows from the reservoirs in the analysis channel and travels over the electrodes in the following order: positive control working electrode **376**, reference electrode **339''**, counter electrode **341''**, working electrode **340''**, and negative control working electrode **342''**.

[0250] Sensor **338''** may be an electrochemical sensor that forms an electrochemical cell within the analysis channel. Positive control working electrode **376** may have affinity molecules pre-bound to the surface of positive control working electrode **376** to achieve localization of oxidizing enzymes, or other signaling agents, over positive control working electrode **376**. The affinity molecules may be surface bound antibodies. Positive control working electrode **376** may be configured to detect current generated by reactions between an oxidizing enzyme, or other signaling agents, indirectly bound to the affinity molecules and an appropriate chemical substrate introduced into the analysis channel, e.g., from the substrate reservoir. In such embodiments, positive control working electrode **376** releases electrons to replenish electrons stripped from the substrate by the oxidizing enzyme in a quantity proportional to the amount of oxidizing enzyme present. The release of electrons from positive control working electrode **376** is a current which may be detectable as a signal within a circuit connected to sensor **338''**. A processor, for example, within the reader device described

below, may process the signal to determine if the signal indicates a quantity of oxidizing enzyme, or other signaling agent, within a predetermined range which may be stored in memory of the cartridge and/or reader device. If the detected quantity is within the range, the processor may verify the cartridge and continue processing signals to determine the presence, absence, and/or quantity of one or more target analytes within the sample. The signal from positive control working electrode **376** may be used for error detection and diagnosis of an improperly functioning cartridge, e.g., as evidenced by the signal from positive control working electrode **376** having electrical value(s) outside a predetermined range(s). For example, the processor of the reader may generate an error alert if the signal from the positive working control electrode **376** is outside a predetermined range and/or may consider a reading from working electrode **340** acceptable if within the predetermined range.

[0251] Reference electrode **339** may be configured to create a voltage differential between itself and working electrode **340**. Counter electrode **341** may provide electrons which gather on working electrode **340** when the electrical environment created by reference electrode **339** and working electrode **340** results in a positive charge over working electrode **340**. Reference electrode **339** and/or counter electrode **341** may have a surface chemistry formed of a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol for added stability. The hydrophilic nature of the head groups of such surface chemistry facilitates flow and protein resistance. Additionally or alternatively, the surface of reference electrode **339** and/or counter electrode **341** may be backfilled with mercaptoundecanoic acid, mercaptohexanol, or other suitable backfiller.

[0252] Sensor **338** may be used when magnetic particles are not present in the cartridge. For example, working electrode **340** may have affinity molecules pre-bound to the surface of working electrode **340** to achieve localization of oxidizing enzymes, or other signaling agents, over working electrode **340**. The affinity molecules may be surface bound antibodies. Working electrode **340** may be configured to detect current generated by reactions between an oxidizing enzyme, or other signaling agents, indirectly bound to the affinity molecules and an appropriate chemical substrate introduced into the analysis channel, e.g., from the substrate reservoir. In such embodiments, working electrode **340** releases electrons to replenish electrons stripped from the substrate by the oxidizing enzyme in a quantity proportional to the amount of oxidizing enzyme present. The release of electrons from working electrode **340** is a current which may be detectable as a signal within a circuit connected to sensor **338**. Sensor **338** can thereby indirectly detect the presence, absence, and/or quantity of oxidizing enzymes localized in the analysis zone. A processor, for example, within the reader device described below, can then correlate the presence, absence, and/or quantity of one or more target analytes to the presence, absence, and/or quantity of oxidizing enzymes. The functions of such a processor are described in more detail below.

[0253] Working electrode **340** illustratively does not include a plurality of striations as sensor **338** may be used to detect the presence, absence, and/or quantity of one or more target analytes without the use of magnetic particles.

[0254] Negative control working electrode **342** may be ambient electrochemical noise sensors spaced within the analysis channel away from the site of localization. Negative control working electrode **342** may be used to quantify background noise downstream of working electrode **340**. Such noise may be due to, for example, the presence of non-specifically bound enzyme. During processing of the detection results, a processor at reader **400** may apply an algorithm to remove the negative control working electrode signal(s) (from negative control working electrode **342**) from the detection sensor signal (from working electrode **340**) to account for and/or eliminate system noise and to thereby allow for proper quantification or detection of the one or more target analytes. The signal from negative control working electrode **342** may be used for error detection and diagnosis of an improperly functioning cartridge, e.g., as evidenced by the signal from negative control working electrode **342** having electrical value(s) outside a predetermined range(s). For

example, the processor of the reader may generate an error alert if the signal from the negative working control electrode **342''** is above a threshold and/or may consider a reading from working electrode **340''** acceptable if below the threshold.

[0255] Negative control working electrode **342''** may have a surface chemistry formed of a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol for added stability. The hydrophilic nature of the head groups of such surface chemistry facilitates flow and protein resistance. Additionally or alternatively, the surface of Negative control working electrode **342''** may be backfilled with mercaptoundecanoic acid, mercaptohexanol, or other suitable backfiller.

[0256] Referring now to FIG. 7E, sensor **338'** may be used when magnetic particles are present in the cartridge. Like sensor **338''**, sensor **338'** includes positive control working electrode **376'**, reference electrode **339'**, counter electrode **341''**, and negative control working electrode **342'''** structured similarly to the like-primed components of FIG. 7D described above. Working electrode **340''** may be structurally similar to working electrode **340** described above with respect to FIG. 6A. In this manner, sensor **338'** is particularly well suited for target analyte detection using one or more magnetic fields to facilitate localization of the enzymes or other signaling agents within the analysis zone.

[0257] Referring now to FIG. 7F, an alternative exemplary sensor that may be used in the cartridges described herein is provided. Sensor **338''** may be structured the same manner as sensor **338'** except that negative control working electrode **342''** may be positioned between positive control working electrode **376''** and reference electrode **339''**. Preferably, fluid flows from the reservoirs in the analysis channel and travels over the electrodes in the following order: positive control working electrode **376''**, negative control working electrode **342''**, reference electrode **339''**, counter electrode **341''**, and working electrode **340''**. In one embodiment, positive control working electrode **376''**, negative control working electrode **342''**, reference electrode **339''**, counter electrode **341''**, and/or working electrode **340''** have surface-bound affinity molecules pre-bound to surface chemistry of a self-assembled monolayer.

[0258] FIG. 7G are images generated by Atomic Force Microscopy (AFM) of exemplary sensor surfaces that may be used on the one or more sensors within the housing of the cartridge device, including the surface of any of the electrodes described above in FIGS. 6A, 7A, 7D, 7E, and/or 7F. The upper left image shows a gold unmodified sensor surface. The upper right image shows the sensor surface after application of the self-assembled monolayer. The self-assembled monolayer covers the surface and is very stable. In the image, the self-assembled monolayer laid down on the gold electrode surface is hexaethylene glycol dithiol (HEGT). The bottom two images show the sensor surface after a backfiller backfills the self-assembled monolayer matrix. Preferably, the backfiller is highly responsive to the self-assembled monolayer. In the images, the backfiller is mercaptoundecanoic acid (MUDA). The backfilled self-assembled monolayer sensor surface is very stable and highly responsive. As a result, the sensor is capable of passing current from the nanoamp to the microamp range.

[0259] FIG. 7H is a graph showing measured current in nA before and after self-assembled monolayer treatment for Electrode 1 and Electrode 2. As is shown, the self-assembled monolayer treatment controls background noise at the sensor. The graph illustrates that before self-assembled monolayer treatment, there is high, fluctuating background noise on the electrodes. After the self-assembled monolayer treatment, the signal-to-noise ratio (SNR) greatly improves, thereby increasing consistency for sensor readings.

[0260] FIG. 7I is a graph showing measured current in nA versus CRP concentration (mg/L) for a new sensor with a self-assembled monolayer prepared as described herein and a sensor with a self-assembled monolayer aged the equivalent of 18 months in an aging chamber. The graph illustrates that the sensor with the self-assembled monolayer is stable over time as the new sensor and the aged sensor record signal per CRP concentration nearly identically. As such, the sensors described

herein with the self-assembled monolayer are stable across a wide range of temperatures and environmental conditions.

[0261] Referring to FIG. 8A, internal component **316** is coupled to circuit board **331**, e.g., via layer **329** positioned therebetween. Internal component **316** may include sample preparation reservoir **317** having outlet **353** with valve **354** positioned adjacent to heating element **333**, wash reservoir **318** having outlet **355** with valve **356** positioned adjacent to heating element **334**, and substrate reservoir **319** having outlet **357** with valve **358** positioned adjacent to heating element **335**. Each of reservoirs **317**, **318**, and **319** is, at least at times, in fluid communication with analysis channel **330** such that fluid exiting the reservoirs, e.g., via their respective outlets when the respective valves are opened, flows into analysis channel **330**. In addition, fluidic isolator **359** may be positioned adjacent to heating element **336** and in fluidic communication with analysis channel **330**.

[0262] Valves **354**, **356**, and **358** may be located within outlets **353**, **355**, and **357**, respectively, at the bottom of reservoirs **317**, **318**, and **319**, respectively, of cartridge **300**. Outlets **353**, **355**, and **357** may each be formed of a hole within a bottom wall of internal component **316** above analysis channel **330**. Valves **354**, **356**, and **358** may each be formed of a heat-sensitive, phase-changeable material, such as, for example, a hydrophilic wax. Prior to actuation, the wax or other heat-sensitive material of the valve is in a solid or semi-solid state and is sized and shaped to fill an entire cross-section of the outlet such that no fluid can escape from the respective reservoir into analysis channel **330**. Valves **354**, **356**, and **358** may be aligned directly above one or more heating elements (with a solder mask therebetween) or other localized heat-conductive element. Such alignment allows for the localized application of heat to induce a phase change in the valve without causing a phase change of any neighboring valves. In various embodiments, the phase change melts or otherwise transforms the heat-sensitive material such that it no longer causes full occlusion of the outlet, but instead permits fluid in the respective reservoir to flow into analysis channel **330**.

[0263] In addition, fluidic isolator **359** may be formed of a heat-sensitive, phase-changeable material, such as, for example, a hydrophilic wax. Prior to actuation, the wax or other heat-sensitive material is in a solid or semi-solid state and is disposed out of the flow path between the outlets of the respective reservoirs and sensor **338** within analysis channel **330**. Fluidic isolator **359** may be sized and shaped to block the flow path on analysis channel **330** between an outlet of one reservoir, e.g., sample preparation reservoir **317**, and the outlet of another reservoir, e.g., substrate reservoir **319**, when fluidic isolator is activated, e.g., by heating heating element **336**. Fluidic isolator **359** may be aligned directly above heating element **336** (with a solder mask therebetween) or other localized heat-conductive element. Such alignment allows for the localized application of heat to induce a phase change in fluidic isolator **359** without causing a phase change of any neighboring valves. In various embodiments, the phase change melts or otherwise transforms the heat-sensitive material such that it flows into analysis channel to block outlet **353** of sample preparation reservoir **317** from analysis channel **330**. In this manner, fluid from substrate reservoir **319**, when released into analysis channel **330**, cannot enter into sample preparation reservoir **317** and cannot interact with leftover signaling agents from sample preparation reservoir **317**.

[0264] The wax material disposed upon the via or a solder mask over via, and which occludes the opening of the respective reservoir or isolates the analysis channel, may be a hydrophilic material such as hexadecanol or octodecanol. This advantageously promotes, rather than obstructs the flow of fluid past any wax bits that harden within any area of the analysis channel after actuation. These materials also preferably have a melting temperature between 50 and 100 degrees Celsius, which allows for actuation with reasonable power-consumption for a battery-operated device, yet remains unactuated in general handling and storage environments and/or during a sonication protocol. The amount of wax per valve may be below 1 microliter in its liquid state, the amount may be less than or equal to 0.5 microliters, and the amount may be greater than 2 nanoliters. Using minimal amount of wax in the valves is one way to reduce any occlusion of the analysis channel and maximize full valve actuation when heat is applied. The valve also may have a feedback-and-control system that

allows for a consistent thermal profile to be achieved at the via for consistent valve actuation. Furthermore, this feedback-and-control system may incorporate sensing elements to enable the system to confirm that each valve has properly actuated.

[0265] As shown in FIG. 8A, circuit board **331** may have exposed leads **360** at electrical connector **312**. In one embodiment, leads **360** are exposed only on the bottom surface of circuit board **331** although leads **360** may be exposed on the top surface of circuit board **331** but are preferably traceless as shown. As described above, electrical connector **312** permits electrical connection with a corresponding electrical connector of reader device **400** such that cartridge device **300** may transmit signals indicative of the presence, absence, and/or quantity of one or more target analytes within a collected sample sensed by the sensor of cartridge device **300**.

[0266] Internal component **316** may include absorber pad housing **361** sized and shaped to hold absorber pad **328**. Absorber pad housing **361** may include a plurality of vent holes **362** to permit exposure of absorber pad **328** within absorber pad housing **361** to the environment within housing **304** of cartridge device **300**.

[0267] Input tunnel **301** of cartridge device **300** may include slot **363** configured to permit seal piercer **321** to be at least partially disposed within input tunnel **301**. Slot **363** may be at the proximal end of input tunnel component **326** as illustrated. Slot **363** may be sized and shaped to permit engager **324** of slider **321** to be disposed within input tunnel **301**. In addition, slot **363** may have a length sufficient to permit slider **322** to slide, when engager **324** contacts the sample collection device within input tunnel **301**, distally from a pre-venting position to a venting position.

[0268] Referring now to FIGS. 8B and 8C, a close-up view of certain components of circuit board **331** and a valve are shown. FIG. 8B depicts conductor **364** coupled to resistor **365**, e.g., an aluminum resistor, which is coupled to heating element **333** and FIG. 8C further shows mask **348** disposed between heating element **333** and valve **354**. As will be clear to one skilled in the art, while the details of heating element **333**, mask **348**, and valve **354** are illustrated, such a configuration may be utilized with respect to heating elements **334**, **335**, **336**, and **337** with their respective masks and valves, fluidic isolators, or air pockets. Current from reader device **400** may pass from conductor **364** through resistor **365** to generate heat through Joule heating. The heat is conducted to heating element **333** due to physical contact between resistor **365** and heating element **333**. Heating element **333** generates heat through mask **348** to cause a phase change of the phase-changeable material of valve **354** while promoting electrical isolation from sensor **338** to minimize interference by heating element **333** with electrical signals sensed by sensor **338**.

[0269] Referring now to FIG. 8D, a close-up view of alternative components of circuit board **331'** and a valve are shown. FIG. 8D depicts conductors **364'**, e.g., solder pads, coupled to resistor **365'**, e.g., an aluminum resistor. Unlike the configuration shown in FIGS. 8B and 8C, conductor **364'** is not coupled to heating element **333'** (which includes a via in circuit board **331'**) such that a mask disposed between heating element **333'** and valve **354'** is not needed. As will be clear to one skilled in the art, while the details of heating element **333'** and valve **354'** are illustrated, such a configuration may be utilized with respect to heating elements **334**, **335**, **336**, and **337** with their respective valves, fluidic isolators, or air pockets. Current from reader device **400** may pass from conductor **364'** through resistor **365'** to generate heat through Joule heating. The via of heating element **333'** is disposed between, and electrically isolated from, conductors **364'** coupled to resistor **365'**. The heat is conducted to heating element **333'** through indirect contact between resistor **365'** and heating element **333'**. Heating element **333'** generates heat to cause a phase change of the phase-changeable material of valve **354'** while promoting electrical isolation from the sensor to minimize interference by heating element **333'** with electrical signals sensed by the sensor.

[0270] FIGS. 9A and 9B illustrate various shuttles that may be disposed within the input tunnel of the cartridge device. Referring to FIG. 9A, shuttle **324** may include first end **366**, reagent ball compartment **367**, compartment divider **368** having slot **369**, sample compartment **370**, second end

371 having opening **372** therethrough, and/or beams **373**, **374**.

[0271] Shuttle **324** is configured to be disposed within the cartridge housing, preferably within the input tunnel between the sample preparation reservoir and the aperture defining the opening of the input tunnel in a pre-mixing state. In such a pre-mixing state, first end **366** may form a wall of the sample preparation reservoir to seal fluid within the reservoir. First end **366** may include one or more sealing members, e.g., O-rings, to enhance a liquid tight seal that may be comprised in part of chlorobutyl. The area between first end **366** and compartment divider **368** may be referred to as reagent ball compartment **367**. Reagent ball compartment **367** is configured to house one or more reagent balls, e.g., reagent ball **375**. In the pre-mixing state, reagent ball compartment **367** is preferably sealed off from the fluid in the sample preparation reservoir. Compartment divider **368** may be used to divide compartments in shuttle **324**, e.g., reagent ball compartment **367** and sample compartment **370**. Compartment divider **368** may include slot **369** which permits fluid to flow through compartment divider **368** when compartment divider **368** is disposed within the fluid of the sample preparation reservoir in an mixing state. Permitting fluid to flow through slot **369** may enhance mixing of the sample, the reagent ball, and the fluid in the sample preparation reservoir. In addition, slot **369** permits enhanced fluidic communication between reagent ball **375** and desiccant **325** disposed within the cartridge housing in the pre-mixing state. It should be understood without being explicitly stated that the compartment may contain reagent balls that are the same or different from each other, and can be preselected for detection and/or quantification of the target analyte.

[0272] In the pre-mixing state, second end **371** may be disposed between first end **366** and the aperture defining the opening of the input tunnel. Second end **371** may include opening **372** which is configured to permit the collected sample to be inserted therethrough and into sample compartment **370**. For example, opening **372** may be sized slightly larger than the tip of the sample collection device such that the tip can travel through opening **372** while opening **372** wipes excess sample from the tip. In this manner, at most a predetermined volume of sample is inserted in sample compartment **370**. The area between second end **371** and compartment divider **368** may be referred to as sample compartment **370**. Shuttle **324** may include one or more beams **373**, **374** configured to couple first end **366** to second end **371** and to be coupled to compartment divider **368**. Beams **373**, **374** are preferably positioned closer to the top surface of the cartridge housing when in the input tunnel to avoid interference with fluid mixing in the mixing state.

[0273] Shuttle **324** may move from a first position in a pre-mixing state to a second position in a mixing state, e.g., responsive to application of a force beyond a threshold force exerted on shuttle **324** by the sample collection device. In the mixing state, first end **366**, reagent ball compartment **367**, and sample compartment **370** may be disposed within the sample preparation reservoir. Accordingly, the sample and the reagent ball(s) may be mixed within fluid in the sample preparation reservoir. Second end **371** of shuttle **324** may re-form the wall of the sample preparation reservoir to seal fluid therein when the sample and/or the one or more reagent ball are in the sample preparation reservoir. Preferably, the distal portion of the sample collection device fluidically seals opening **372** such that fluid cannot escape the sample preparation reservoir into the input tunnel in the mixing state. Second end **371** may include one or more sealing members, e.g., O-rings, to enhance a liquid tight seal. In this manner, the sample preparation reservoir remains fluidically sealed in the pre-mixing state by first end **366** and in the mixing state by second end **371** and the sample collection device and during movement from the pre-mixing state to the mixing state. In addition, unlike a breakable membrane housing reagents, shuttle **324** may remain intact as the sample is moved into sample preparation reservoir **317** for analysis.

[0274] Reagent ball **375** may include one or more of magnetic particles, affinity molecules, connection molecules, signaling agents, competitor binding molecules, competitor molecules, labels, signaling agents, primers, nucleic acid probes, and/or polymerases, and other enzymes or components as described in further detail herein and the components, encapsulation material and dimensions may be the same or different from each other. In one aspect, reagent ball **375** is formed

by the freezing (i.e. lowering the temperature of a volume of liquid (such as between 5 microliters and 30 microliters)) to a temperature to induce a phase change in the liquid. The temperature may vary depending on the components of the liquid. In a further aspect, the liquid additionally comprises excipients known to those of ordinary skill in the art of lyophilization such as lyoprotectants for the functional preservation of the reagents that may be temperature sensitive, e.g., nucleic acids and/or protein components, within the liquid volume as it undergoes the process of freezing and drying to become reagent ball **375**. Stabilizers such as disaccharides like sucrose and trehalose or other lyoprotectants such as polyethylene glycol of various molecular weights and bulking or caking agents such as mannitol, glycine, povidone, and others known in the art can comprise some of the final constituents of reagent ball **375** in addition to the reagents noted herein. The (w/v) percentage of the excipients within the volume of liquid to be freeze dried to form reagent ball **375** can vary widely from about 0.1% to about 30% and be combined in various ways, often with a combination of a disaccharide as a lyoprotectant and a caking agent or bulking agent to add structure. Reagent ball **375** can be of many sizes, non-limiting examples of such include having an diameter of between about 1 mm to about 7 mm, or alternatively from about 2 mm to about 5 mm, or alternatively about 3 mm, or alternatively less than about 7 mm, or alternatively less than about 5 mm, or alternatively less than about 4 mm. While reagent ball **375** is illustrated as a sphere, the disclosure is not limited thereto and many shapes may be used and multiple reagent balls each containing the same or different reagents also may be used. As is ordinary in the art, such liquid when formed containing the components and excipients can be frozen through flash freezing in liquid nitrogen or through shelf freezing within a lyophilizer machine. After freezing, said frozen volume is subjected potentially to annealing treatments for crystallization of caking agents, primary drying and secondary drying wherein water is removed until a sufficiently small percentage (e.g. <8%, preferably <5% and preferably around 1%) remains in the final freeze dried product, which is reagent ball **375**.

[0275] Referring now to FIG. **9B**, shuttle **324'** is constructed similarly to shuttle **324** of FIG. **9A**, wherein like components are identified by like-primed reference numbers, except compartment divider **368'** is solid without a slot and beams **377**, **378**, and **379** are positioned in different orientations on shuttle **324'**.

[0276] As will be readily apparent to one skilled in the art, while FIGS. **9A** and **9B** illustrate shuttles having one sample compartment and one reagent ball compartment, the disclosure is not limited thereto and the shuttles may define one or more compartments configured to receive the collected sample from the sample collection device when inserted in the input tunnel **301** and one or more additional compartments configured to house one or more reagents balls.

[0277] Referring now to FIGS. **10A** through **10P**, insertion of a sample collected by a sample collection device into a cartridge device is described. Prior to insertion within input tunnel **301** of cartridge device **300**, sample collection device **200** is exposed to a sample, e.g., within an inner cheek, the throat, the mouth, a nasal passageway, an ear, from urine, from blood, from plasma, from saliva, etc. Tip **204** of sample collection device **200** is designed to retain some of the sample to permit analysis of the presence, absence, and/or quantity of one or more target analytes within the sample using cartridge device **300** and reader device **400**. Cartridge device **300** may be electrically coupled to reader device **400** before or after sample collection device **200** is inserted in cartridge device **300**.

[0278] Referring to FIG. **10A**, the distal end of sample collection device **200** is first inserted into the aperture defining the opening of input tunnel **301**. Shuttle **324** is disposed within input tunnel **301** in a pre-mixing position wherein first end **366** of shuttle **324** forms a wall of sample preparation reservoir **317** to fluidically seal fluid within sample preparation reservoir **317**. In this pre-mixing position, reagent ball **375** housed by shuttle **324** is within input tunnel **301** and not exposed to fluid within sample preparation reservoir **317**. As the distal end of sample collection device **200** moves distally into input tunnel **301**, sample collection device **200** may contact engager **380** of

slider **322**. For example, distal sealing zone **208** may contact engager **380** as illustrated. Distal sealing zone **208** may be angled to facilitate movement of engager **380** past distal sealing zone **208** as sample collection device **200** is moved more distally or distal sealing zone **208** may be a shoulder sized to cause movement of slider **322**.

[0279] As shown in FIG. **10B**, tip **204** (having sample thereon and/or within tube **205**) enters sample compartment **370** of shuttle **324** through opening **372** of second end **371**. Preferably, as tip **204** enters sample compartment **370**, shuttle **324** remains substantially in position within input tunnel **301** and first end **366** of shuttle **324** continues to form a wall of sample preparation reservoir **317** to seal fluid within sample preparation reservoir **317**. In addition, first end **366** of shuttle **324** may continue to form a wall of sample preparation reservoir **317** to seal fluid within sample preparation reservoir **317** (and shuttle **324** may remain substantially in pre-mixing position) until distal sealing portion **208** contacts second end **371** to fluidically seal opening **372**. Application of a force greater than a threshold force on shuttle **324** (e.g., at second end **371**) by sample collection device **200** (at distal sealing portion **208**) may move shuttle **324** from the pre-mixing position to the mixing position wherein the sample on sample collection device **200** and reagent ball **375** are mixed within fluid of sample preparation reservoir **317**.

[0280] In addition, as sample collection device **200** (e.g., tip **204**) is inserted into shuttle **324** (e.g., via opening **372**), shuttle **324** may wipe excess sample from sample collection device **200**, thereby preventing the wiped sample from entering shuttle **324**. For example, the walls of second end **371** that define opening **372** may wipe excess sample from tip **204** as tip **204** is inserted through opening **372**. All or substantially all of the sample on the outer surface of tip **204** may be wiped away leaving only sample disposed within tube **205**. Tube **205** may hold, at most, a predetermined volume of sample, e.g., about 2 μ l. Wiping excess sample from sample collection device **200** may enhance precision, accuracy, and/or consistency of analysis as, at most, a predetermined volume of sample is inserted in sample preparation reservoir **317** in the mixing position.

[0281] Referring now to FIGS. **10C** through **10K**, an exemplary process is described for piercing the sealing material disposed over one or more reservoirs in the cartridge device via interaction between the sample collection device and the seal piercer within the cartridge device. As described above, seal piercer **321** may include slider **322** and piercer **323**.

[0282] FIG. **10C** is a top view of a portion of cartridge device **300** illustrating a possible orientation of piercing elements over the reservoirs. Piercer **323** may have first piercing element **381** having a piercing end disposed over sample preparation reservoir **317**, second piercing element **382** having a piercing end disposed over wash reservoir **318**, and/or third piercing element **383** having a piercing end disposed over substrate reservoir **319**. In the pre-mixing state, piercing elements **381**, **382**, and **383** are disposed over their respective reservoirs and have not pierced the sealing material sealing fluid within the respective reservoirs. Piercing elements **381**, **382**, and **383** may be coupled (e.g., at an end opposite the piercing end) to the housing of the cartridge device.

[0283] FIG. **10D** is a perspective view of sample collection device **200** within the input tunnel of cartridge device **300**, with half the cartridge housing removed to show slider **322** for clarity. Slider **322** may include first track **384** configured to engage and move first piercer **381** into a piercing position, second track **385** configured to engage and move second piercer **382** into a piercing position, and/or third track **386** configured to engage and move third piercer **383** into a piercing position. As sample collection device **200** is moved distally through input tunnel **301**, preferably slider **322** does not move within the cartridge housing, and remains in a pre-vent position, until sample collection device **200** securely engages slider **322**. Slider **322** may securely engage sample collection device **200** by temporarily or permanently coupling engager **380** of slider **322** to engagement zone **209** of sample collection device **200**. Engager **380** is disposed in input tunnel **301** (e.g., hanging below, at least at times, slot **363** of FIG. **8A**). Engager **380** may be sized to fit within a groove of engagement zone **209**, e.g., protrusions and/or U-shaped, or to receive a protrusion of engagement zone **209**.

[0284] FIG. 10E is a cross-sectional side view depicting the seal piercer in the pre-piercing position and the shuttle in the pre-mixing position. As shown, sample collection device 200 has been moved distally in input tunnel 301 such that engager 380 of the slider has engaged engagement zone 209 of sample collection device 200 and distal sealing zone 208 of sample collection device 200 has fluidically sealed the opening of second end 371 of shuttle 324. Preferably, first end 366 continues to fluidically seal fluid within sample preparation reservoir 317 at least until sample collection device 200 fluidically seals second end 371 of shuttle 324. As such, the one or more reagent balls housed by shuttle 324 and the collected sample within shuttle 324 remain out of fluidic contact with the fluid in sample preparation reservoir 317. Slider 322 and shuttle 324 may be positioned within the cartridge housing such that engager 380 of slider 322 engages engagement zone 209 of sample collection device 200 at the same time, or approximately the same time, that distal sealing zone 208 of sample collection device contacts and fluidically seals second end 371 of shuttle 324. In such an embodiment, slider 322 and/or shuttle 324 may not yet have been caused to move within cartridge device 300 at this time.

[0285] FIGS. 10F and 10G further illustrate positioning of the slider and the piercer in the pre-venting position shown in FIG. 10E. As shown in FIG. 10F, when engager 380 of slider 322 engages engagement zone 209 of sample collection device 200 in the pre-venting position, piercing elements 381, 382, and 383 have not yet been deflected downwardly toward their respective reservoirs by slider 322. In such a pre-venting position, tracks 384, 385, and 386 may not yet contact piercing elements 381, 382, and 383, respectively. As shown in FIG. 10G, in the pre-venting position, piercing element 381 is disposed above, but has not yet pierced, sealing material 320 which seals fluid within sample preparation reservoir 317. In FIG. 10G, track 384 of slider 322 has not yet contacted piercing element 381 of piercer 323.

[0286] Referring now to FIG. 10H, sample collection device 200 is moved further distally within input tunnel 301 from the pre-mixing and pre-venting positions towards the mixing and venting positions. As collector pushes sample collection device 200 distally, shuttle 324 is partially moved within sample preparation reservoir 317. Shuttle 324 may be caused to move within the cartridge by, for example, application of a force greater than a threshold force by sample collection device 200 (e.g., at distal sealing zone 208) on shuttle 324 (e.g., at second end 371). As shuttle 324 moves distally, first end 366 may unseal such that one or more reagent balls within shuttle 324 and the collected sample within shuttle 324 are exposed to fluid within sample preparation reservoir 317. Advantageously, before first end 366 of shuttle 324 is unsealed, second end 371 of shuttle 324 is fluidically sealed by sample collection device 200 such that fluid from sample preparation reservoir 317 does not leak into input tunnel 301 beyond second end 371. In addition, as distal movement of sample collection reservoir 200 causes the transition from the pre-mixing position to the mixing position, second end 371 of shuttle continuously fluidically seals the fluid in sample preparation reservoir 317 from input tunnel 301.

[0287] As collector pushes sample collection device 200 distally, seal piercer 321 also may be moved from the pre-venting position towards the venting position. Seal piercer 321 may be caused to move within the cartridge by, for example, application of a force greater than a threshold force by sample collection device 200 (e.g., at engagement zone 209) on seal piercer 321 (e.g., at engager 380 of slider 322). As sample collection device 200 is moved distally, seal piercer 321 causes the sealing material over sample preparation reservoir 317, wash reservoir 318, and/or substrate reservoir 319 to be pierced to vent the fluid within the reservoir(s). Seal piercer 321 may be caused to move in a first direction, e.g., generally laterally in a direction generally parallel with movement of sample collection device 200 within input tunnel 301, responsive to insertion of sample collection device 200 in input tunnel 301 and in a second (different) direction, e.g., downwardly in a generally vertical manner toward the respective reservoir, to pierce into sealing material 320. For example, slider 322 may move in the first direction and piercer 323 may move in the second direction.

[0288] As shown in FIG. 10I, as sample collection device **200** causes slider **322** to move distally in a direction generally parallel to movement of sample collection device **200**, slider **322** may contact piercer **323** to cause piercer **323** to move in a different direction, e.g., generally perpendicular to movement of slider **322**. As slider **322** moves distally, tracks **384**, **385**, and **386** of slider **322** may contact piercing elements **381**, **382**, and **383**, respectively, of piercer **323**. Distal movement of slider **322** may cause piercer **323** to pierce the sealing material to vent the reservoir(s).

[0289] FIG. 10J illustrates piercing element **381** piercing sealing material **320** over sample preparation reservoir **317** to vent sample preparation reservoir **317**. As shown, track **384** contacts and moves piercing element **381** downwardly into the piercing position.

[0290] FIG. 10K shows shuttle **324** in the mixing position and the seal piercer in the venting position. In the mixing position, first end **366** of shuttle **324**, the one or more reagent ball compartments of shuttle housing one or more reagent balls, and/or the one or more sample compartments housing the sample(s) to be analyzed may be disposed within sample preparation reservoir **317**. As explained above, shuttle **324** (e.g., at second end **371**) and sample collection device **200** (e.g., via tip **204** inserted in opening **372** and distal sealing zone **208**) also may fluidically seal sample preparation reservoir **317** in the mixing position such that the collected sample(s), the reagent ball(s), and the fluid within sample preparation reservoir **317** are sealed within the reservoir. In this manner, the collected sample(s), the reagent ball(s), and the fluid may be mixed within sample preparation reservoir **317**.

[0291] Advantageously, a configuration where insertion of sample collection device **200** causes sample preparation reservoir **317**, wash reservoir **318**, and/or substrate reservoir **319** to be vented ensures that the reservoir(s) remain fluidically sealed prior to sample collection device **200** insertion and facilitates drainage of the reservoir(s) into the analysis channel when the outlet of the respective reservoir permits fluid flow therethrough.

[0292] In the mixing position, seal piercer **321** may move out of the pierced holes to open the pierced holes and facilitate venting. As shown in FIG. 10K, engagement zone **209** of sample collection device **200** may be moved distally past engager **380** such that engager **380** disengages engagement zone **209** of sample collection device **200** in the venting position. In addition, in the mixing position, proximal sealing zone **207** of sample collection device **200** is configured to seal input tunnel **301** at aperture **302**. In this manner, proximal sealing zone **207** provides additional structure to minimize or eliminate liquid leakage from cartridge device **300**, e.g., at aperture **302**.

[0293] Cartridge device **300** may further include locking member **387** configured to irreversibly lock sample collection device **200** within cartridge device **300**. Locking member **387** may be biased inwardly in input tunnel **301** such that a locking end of locking member **387** engages sample collection device **200** in the mixing position. The locking end may lock to engagement zone **209**. The locking end may be a protrusion sized to fit within a groove of engagement zone **209**, as illustrated. Locking member **387** also may define a portion of input channel **301**, as illustrated, and may be coupled to the cartridge housing at the end opposite its locking end. Advantageously, locking sample collection device **200** (e.g., longitudinally and/or axially) within input tunnel **301** promotes sealing of sample preparation reservoir **317** over time as sample collection device **200** cannot be retracted once locked to facilitate safe disposability and consistency of testing because a user cannot pull out sample collection device **200** from cartridge device **300** inadvertently once the test has started.

[0294] FIGS. 10L and 10M further depict sample collection device **200** in the mixing position within cartridge device **300** (FIG. 10L) and internal component **316** (FIG. 10M) for clarity.

[0295] Referring now to FIGS. 10N, 10O, and 10P, a process for enhanced mixing of the contents within sample preparation reservoir **317** is described. In the mixing position, the fluid within sample preparation reservoir **317** is mixed with the collected sample and one or more reagent balls (if provided). Mixing may be enhanced via sonicator element **327** which may be a piezoelectric transducer such as a piezoceramic disc. Sonicator element **327** is configured to vibrate responsive

to electrical signals (e.g., transmitted from the reader device) to further mix the contents within sample preparation reservoir **317**. For example, sonicator element **327** may facilitate mixing of fluid held in sample preparation reservoir **317**, which may be pre-filled and/or filled during the mixing process, with reagent ball **375** and the collected sample. Sonicator element **327** may cause the fluid to flow in a wave pattern as shown in FIGS. **10O** and **10P**. Such a wave pattern may be between compartments defined by the shuttle, e.g., between reagent ball compartment **367** and sample compartment **370**. For example, sonicator element **327** may cause the fluid to flow in one or more directions (e.g., generally up and down as shown in FIG. **10O**) within reagent ball compartment **367** during portions of the wave cycle. Such flow is expected to speed up, and enhance, mixing within sample preparation reservoir **317**. The wave motion of fluid within sample preparation reservoir **317** also may facilitate removal of sample from the distal portion of sample collection device **200** to enhance mixing and homogenization.

[0296] Sonicator element **327** may be configured to emit acoustic waves to move the fluid in sample preparation reservoir **317** in the wave pattern between reagent ball compartment **367** and sample compartment **370** to mix the fluid in sample preparation reservoir **317**. Such mixing may create a fluid mix of the sample, fluid from the reservoir, and the dissolved reagent ball(s). The acoustic emissions from sonicator element **327** may both heat fluid within sample preparation reservoir **317** and mix the contents of sample preparation reservoir **317** at the macro and the micro level for amplification such as isothermal amplification. The reagent ball(s) may include, for example, polymerases, primers, and signaling agents for isothermal amplification. Shuttle **324** may have compartment divider **368**, which may be a flange, configured to divide reagent ball compartment **367** and sample compartment **370**. The fluid flowing around compartment divider **368** may facilitate formation of the wave pattern. Compartment divider **368** may have a slot configured to permit the fluid to flow through compartment divider **368** via the slot during mixing. Sonicator element **327** may form a wall, e.g., part of the bottom wall, of sample preparation reservoir **317** and may be positioned off-center of sample preparation reservoir **317** to facilitate mixing of the fluid within sample preparation reservoir **317**. For example, sonicator element **327** may be positioned off-center relative to a center axis of sample preparation reservoir **317** running perpendicular to a longitudinal axis running through the center of the input tunnel of the cartridge. Such off-center positioning of sonicator element **327** also facilitates enhanced mixing. Sonicator element **327** may be electrically coupled to a printed circuit board via one or more spring contacts as described in detail below.

[0297] Referring now to FIGS. **11A** through **11E**, an alternative seal piercer is shown. FIGS. **11A** and **11B** show seal piercer **321'** in a pre-venting and pre-contact position. Seal piercer **321'** includes slider **322'** slidably coupled to piercer **323'** throughout the piercing process. Seal piercer **321'** may further include post **388** configured to press contact switch **389** of circuit board **331'**. Depression of contact switch **389** may complete a circuit such that electrical signals may be transmitted, e.g., to the reader device and/or the computing device running the software-based user interface system. In this manner, proper insertion of the tip **204'** of the sample collection device **200'** within the input tunnel generates an electrical signal that may be transmitted to the reader device and/or computing device to notify the reader device and/or computing device of such proper insertion. Post **388** may be coupled to piercer **323'** or may be integral with piercer **323'**. As sample collection device **200'** causes slider **322'** to move distally in a direction generally parallel to movement of sample collection device **200'** (e.g., by force applied by engagement zone **209'** and/or engagement zone **210** to engager **380'**), slider **322'** may cause piercer **323'**/post **388** to move in a different direction, e.g., generally perpendicular to movement of slider **322'**. Slider **322'** may have angled face **390** configured to contact angled face **391** of piercer **323'** as distal movement of sample collection device **200'** causes distal movement of slider **322'**. Continuing distal movement of sample collection device **200'**, and thereby distal movement of slider **322'**, causes piercer **323'** and post **388** to be moved downwardly such that piercer **323'** pierces the sealing material on the respective

reservoir(s) (e.g., the sample preparation reservoir **317'**) and post **388** activates contact switch **389** as shown in FIGS. **11D** and **11E** where the locking member **387'** engages the sample collection device **200'** in the contact position.

[0298] After contact with contact switch **389** and piercing of the sealing material, seal piercer **321'** may move such that post **388** no longer depresses contact switch **389** and piercer **323'** moves out of the hole(s) pierced in the sealing material to vent the respective reservoir(s). Insertion of the sample collection device within the input tunnel may cause the seal piercer to pierce the sealing material before a shuttle begins movement from the pre-mixing position to the mixing position or during movement from the pre-mixing position to the mixing position.

[0299] FIGS. **11A** to **11E** illustrate sonicator element **327'** coupled to spring contact **392**. Spring contact **392** is coupled to circuit board **331'** and is a conductor such that sonicator element **327'** is electrically coupled to circuit board **331'** via spring contact **392**. Beneficially, spring contact **392** absorbs movement of sonicator element **327'** such that circuit board **331'** vibrates minimally in a suitable manner when sonicator element **327'** is activated, e.g., responsive to signals transmitted by the reader device, and spring contact **392** permits ease of assembly and reproducibility compared to soldering which may adversely impact sonicator element **327'**.

[0300] As will be readily understood to one skilled in the art, while FIGS. **11A** and **11D** do not depict a shuttle and/or reagent balls within the input tunnel, such features may be included in these embodiments.

[0301] Referring now to FIGS. **12A** through **12E**, an alternative configuration for collecting and analyzing a fluid sample is described. Cartridge device **300''** may be constructed similarly to cartridge device **300** described above except cartridge device **300''** may be modified for enhanced collection of relatively large amounts of fluid. In addition, sample collection device **200''** may be constructed similarly to sample collection device **200** described above except sample collection device **200''** may have a modified collection area for enhanced collection of relatively large amounts of fluid. For example, sample collection device **200''** and cartridge device **300''** may be particularly useful for collecting and analyzing saliva, blood, plasma, urine, or the like.

[0302] Sample collection device **200''** may include modified distal portion **201''** for enhanced collection of relatively large amounts of fluid (e.g., about 10-100 microliters) that may include distal sealing zone **208''**, wicking portion **211**, intermediate sealing zone **212**, and shroud **213**. Distal portion **201''** of sample collection device **200''** is adapted to be exposed to a sample, preferably a liquid sample, to absorb at least a portion of the sample, and to be compressed to expel the collected sample from distal portion **201''** into cartridge device **300''** for analysis of the expelled sample. Wicking portion **211** is configured to wick and absorb a sample and may be formed from a wicking material. Wicking portion **211** may be coupled to intermediate sealing zone **212**.

Intermediate sealing zone **212** may be slidably disposed within shroud **213** and may include one or more sealing members, e.g., O-rings, configured to create a liquid tight seal that reduces or prevents fluid absorbed on wicking portion **211** from moving proximally within shroud **213** beyond intermediate sealing zone **212**. Wicking portion **211** is at least partially disposed within shroud **213** and may slide within a lumen of shroud **213**. Wicking portion **211** may be configured to become transparent when exposed to fluid such that increasing amounts of wicking portion **211** become transparent as increasing amounts of sample fluid are collected. Sample collection device **200''** may include sample collection indicator **214** configured to visually alert a collector based on a volume of sample fluid that has been collected. In one embodiment, sample collection indicator **214** visually alerts a collector as the volume of sample collected increases and, optionally, that at least a predetermined volume of sample fluid has been collected. For example, sample collection indicator **214** may change color when the predetermined volume of sample, or more, has been collected. As another example, an increasing amount of sample collection indicator **214** may become visible as the volume of sample collected increases. For example, sample collection indicator **214** may be a colored thread embedded in wicking portion **211** that becomes increasingly visually exposed as

increasing volumes of collected fluid sample cause the surrounding wicking material to turn more transparent such that a collector may monitor progress of fluid collection and determine when a sufficient volume of sample has been collected. In one embodiment, sample collection indicator **214** includes a transparent area on shroud **213**. In addition, or alternatively, sample collection indicator **214** may change color as the volume of sample collected increases.

[0303] Shuttle **324"** may include first end **366"**, reagent ball compartment **367"**, compartment divider **368"**, and sample compartment **370"** constructed similarly to those respective components described above. Preferably, compartment divider **368"** does not have a slot, similar to compartment divider **368'**, such that compartment divider **368"** fluidically seals reagent ball compartment **367"** from sample compartment **370"** in the pre-mixing position. Second end **371"** of shuttle **324"** may be modified to include distal flange **393** and proximal flange **394** having cavity **395**. In addition, unlike opening **372** of shuttle **324**, opening **372"** is configured to permit flow of expelled sample compressed from sample collection device **200"** to sample compartment **370"**, but not a portion of sample collection device **200"**. Second end **371"** (e.g., at distal flange **393**) may be configured to fluidically seal sample compartment **370"** in both the pre-mixing position and the mixing position and continuously during the transition therebetween. Distal flange **393** may include one or more sealing members, e.g., O-rings, configured to create a liquid tight seal that reduces or prevents fluid from flowing proximally out of sample compartment **370"**. In addition, compartment divider **368"** may be configured to fluidically seal sample compartment **370"** in the pre-mixing position. Compartment divider **368"** may include one or more sealing members, e.g., O-rings, configured to create a liquid tight seal that reduces or prevents fluid from flowing distally out of sample compartment **370"**.

[0304] Prior to insertion within input tunnel **301"** of cartridge device **300"**, sample collection device **200"** is exposed to a sample, e.g., within an inner cheek, the throat, the mouth, a nasal passageway, an ear, from urine, from blood, from plasma, from saliva, etc. Wicking portion **211** of sample collection device **200"** is designed to retain some of the sample to permit analysis of the presence, absence, and/or quantity of one or more target analytes within the sample using cartridge device **300"** and the reader device. Cartridge device **300"** may be electrically coupled to the reader device before or after sample collection device **200"** is inserted in cartridge device **300"**.

[0305] Referring to FIG. **12A**, the distal end of sample collection device **200"** is first inserted into aperture **302"** defining the opening of input tunnel **301"**. Shuttle **324"** is disposed within input tunnel **301"** in a pre-mixing position wherein first end **366"** of shuttle **324"** forms a wall of sample preparation reservoir **317"** to fluidically seal fluid within sample preparation reservoir **317"**. In this pre-mixing position, reagent ball **375"** housed by shuttle **324"** is within input tunnel **301"** and not exposed to fluid within sample preparation reservoir **317"**.

[0306] Referring to FIG. **12B**, as sample collection device **200"** moves distally in input tunnel **301"**, sample collection device **200"** may contact shuttle **324"**. For example, the distal end and/or distal sealing zone **208"** may contact second end **371"** (e.g., at cavity **395** and/or proximal flange **394**) as illustrated. Cavity **395** may be sized slightly larger than the outer surface of wicking portion **211** such that the distal end of wicking portion **211** fits snugly within cavity **395**. Distal sealing zone **208"** may be configured to fluidically seal sample collection device **200"** to shuttle **324"** such that fluid expelled from sample collection device **200"** travels into sample compartment **370"**. In this pre-mixing position, sample compartment **370"** is within input tunnel **301"** and not exposed to fluid within sample preparation reservoir **317"**.

[0307] As shown in FIG. **12C**, as sample collection device **200"** moves further distally in input tunnel **301"**, sample collection device **200"** may expel collected fluid sample into sample compartment **370"** of shuttle **324"**, e.g., through opening **372"** of second end **371"**. As sample collection device **200"** is moved distally, wicking portion **211** may be compressed to expel collected sample and the distal end of wicking portion **211** may remain substantially in position during compression. Intermediate sealing zone **212** and/or shroud **213** may move distally in

proportion to movement of the handle of sample collection device **200** during such compression. Preferably, as wicking portion **211** is compressed and sample is expelled therefrom, the expelled sample travels through opening **372** into sample compartment **370**. During compression, shuttle **324** may remain substantially in position within input tunnel **301** and first end **366** of shuttle **324** may continue to form a wall of sample preparation reservoir **317** to seal fluid within sample preparation reservoir **317**. Cartridge device **300** may include proximal ledge **396** configured to hold shuttle **324** in the pre-mixing position during compression of wicking portion **211**. Proximal ledge **396** may engage proximal flange **394** to hold shuttle **324** in position.

[0308] A predetermined volume of sample, at most, may be configured to be held in sample compartment **370**. Cartridge device **300** may include overflow compartment **397** and overflow lumen **398**. Overflow compartment **397** and overflow lumen **398** may be part of the cartridge housing or the internal component within the cartridge. If the amount of sample introduced into sample compartment **370** exceeds the predetermined volume, e.g., exceeds about 20 μl , the excess sample may travel to overflow compartment **397** fluidically connected to sample compartment **370**, e.g., via overflow lumen **398**. Limiting the volume of sample within sample compartment **370** may enhance precision, accuracy, and/or consistency of analysis as, at most, a predetermined volume of sample is inserted in sample preparation reservoir **317** in the mixing position. Overflow compartment **397** may be otherwise sealed to prevent or reduce leakage of excess sample within overflow compartment **397**.

[0309] FIG. **12D** shows sample collection device **200** and cartridge device **300** in the mixing position and FIG. **12E** shows a close up view of a portion of FIG. **12D** for clarity. As collector pushes sample collection device **200** distally from the pre-mixing position to the mixing position, shuttle **324** is partially moved within sample preparation reservoir **317**. Application of a force greater than a threshold force on shuttle **324** (e.g., at second end **371** and preferably within cavity **395**) by sample collection device **200** (at distal sealing portion **208** and/or the distal end of wicking material **211**) may cause shuttle **324** to move from the pre-mixing position to the mixing position wherein the expelled sample in sample compartment **370** and reagent ball **375** are mixed within fluid of sample preparation reservoir **317**. For example, the threshold force may be the force required to push shuttle **324** distally past proximal ledge **396**. As shuttle **324** moves distally, first end **366** may unseal such that one or more reagent balls within shuttle **324** and the collected sample within shuttle **324** are exposed to fluid within sample preparation reservoir **317**.

Advantageously, before first end **366** of shuttle **324** is unsealed, second end **371** of shuttle **324** is fluidically sealed by sample collection device **200** such that fluid from sample preparation reservoir **317** does not leak into input tunnel **301** beyond second end **371** and/or intermediate sealing zone **212**. In addition, as distal movement of sample collection device **200** causes the transition from the pre-mixing position to the mixing position, distal flange **393** of shuttle **324** continuously fluidically seals the fluid in sample preparation reservoir **317** from input tunnel **301**. Cartridge device **300** may include distal ledge **399** configured to hold shuttle **324** in the mixing position and to prevent further distal movement. Distal ledge **399** may engage proximal flange **394** to hold shuttle **324** in position.

[0310] In the mixing position, first end **366** of shuttle **324**, the one or more reagent ball compartments of shuttle housing one or more reagent balls, and/or the one or more sample compartments housing the sample(s) to be analyzed may be disposed within sample preparation reservoir **317**. As explained above, shuttle **324** (e.g., at second end **371**) and sample collection device **200** (e.g., via distal sealing zone **208** inserted in cavity **395**) also may fluidically seal sample preparation reservoir **317** in the mixing position such that the collected sample(s), the reagent ball(s), and the fluid within sample preparation reservoir **317** are sealed within the reservoir. In this manner, the collected sample(s), the reagent ball(s), and the fluid may be mixed within sample preparation reservoir **317**.

[0311] Cartridge device **300** may further include locking member **387** configured to irreversibly

lock sample collection device **200** within cartridge device **300**. Locking member **387** may be biased inwardly in input tunnel **301** such that a locking end of locking member **387** engages sample collection device **200** in the mixing position. The locking end may lock to an engagement zone of sample collection device **200**. The locking end may be a protrusion sized to fit within a groove on the shaft of sample collection device **200** or within a groove on shroud **213**, as illustrated. Locking member **387** also may define a portion of input channel **301**, as illustrated, and may be coupled to the cartridge housing at the end opposite its locking end. Advantageously, locking sample collection device **200** (e.g., longitudinally and/or axially) within input tunnel **301** promotes sealing of sample preparation reservoir **317** over time as sample collection device **200** cannot be retracted once locked to facilitate safe disposability and consistency of testing because a user cannot pull out sample collection device **200** from cartridge device **300** inadvertently once the test has started.

[0312] As will be readily understood to one skilled in the art, while FIGS. **12A** through **12E** do not depict a seal piercer in the cartridge device, a seal piercer may be included in these embodiments. For example, sample collection device **200** (e.g., at distal sealing zone **208**) may contact the seal piercer to move the seal piercer from the pre-venting position to the venting position in a manner described above with respect to FIGS. **10A** to **10J** and/or FIGS. **11A** to **11E**. Sample collection device **200** may contact and move the seal piercer from the pre-venting to the venting position before, during, and/or after contacting shuttle **324** and moving shuttle **324** from the pre-mixing to the mixing position. In addition, insertion of sample collection device **200** may cause activation of a contact switch as described above with respect to FIGS. **11A** to **11E**. For example, sample collection device **200** may cause movement of a seal piercer which in turn causes activation of the contact switch.

[0313] Referring now to FIG. **13A**, an alternative cartridge for analyzing a sample is described. Cartridge device **300** may be constructed similarly to cartridge device **300** and/or cartridge device **300** described above wherein like components are identified by like-primed reference numbers. Cartridge device **300** is a universal configuration that includes components that may be used for different types of samples. For example, many components within cartridge device **300** may be used for various types of samples without modification and, in some embodiments, only the shuttle, collet, and reagent ball(s) may be varied for analysis of different indications. In this manner, cartridge device **300** may be universal and the shuttle, collet, and/or reagent ball(s) may be selected for use in cartridge device **300** based on the target analyte(s) to be analyzed. For example, cartridge device **300** may be fitted with a shuttle designed for relatively small sample collection, e.g., a nasal passageway, an ear, blood, such as shuttle **324** or **324'** described above with respect to FIGS. **9A** and **9B**, and a collet also designed for relatively small sample collection or cartridge device **300** may be fitted with a shuttle designed for relatively large fluid sample collection, e.g., saliva, blood, plasma, urine, such as shuttle **324** described above with respect to FIGS. **12A** through **12E** and a collet also designed for relatively large fluid sample collection. Cartridge device **300** may be further fitted with a reagent ball(s) intended to identify different target analytes that may be indicative of, for example, inflammation, influenza, testosterone, fertility, HIV, or Vitamin D. Accordingly, cartridge device **300** is highly interchangeably for different types of samples and indications which reduces costs and manufacturing burdens.

[0314] In FIG. **13A**, an exploded view of cartridge device **300** is shown. Cartridge device **300** may include internal component **316**—which may include sample preparation reservoir **317**, wash reservoir **318**, substrate reservoir **319**, input tunnel component **326**, overflow compartment **397**, and/or posts **610** and **612**—sealing material **320**, seal piercer **321**, shuttle **324** having first end **366** and sealing members **614**, desiccant **325**, sonicator element **327**, absorbent pad **328**, layer **329**, analysis channel **330**, circuit board **331**, memory **332**, sensor **338**, heating elements, contact switch **389**, spring contacts **392**, reagent ball **375**, temperature sensor **616**, and/or collet **618**. The internal components may be disposed within housing **304**, e.g.,

between first and second cover components **310'''** and **311'''**. Alternatively, one or more internal components may be disposed within one housing while other internal components may be disposed within another housing(s). In the case of multiple housings, such separate housing may be configured to couple to one another.

[0315] In FIG. **13A**, shuttle **324'''** is illustrated as being substantially similar to shuttle **324''** of FIGS. **12A** through **12E** although a shuttle like that of shuttle **324** in FIG. **9A** or shuttle **324'** in FIG. **9B** may be used depending on the type of sample to be collected. Collet **618** also may be substituted for collet **618'** described below based on the type of sample to be collected. Various reagent ball(s) **375'** may be substituted in cartridge device **300'''''** as well.

[0316] Posts **610** and **612** are configured to be coupled to seal piercer **321'''** and to permit seal piercer **321'''** to be moved from the pre-venting position to the venting position. Posts **610** and **612** may be integrally formed with internal component **316'''** or may be separate pieces.

[0317] Temperature sensor **616** may be configured to sense temperature indicative of temperature of fluid in a reservoir, e.g., sample preparation reservoir **317'''**. For example, temperature sensor **616** may sense temperature changes adjacent the reservoir that are indicative of temperature in the reservoir. Temperature sensor **616** may be a thermistor and may be disposed on circuit board **331'''** to permit electrical coupling with the reader device via one or more leads in circuit board **331'''**. Preferably, temperature sensor **616** is positioned adjacent sonicator element **327'''** on circuit board **331'''** such that temperature sensor **616** senses temperature during mixing within sample preparation reservoir **317'''** via sonicator element **327'''**. Advantageously, temperature indicative of temperature in sample preparation reservoir **317'''** may be monitored during mixing of the fluid in the reservoir with reagents from reagent ball(s) and the sample, to ensure that temperature within sample preparation reservoir **317'''** is within a predetermined range. If outside the predetermined range, emission of acoustic waves into sample preparation reservoir **317'''** via sonicator element **327'''** may be modified, e.g., responsive to electrical signals sent by the reader to the sonicator element **327'''**. Temperature sensor **616** may generate a signal indicative of temperature of the fluid in the reservoir which may be transmitted to the reader for processing via leads in circuit board **331'''**. Temperature sensor **616** may be positioned beneath sonicator element **327'''** on circuit board **331'''** and adjacent to one or more spring contacts, e.g., between first and second spring contacts **392'''**.

[0318] Collet **618** is preferably disposed in input tunnel **301'''** between aperture **302'''** and sample preparation reservoir **317'''**. Collet **618** also may be disposed at least partially proximal to shuttle **324'''** in input tunnel **301'''**. For example, in the pre-mixing position, an end, e.g., the second end, of the shuttle may be disposed within collet **618**. Collet **618** also may be configured to hold shuttle **324'''** in the pre-mixing position and to decouple from shuttle **324'''** during insertion of the sample collection device in input tunnel **301'''**. In this manner, collet **618** may hold shuttle **324'''** in the pre-mixing state until force applied from the sample collection device decouples collet **618** from shuttle **324'''** such that shuttle **324'''** moves from the pre-mixing position to the mixing position while collet **618** remains in place within input tunnel **301'''**. Collet **618** has a lumen therethrough sized to permit insertion of the distal portion of the sample collection device. Collet **618** may have a generally tubular shape as illustrated in FIG. **13A**. Collet **618** also may be configured to activate contact switch **389'''**. For example, collet **618** may activate contact switch **389'''** responsive to a force applied on collet **618** by the sample collection device during insertion of the sample collection device in input tunnel **301'''**.

[0319] Referring now to FIGS. **13B** through **13SS**, various exemplary sample collection devices are illustrated that may be used in detection system **100**.

[0320] Referring to FIG. **13B**, sample collection device **200'''** may be constructed similarly to sample collection device **200''** described above except sample collection device **200'''** may have a modified shaft for locking sample collection device **200'''** within a cartridge during partial and full insertion of sample collection device **200'''**. For example, sample collection device **200'''**

illustratively includes engagement zone **209'''** having a plurality of grooves and protrusions distal to, and spaced apart from, proximal sealing zone **207'''**. The plurality of grooves and protrusions are configured for engagement with one or more components of the cartridge device for irretractability of sample collection device **200'''** during partial and full insertion of sample collection device **200'''** within the cartridge. For example, the cartridge may sequentially engage grooves of the multiplicity of grooves in engagement zone **209'''** in a distal to proximal direction as sample collection device **200'''** is moved distally in the input tunnel. Engagement zone **209'''** may be configured to be coupled, permanently or temporarily, to the seal piercer of the cartridge device to move the seal piercer within the cartridge device responsive to movement of sample collection device **200'''**. In addition or alternatively, engagement zone **209'''** may be configured to activate a contact switch during sample collection device insertion. For example, shoulder **220** at the distal end of engagement zone **209'''** may be configured to be coupled to the seal piercer and/or to activate the contact switch.

[0321] Like sample collection device **200''**, sample collection device **200'''** may include modified distal portion **201'''** for enhanced collection of relatively large amounts of fluid (e.g., about 10-100 microliters, preferably about 20 microliters) that may include distal sealing zone **208'''**, wicking portion **211'''**, intermediate sealing zone, and/or shroud **213'''**. Sample collection device **200'''** also may include proximal portion **202'''**, shaft **203'''** extending between distal portion **201'''** and proximal portion **202'''**, handle **206'''**, proximal sealing zone **207'''**, and/or engagement zone **209'''** having shoulder **220** similar to the like primed references described above. Sample collection device **200'''** is configured for full or partial insertion within cartridge device **300'''** after sample collection. Sample collection device **200'''** and cartridge device **300'''** may be particularly useful for collecting and analyzing saliva, blood, plasma, urine, or the like. Shoulder **220** of engagement zone **209'''** preferably extends further from the longitudinal axis of sample collection device **200'''** than a shoulder of distal sealing zone **208'''** such that shoulder **220** contacts and moves the seal piercer to vent one or more reservoirs within cartridge device **300'''** and/or contacts the collet to activate the contact switch while the shoulder of distal sealing zone **208'''** may be sized to move distally through the input tunnel without moving the seal piercer and/or without activating the contact switch.

[0322] FIGS. **13C**, **13D**, **13E**, **13F**, **13G**, and **13H** are back, side, front, back, side, and front views, respectively, of sample collection device **200'''**.

[0323] Referring to FIG. **13I**, sample collection device **200''''** may be constructed similarly to sample collection device **200**, wherein like components are identified by like-primed reference numbers, described above except sample collection device **200''''** may have engagement zone **209''''** similar to engagement zone **209'''** of FIG. **13B** and tip **204''''** does not include a tube. Tip **204''''** may have a rounded end as illustrated and may be configured to collect a sample from any desired region or location, although tip **204''''** may be particularly useful when collecting a sample from a nasal area. FIGS. **13J**, **13K**, **13L**, **13M**, **13N**, and **13O** are back, side, front, back, side, and front views, respectively, of sample collection device **200''''**.

[0324] Referring to FIG. **13P**, sample collection device **200'''''** may be constructed similarly to sample collection device **200''''** shown in FIG. **13I**, wherein like components are identified by like-primed reference numbers, except tip **204'''''** of sample collection device **200'''''** includes tube **205'''''** (like the tube shown in FIGS. **2A** and **2B**). Distal portion **201'''''**, including tip **204'''''**, is configured to be exposed to a sample such that, at most, a predetermined volume of the sample (e.g., less than 10 microliters, preferably about 2 microliters) is disposed in tube **205'''''** for analysis. Collection of a predetermined volume of the sample is expected to promote accuracy of analyte analysis as a substantially known quantity of the sample will be analyzed. Tip **204'''''** may have a rounded end as illustrated and may be configured to collect a sample from any desired region or location, although tip **204'''''** may be particularly useful when collecting a sample of blood. FIGS. **13Q**, **13R**, **13S**, **13T**, **13U**, **13V**, and **13W** are side, back, side, front, back, side, and front views,

respectively, of sample collection device **200''''''**.

[0325] Referring to FIG. **13X**, sample collection device **200''''''** may be constructed similarly to sample collection device **200''''''** shown in FIG. **13P**, wherein like components are identified by like-primed reference numbers, except tip **204''''''** of sample collection device **200''''''** includes slot **222** rather than a tube. Distal portion **201''''''**, including tip **204''''''**, is configured to be exposed to a sample such that, at most, a predetermined volume of the sample (e.g., less than 10 microliters, preferably about 5 microliters) is disposed in slot **222** for analysis. Collection of a predetermined volume of the sample is expected to promote accuracy of analyte analysis as a substantially known quantity of the sample will be analyzed. Tip **204''''''** may have a rounded end as illustrated and may be configured to collect a sample from any desired region or location, although tip **204''''''** may be particularly useful when collecting a sample of blood. FIGS. **13Y**, **13Z**, **13AA**, **13BB**, **13CC**, **13DD**, and **13EE** are side, back, side, front, back, side, and front views, respectively, of sample collection device **200''''''**.

[0326] Referring to FIG. **13FF**, sample collection device **200''''''''** may be constructed similarly to sample collection device **200''''''** shown in FIG. **13P**, wherein like components are identified by like-primed reference numbers, except tip **204''''''''** of sample collection device **200''''''''** includes ring **224** rather than a tube. Distal portion **201''''''''**, including tip **204''''''''**, is configured to be exposed to a sample such that, at most, a predetermined volume of the sample (e.g., less than 10 microliters, preferably about 2 microliters) is disposed in a groove formed by ring **224** for analysis. Collection of a predetermined volume of the sample is expected to promote accuracy of analyte analysis as a substantially known quantity of the sample will be analyzed. Tip **204''''''''** may have a rounded end as illustrated and may be configured to collect a sample from any desired region or location, although tip **204''''''''** may be particularly useful when collecting a sample of blood. FIGS. **13GG**, **13HH**, **13II**, **13JJ**, **13KK**, and **13LL** are back, side, front, back, side, and front views, respectively, of sample collection device **200''''''''**.

[0327] Referring to FIG. **13MM**, sample collection device **200''''''''''** may be constructed similarly to sample collection device **200''''''** shown in FIG. **13P**, wherein like components are identified by like-primed reference numbers, except tip **204''''''''''** of sample collection device **200''''''''''** includes first ring **226** and second ring **228** rather than a tube. Distal portion **201''''''''''**, including tip **204''''''''''**, is configured to be exposed to a sample such that, at most, a predetermined volume of the sample (e.g., less than 10 microliters, preferably about 5 microliters) is disposed in a groove formed by first ring **226** and a groove formed by second ring **228** for analysis. Collection of a predetermined volume of the sample is expected to promote accuracy of analyte analysis as a substantially known quantity of the sample will be analyzed. Tip **204''''''''''** may have a rounded end as illustrated and may be configured to collect a sample from any desired region or location, although tip **204''''''''''** may be particularly useful when collecting a sample of blood. FIGS. **13NN**, **1300**, **13PP**, **13QQ**, **13RR**, and **13SS** are back, side, front, back, side, and front views, respectively, of sample collection device **200''''''''''**.

[0328] Referring now to FIGS. **14A** and **14B**, an exemplary collet for use in a cartridge is described. Collet **618** may be specifically designed for relatively large fluid sample collection, e.g., saliva, blood, plasma, urine, such as when the sample is compressed from the distal portion of the sample collection device as described above with respect to FIGS. **12A** through **12E**, FIG. **13B**, and below. Collet **618** may include proximal end **620**, distal end **622**, and lumen **624** extending between ends **620** and **622**. Lumen **624** may be sized to permit insertion of the distal portion of a sample collection device into lumen **624**. Collet **618** is positioned in the input tunnel such that the distal portion of the sample collection device first enters lumen **624** at proximal end **620**. Collet **618** also may include slot **626**, e.g., at upper surface of collet **618**. Slot **626** is sized to receive a portion of the seal piercer therethrough. For example, the engager of the seal piercer may extend through the slot into the input tunnel to permit contact between the seal piercer and the sample collection device.

[0329] Lumen **624** of collet **618** also may be sized to receive a portion of the shuttle therein. For example, the proximal end of the shuttle may be disposed in lumen **624** through distal end **622** of collet **618**. Collet **618** also may be configured to hold the shuttle in the input tunnel in the pre-mixing position. For example, collet **618** may include one or more locking arms configured to couple collet **618** to the shuttle in the pre-mixing position. Illustratively, collet **618** includes first locking arm **628** and second locking arm **630** at opposing lateral sides of collet **618**.

Advantageously, when using a sample collection device having a compressible distal portion for collecting a fluid sample, collet **618** may hold the shuttle in place in the input tunnel during compression of the distal portion to expel sample fluid into the shuttle. Each locking arm may include a ramp and a protrusion, shown as ramp **632** and protrusion **634** for locking arm **630** in FIG. **14B**. Protrusion **634** may be coupled to the shuttle to hold the shuttle in place in the pre-mixing position. For example, the protrusions may hold the proximal flange of the shuttle during compression of the distal portion of the sample collection device. First and second locking arms **628** and **630** also may decouple from the shuttle during insertion of the sample collection device in the input tunnel **301**. First and second locking arms **628** and **630** may be deflected to decouple first and second locking arms **628** and **630** from the shuttle responsive to a force applied on first and second locking arms **628** and **630** by the sample collection device during insertion of the sample collection device in the input tunnel. For example, the shoulder of the sample collection device may contact the ramp(s) of the locking arm(s) and cause the protrusion(s) to deflect outwardly as the sample collection device moves distally along the ramp(s) and in the input tunnel. The ramp(s) are shaped such that the protrusion(s) decouple from the proximal flange of the shuttle to unlock the shuttle and permit the shuttle to move from the pre-mixing position to the mixing position where the shuttle is partially disposed within the sample preparation reservoir.

[0330] Collet **618** may include deflector portion **636** configured to deflect to activate a contact switch within the cartridge. Preferably, deflector portion **636** is disposed on a bottom face of collet **618** and positioned above the contact switch in the input tunnel. Deflector portion **636** may deflect to activate the contact switch responsive to a force applied on deflector portion **636** by the sample collection device during insertion of the sample collection device in the input tunnel. For example, the shoulder of the sample collection device may contact deflector portion **636** and force deflector portion **636** downwardly as the sample collection device moves distally in the input tunnel to activate the contact switch. Illustratively, deflector portion **636** is an arm configured to deflect downwardly.

[0331] Referring now to FIGS. **14C** and **14D**, an alternative collet for use in a cartridge is described. Collet **618'** may be specifically designed for relatively small sample collection, e.g., a nasal passageway, an ear, blood, such as when the sample need not be compressed from the distal portion of the sample collection device. As will be appreciated when comparing FIGS. **14C** and **14D** to FIGS. **14A** and **14B**, like components are identified by like-primed reference numbers, where collet **618'** is similar to collet **618** except collet **618'** does not have locking arms. Collet **618'** has one or more protrusions **637** disposed at distal end **622'** and configured to contact the proximal end of the shuttle in the pre-mixing position. For example, the one or more protrusions **637** may contact a sealing member, e.g., O-ring, at second end **371**, **371'** of shuttle **324**, **324'** to retain the sealing member in position. The one or more protrusions may have lead-in angles configured to guide the distal portion of the sample collection device into the opening at the second end of the shuttle.

[0332] Referring now to FIGS. **15A** through **15D**, cartridge **300''''** is shown in various positions with the upper surface of the housing removed for clarity. FIGS. **15A** and **15B** are cross-sectional views through the center of the input tunnel for additional clarity. In FIG. **15A**, cartridge **300''''** is shown in the pre-mixing, pre-venting, storage position where a sample collection device has not yet been inserted in input tunnel **301'''**. As shown, seal piercer **321'''** is in the pre-venting position and has not yet pierced seal material **320'''** over the reservoirs, the proximal end of shuttle **324'''** is

disposed within the distal end of collet **618** while the distal end of shuttle **324'''** forms a wall of sample preparation reservoir **317'''**, and deflector portion **636** of collet **618** is in the pre-deflection position where deflector portion **636** does not activate contact switch **389'''**. In FIG. **15B**, cartridge **300'''** is shown in the mixing, venting, analysis position where the sample collection device is fully inserted in input tunnel **301'''**. As shown, seal piercer **321'''** is in the venting position and has pierced seal material **320'''** over each of the reservoirs, shuttle **324'''** has moved distally from collet **618** such that the sample and reagent ball **375'''** are mixing with the fluid in sample preparation reservoir **317'''**, deflector portion **636** of collet **618** is in the deflection position where deflector portion **636** has activated contact switch **389'''**, and locking members **387'''** have locked the sample collection device in input tunnel **301'''**. In FIG. **15C**, cartridge **300'''** is still in the pre-mixing, preventing position as the sample collection device has only been partially inserted in input tunnel **301'''**. FIG. **15D** shows cartridge **300'''** in the venting position.

[0333] Referring back to FIG. **15A**, an exemplary process is described for piercing the sealing material disposed over one or more reservoirs in the cartridge device via interaction between the sample collection device and the seal piercer within the cartridge device.

[0334] Seal piercer **321'''** is configured to pierce sealing material **320'''** to vent the fluid in sample preparation reservoir **317'''**, wash reservoir **318'''**, and/or substrate reservoir **319'''**. Preferably, seal piercer **321'''** is configured to pierce sealing material **320'''** over the reservoirs sequentially to reduce resistance on the sample collection device during piercing. Seal piercer **321'''** may be configured to be contacted by the distal portion, e.g., at a shoulder, of a sample collection device within input tunnel **301'''** and to move within housing **304'''**, responsive to force applied by the sample collection device, to cause sealing material **320'''** to be pierced to vent the fluid in sample preparation reservoir **317'''**, wash reservoir **318'''**, and/or substrate reservoir **319'**. Illustratively, seal piercer **321'''** is a single piece. Seal piercer **321'''** is disposed within housing **304'''** and may be partially disposed within input tunnel **301'''**. For example, engager **380'''** of seal piercer **321'''** may be disposed within input tunnel **301'''**, e.g., through slot **626** of collet **618**. Seal piercer **321'''** has one or more piercing elements with ends sufficiently sharp to cut open sealing material **320'''**. Illustratively, seal piercer **321'''** has first piercing element **381'''** having a piercing end disposed adjacent sample preparation reservoir **317'''**, second piercing element **382'''** having a piercing end disposed adjacent wash reservoir **318'''**, and third piercing element **383'''** having a piercing end disposed adjacent substrate reservoir **319'''**.

[0335] Seal piercer **321'''** also may include slots **638** and **640** configured to receive a portion of posts **610** and **612** respectively. Accordingly, seal piercer **321'''** may move within housing **304'''** while the portions of posts **610** and **612** remain in slots **638** and **640**. Cartridge **300'''** also may include one or more ramps configured to deflect the one or more piercing elements toward sealing material **320'''** to pierce sealing material **320'''**. The one or more ramps may be directly coupled to housing **304'''** of cartridge **300'''**. The peak(s) of the one or more ramps may be positioned such that the one or more piercing elements travel past the peak(s) when the sample collection device is fully inserted in input tunnel **301'''** to facilitate venting of the reservoirs. Illustratively, cartridge **300'''** has first ramp **642** disposed adjacent sample preparation reservoir **317'''**, second ramp **644** disposed adjacent wash reservoir **318'''**, and third ramp **646** disposed adjacent substrate reservoir **319'''**. The distance between each ramp and each piercing element in the pre-venting position may be different such that the reservoirs are pierced sequentially. Each ramp may have a break in the middle to fit the portion of seal piercer **321'''** adjacent the piercing elements in the venting and mixing positions, as shown in FIG. **15B**.

[0336] As sample collection device **200'''** is moved distally through input tunnel **301'''**, preferably seal piercer **321'''** does not move within the cartridge housing, and remains in a pre-vent position, until sample collection device **200'''** securely engages seal piercer **321'''**. Seal piercer **321'''** may securely engage sample collection device **200'''** by temporarily or permanently coupling engager **380'''** of seal piercer **321'''** to sample collection device **200'''**, e.g., at shoulder **220**, once sample

collection device **200'''** is sufficiently inserted in input tunnel **301'''**.

[0337] As collector pushes sample collection device **200'''** distally, seal piercer **321'''** is moved from the pre-venting position towards the venting position. Seal piercer **321'''** may move within the cartridge by, for example, application of a force greater than a threshold force by sample collection device **200'''** (e.g., at shoulder **220** which may be at the distal end of shroud **213'''** and/or at the distal end of engagement zone **209'''**) on seal piercer **321'''** (e.g., at engager **380'''**). As sample collection device **200'''** is moved distally, the piercing element the shortest distance from its ramp first pierces the sealing material above that reservoir. For example, as sample collection device **200'''** is moved distally, seal piercer **321'''** moves generally parallel to movement of sample collection device **200'''** until second piercing element **382'''** contacts second ramp **644** and second ramp **644** deflects second piercing element **382'''** downwardly to pierce sealing material **320'''** over wash reservoir **318'''**. As seal piercer **321'''** continues to move distally second piercing element **382'''** moves past the peak of second ramp **644** and moves upward out of the hole pierced above wash reservoir **318'''** and third piercing element **383'''** contacts third ramp **646** and third ramp **646** deflects third piercing element **383'''** downwardly to pierce sealing material **320'''** over substrate reservoir **319'''**. As seal piercer **321'''** continues to move distally third piercing element **383'''** moves past the peak of third ramp **646** and moves upward out of the hole pierced above substrate reservoir **319'''** and first piercing element **381'''** contacts first ramp **642** and first ramp **642** deflects first piercing element **381'''** downwardly to pierce sealing material **320'''** over sample preparation reservoir **317'''**. As seal piercer **321'''** continues to move distally first piercing element **381'''** moves past the peak of first ramp **642** and moves upward out of the hole pierced above sample preparation reservoir **317'''**. As will be understood to one skilled in the art, the order of piercing may be varied for sequential piercing.

[0338] Advantageously, a configuration where insertion of sample collection device **200'''** causes sample preparation reservoir **317'''**, wash reservoir **318'''**, and/or substrate reservoir **319'''** to be vented sequentially ensures that the reservoir(s) remain fluidically sealed prior to sample collection device **200** insertion, eases resistive forces during insertion of the sample collection device, and facilitates drainage of the reservoir(s) into the analysis channel when the outlet of the respective reservoir permits fluid flow therethrough.

[0339] Referring now to FIGS. **16A** through **16J**, a configuration for collecting and analyzing a sample using cartridge **300'''** is described. Illustratively, sample collection device **200'''** is inserted in cartridge **300'''**, although it should be understood that any sample collection device described herein may be used as cartridge **300'''** is designed for universal application for different types of sample and different types of indications, although certain internal components may be substituted such as the shuttle, reagent ball(s), and/or collet based on the application. Sample collection device **200'''** is described above with respect to FIG. **13B**. Sample collection device **200'''** illustratively includes engagement zone **209'''** having a plurality of grooves and protrusions distal to, and spaced apart from, the sealing zone at the proximal portion of sample collection device **200'''**. The plurality of grooves and protrusions are configured for engagement with one or more components of cartridge device **300'''** for irretractability of sample collection device **200'''** during partial and full insertion of sample collection device **200'''** within cartridge **300'''**. Engagement zone **209'''** may facilitate fixed engagement between sample collection device **200'''** and the cartridge device, e.g., via locking members **387'''**, such that sample collection device **200'''** is mated irreversibly with the cartridge when sample collection device **200'''** is partially inserted a predetermined distance in the input tunnel of the cartridge, e.g., when the distal-most groove of engagement zone **209'''** engages locking members **387'''**. Sample collection device **200'''** continues to be mated irreversibly with the cartridge as insertion continues toward the full insertion position. For example, locking members **387'''** may sequentially engage grooves of the multiplicity of grooves in engagement zone **209'''** in a distal to proximal direction as sample collection device **200'''** is moved distally in input tunnel **301'''**. Locking members **387'''** may engage the proximal-most groove of the multiplicity of grooves

in engagement zone **209'''** when sample collection device **200'''** is fully inserted in input tunnel **301'''**. Preferably, sample collection device **200'''** cannot be retracted after both partial and full insertion, thereby reducing the risk of contamination. Engagement zone **209'''** may be configured to be coupled, permanently or temporarily, to the seal piercer of the cartridge device to move the seal piercer within the cartridge device responsive to movement of sample collection device **200'''**. In addition or alternatively, engagement zone **209'''** may be configured to activate a contact switch during sample collection device insertion. For example, shoulder **220** at the distal end of engagement zone **209'''** may be configured to be coupled to the seal piercer and/or to activate the contact switch.

[0340] Prior to insertion within input tunnel **301'''** of cartridge device **300'''**, sample collection device **200'''** is exposed to a sample, e.g., within an inner cheek, the throat, the mouth, a nasal passageway, an ear, from urine, from blood, from plasma, from saliva, etc. Wicking portion **211'''** of sample collection device **200'''** is designed to retain some of the sample to permit analysis of the presence, absence, and/or quantity of one or more target analytes within the sample using cartridge device **300'''** and the reader device. Cartridge device **300'''** may be electrically coupled to the reader device before or after sample collection device **200'''** is inserted in cartridge device **300'''**.

[0341] Referring to FIG. **16A**, the distal end of sample collection device **200'''** is first inserted into aperture **302'''** defining the opening of input tunnel **301'''**. As shown, seal piercer **321'''** is in the preventing position and has not yet pierced seal material **320'''** over the reservoirs, the proximal end of shuttle **324'''** is disposed within the distal end of collet **618** while the distal end of shuttle **324'''** forms a wall of sample preparation reservoir **317'''**, and deflector portion **636** of collet **618** is in the pre-deflection position where deflector portion **636** does not activate contact switch **389'''**. In this pre-mixing position, reagent ball **375'''** housed by shuttle **324'''** is within input tunnel **301'''** and not exposed to fluid within sample preparation reservoir **317'''**.

[0342] Referring to FIG. **16B**, as sample collection device **200'''** moves distally in input tunnel **301'''**, the distal end of sample collection device **200'''** travels through lumen **624** of collet **618** and contacts shuttle **324'''**. For example, the distal end and/or distal sealing zone **208'''** may contact second end **371'''** (e.g., at cavity **395'''** and/or proximal flange **394'''**) as illustrated. As explained above with respect to FIGS. **12A** through **12E**, cavity **395'''** may be sized slightly larger than the outer surface of wicking portion **211'''** such that the distal end of wicking portion **211'''** fits snugly within cavity **395'''** and distal sealing zone **208'''** may be configured to fluidically seal sample collection device **200'''** to shuttle **324'''** such that fluid expelled from sample collection device **200'''** travels into sample compartment **370'''**. In this pre-mixing position, sample compartment **370'''** is within input tunnel **301'''** and not exposed to fluid within sample preparation reservoir **317'''**.

[0343] As shown in FIG. **16C**, as sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''** may deflect one or more locking members **387'''** of cartridge device **300'''**, e.g., via shoulder **220**, and sample collection device **200'''** may expel collected fluid sample into sample compartment **370'''** of shuttle **324'''**, e.g., through opening **372'''** of second end **371'''**. As is explained above, as sample collection device **200'''** is moved distally, wicking portion **211'''** may be compressed to expel collected sample and the distal end of wicking portion **211'''** may remain substantially in position during compression. Intermediate sealing zone **212'''** and/or shroud **213'''** may move distally in proportion to movement of the handle of sample collection device **200'''** during such compression. Preferably, as wicking portion **211'''** is compressed and sample is expelled therefrom, the expelled sample travels through opening **372'''** into sample compartment **370'''**. During compression, shuttle **324'''** may remain substantially in position within input tunnel **301'''** and first end **366'''** of shuttle **324'''** may continue to form a wall of sample preparation reservoir **317'''** to seal fluid within sample preparation reservoir **317'''**. Collet **618** may hold shuttle **324'''** in the pre-mixing position during compression of wicking portion **211'''**. For example, first and second locking arms of collet **618** may hold shuttle **324'''** in the pre-mixing position.

[0344] Cartridge device **300'''** may include overflow compartment **397'''** and overflow lumen **398'''** configured to receive excess sample if the amount of sample introduced into sample compartment **370'''** exceeds a predetermined volume, e.g., exceeds about 20 μ l, as explained above. Overflow compartment **397'''** is illustratively formed as part of the internal component and is sealed on the top surface by the cartridge housing.

[0345] As shown in FIG. **16D**, as sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''** may deflect one or more locking members **387'''** of cartridge device **300'''** past shoulder **220** such that one or more locking members **387'''** lock to engagement zone **209'''** during partial sample collection device **200'''** insertion. Illustratively, locking members **387'''** engage the distal-most groove of engagement zone **209'''** to accomplish the first lock during insertion. As the distal end of shroud **213'''**, e.g., at shoulder **220**, passes desiccant **325'''**, a liquid safety barrier is formed in input tunnel **301'''** by shroud **213'''** and desiccant **325'''**. During wicking portion **211'''** compression shown in FIG. **16D**, like FIG. **16C**, sealing zone **212'''** and/or shroud **213'''** may move distally to further expel sample into sample compartment **370'''** while shuttle **324'''** may remain substantially in position within input tunnel **301'''**.

[0346] Referring to FIG. **16E**, as sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''** continues to deflect one or more locking members **387'''** of cartridge device **300'''** past protrusions and into grooves of engagement zone **209'''** in a distal to proximal direction such that one or more locking members **387'''** continue to lock to engagement zone **209'''** throughout partial sample collection device **200'''** insertion. During wicking portion **211'''** compression shown in FIG. **16E**, like FIGS. **16C** and **16D**, sealing zone **212'''** and/or shroud **213'''** may move distally to further expel sample into sample compartment **370'''** while shuttle **324'''** may remain substantially in position within input tunnel **301'''**. Once sample collection device **200'''** causes collet **618** to decouple from shuttle **324'''**, shuttle **324'''** may move within input tunnel **301'''** from the pre-mixing position to the mixing position. For example, sample collection device **200'''** may apply a force on collet **618** during insertion of sample collection device **200'''** in input tunnel **301'''** to decouple collet **618** from shuttle **324'''**. In FIG. **16E**, engagement zone **209'''**, e.g., at shoulder **220**, contacts collet **618**, e.g., at first and second locking arms.

[0347] FIG. **16F** is a top view of FIG. **16E** showing partial insertion of sample collection device **200'''** in cartridge **300'''**. Collet **618** is disposed in input tunnel **301'''** and configured to be coupled to shuttle **324'''** in the pre-mixing position. Collet **618** also may be configured to be coupled to shuttle **324'''** during compression of sample collection device **200'''** to allow the distal portion of sample collection device **200'''** to compress while shuttle **324'''** remains in place. Illustratively, second end **371'''** of shuttle **324'''** is disposed in the lumen of collet **618**. In this example, collet **618** includes first locking arm **628** and second locking arm **630** at opposing lateral sides of collet **618**. The protrusion of first locking arm **628** and protrusion **634** of second locking arm **630** hold proximal flange **394'''** of shuttle **324'''** during compression of the distal portion of sample collection device **200'''**. First and second locking arms **628** and **630** also may decouple from shuttle **324'''** during insertion of sample collection device **200'''** in input tunnel **301'''**. First and second locking arms **628** and **630** may be deflected to decouple first and second locking arms **628** and **630** from shuttle **324'''** responsive to a force applied on first and second locking arms **628** and **630** by sample collection device **200'''** during insertion of sample collection device **200'''** in input tunnel **301'''**. In FIG. **16F**, shoulder **220** of sample collection device **200'''** contacts the ramp of first locking arm **628** and ramp **632** of second locking arm **630** during partial sample collection device **200'''** insertion.

[0348] Referring to FIG. **16G**, as sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''**, e.g., at shoulder **220**, applies a force on the ramps of first and second locking arms **628** and **630** to cause the protrusions of first locking arm **628** and protrusion **634** of second locking arm **630** to deflect outwardly. The ramps are shaped such that the protrusions decouple from proximal flange **394'''** of shuttle **324'''** to unlock shuttle **324'''** from collet

618 and permit shuttle **324'''** to move from the pre-mixing position towards the mixing position. [0349] As depicted in FIGS. **16F** and **16G**, sample preparation reservoir **317'''**, wash reservoir **318'''**, and/or substrate reservoir **319'''** may each have symmetric shapes. For example, each of the walls of sample preparation reservoir **317'''** may form a symmetric shape wherein each of the walls of sample preparation reservoir **317'''** meet at an angle greater than a predetermined angle, e.g., 60°, 90°, to facilitate fluid emptying through the outlet of sample preparation reservoir **317'''**. Sonicator element **327'''** may be positioned off-center as illustrated of sample preparation reservoir **317'''** to facilitate mixing of the fluid within sample preparation reservoir **317'''**. Similarly, each of the walls of wash reservoir **318'''** and/or substrate reservoir **319'''** may form a symmetric shape wherein each of the walls of wash reservoir **318'''** and/or substrate reservoir **319'''** meet at an angle greater than a predetermined angle, e.g., 60°, 90°, to facilitate fluid emptying through the respective outlets of wash reservoir **318'''** and/or substrate reservoir **319'''**.

[0350] Referring to FIG. **16H**, as sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''** contacts seal piercer **321'''**. For example, shoulder **220** of sample collection device **200'''** may contact engager **380'''** of seal piercer **321'''** within input tunnel **301'''**. As collector pushes sample collection device **200'''** distally from the pre-mixing position towards the mixing position, shuttle **324'''** may be partially moved within sample preparation reservoir **317'''** as shown in FIG. **16H**. As shuttle **324'''** moves distally, first end **366'''** may unseal such that one or more reagent balls **375'''** within shuttle **324'''** and the collected sample within shuttle **324'''** are exposed to fluid within sample preparation reservoir **317'''**.

Advantageously, before first end **366'''** of shuttle **324'''** is unsealed, second end **371'''** of shuttle **324'''** is fluidically sealed by sample collection device **200'''** such that fluid from sample preparation reservoir **317'''** does not leak into input tunnel **301'''** beyond second end **371'''** and/or intermediate sealing zone **212'''**. In addition, as distal movement of sample collection device **200'''** causes the transition from the pre-mixing position toward the mixing position, sealing members **614** of shuttle **324'''** continuously fluidically seal the fluid in sample preparation reservoir **317'''** from input tunnel **301'''**.

[0351] Referring to FIG. **16I**, as sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''** moves seal piercer from the pre-venting position towards the venting position. As collector pushes sample collection device **200'''** distally, sample collection device **200'''** applies a force to seal piercer **321'''** (e.g., via contact between shoulder **220** and engager **380'''**) to cause the piercing elements to move to sequentially pierce sealing material **320'''** over the reservoirs as described above with respect to FIGS. **15A** through **15D**. In addition, as collector pushes sample collection device **200'''** more distally from the pre-mixing position toward the mixing position, shuttle **324'''** moves more distally within sample preparation reservoir **317'''**. Sample collection device **200'''** also may contact collet **618** to facilitate activation of contact switch **389'''**. For example, shoulder **220** of sample collection device **200'''** may contact deflection portion **636** of collet **618**.

[0352] As sample collection device **200'''** moves further distally in input tunnel **301'''**, sample collection device **200'''** and cartridge device **300'''** are moved into the mixing, venting, analysis positions shown in FIG. **16J** wherein the expelled sample in sample compartment **370'''** and reagent ball **375'''** are mixed within fluid of sample preparation reservoir **317'''**. In FIG. **16J**, sample collection device **200'''** is fully inserted in input tunnel **301'''**. As shown, seal piercer **321'''** is in the venting position and has pierced seal material **320'''** over each of the reservoirs, shuttle **324'''** has moved distally from collet **618** such that the sample and reagent ball **375'''** are mixing with the fluid in sample preparation reservoir **317'''**, deflector portion **636** of collet **618** is in the deflection position where deflector portion **636** has activated contact switch **389'''**, and locking members **387'''** have locked sample collection device **200'''** in input tunnel **301'''**.

[0353] In this mixing position, first end **366'''** of shuttle **324'''**, the one or more reagent ball compartments of shuttle housing one or more reagent balls, and/or the one or more sample

compartments housing the sample(s) to be analyzed may be disposed within sample preparation reservoir **317'''**. As explained above, the components fluidically seal sample preparation reservoir **317'''** in the mixing position such that the collected sample(s), the reagent ball(s), and the fluid within sample preparation reservoir **317'''** are sealed within the reservoir and may be mixed within sample preparation reservoir **317'''**.

[0354] Sample collection device **200'''** may be configured to activate contact switch **389'''** of circuit board **331'''**. For example, insertion of sample collection device **200'''** may cause collet **618** to activate contact switch **389'''**. Illustratively, deflector portion **636** of collet **618** deflects to activate contact switch **389'''** responsive to a force applied on deflector portion **636** by sample collection device **200'''** during insertion of sample collection device **200'''** in input tunnel **301'''**. Shoulder **220** of sample collection device **200'''** may contact deflector portion **636** and force deflector portion **636** downwardly as sample collection device **200'''** moves distally in input tunnel **301'''** to activate contact switch **389'''**. Depression of contact switch **389'''** may complete a circuit such that electrical signals may be transmitted, e.g., to the reader device and/or the computing device running the software-based user interface system. In this manner, proper insertion of sample collection device **200'''** within the input tunnel generates an electrical signal that may be transmitted to the reader device and/or computing device to notify the reader device and/or computing device of such proper insertion. While contact switch **389'''** is positioned within input tunnel **301'''** such that depression of contact switch **389'''** indicates full insertion of sample collection device **200'''** at the mixing position, contact switch **389'''** also could be positioned within input tunnel **301'''** to indicate partial insertion of sample collection device **200'''**. In addition, or alternatively, more than one contact switch may be disposed in the input tunnel such that insertion of the sample collection device within the tunnel may be tracked by sequential depression of the contact switches aligned along the input tunnel.

[0355] In the mixing position, locking members **387'''** are configured to irreversibly lock sample collection device **200'''** within cartridge device **300'''**. Locking members **387'''** may engage the proximal-most groove of the multiplicity of grooves in engagement zone **209'''** when sample collection device **200'''** is fully inserted in input tunnel **301'''** as shown in FIG. 16J. Locking members **387'''** may be biased inwardly in input tunnel **301'''** such that a locking end of locking member **387'''** engages sample collection device **200'''** in the mixing position. Advantageously, locking sample collection device **200'''** (e.g., longitudinally and/or axially) within input tunnel **301'''** during partial insertion and full insertion promotes sealing of sample preparation reservoir **317'''** over time as sample collection device **200'''** cannot be retracted once locked to facilitate safe disposability and consistency of testing because a user cannot pull out sample collection device **200'''** from cartridge device **300'''** inadvertently after partial or full insertion. In addition, in the mixing position, proximal sealing zone **207'''** of sample collection device **200'''** is configured to seal input tunnel **301'''** at aperture **302'''**. In this manner, proximal sealing zone **207'''** provides additional structure to minimize or eliminate fluid leakage from cartridge device **300'''**, e.g., at aperture **302'''**. Proximal sealing zone **207'''** of sample collection device **200'''** is also configured to contact the cartridge housing to resist insertion force by collector once sample collection device **200'''** and cartridge device **300'''** are properly in the mixing position.

[0356] In the mixing position, sonicator element **327'''** may be used to enhance mixing the fluid in sample preparation reservoir **317'''** with the sample and reagent ball(s) as described above with respect to FIGS. 10N through 10P.

[0357] Referring now to FIGS. 17A through 17D, sonicator element **327'''** may be electrically coupled to circuit board **331'''** via first spring contact **392'''** and second spring contact **392'''** as shown. Preferably, sonicator element **327'''** is electrically coupled to circuit board **331'''** only via first and second spring contacts **392'''**, e.g., without a wired connection between sonicator element **327'''** and circuit board **331'''**. Beneficially, spring contacts **392'''** absorb movement of sonicator element **327'''** such that circuit board **331'''** vibrates minimally in a suitable manner when sonicator

element **327'''** is activated, e.g., responsive to signals transmitted by the reader device, and spring contact **392'''** permits ease of assembly and reproducibility compared to soldering which may adversely impact sonicator element **327'''**. Sonicator element **327'''** may form a wall, e.g., part of the bottom wall, of sample preparation reservoir **317'''**. Sonicator element **327'''** may be adhered to internal component **316'''** to form the wall. For example, an annulus of adhesive, e.g., epoxy, may be applied on the top surface of sonicator element **327'''** and UV cured to internal component **316'''** at the bottom of sample preparation reservoir **327'''**. Such an annulus of adhesive permits fluidic sealing between sonicator element **327'''** and sample preparation reservoir **317'''** while permitting sonicator element **327'''** to vibrate during activation. As is explained above, sonicator element **327'''** may be positioned off-center of sample preparation reservoir **317'''** to facilitate mixing of the fluid with the sample and reagent ball(s) within sample preparation reservoir **317'''**.

[0358] Referring now to FIGS. **18A** and **18B**, an alternative cartridge for analyzing a sample is described. Cartridge device **300'''** may be constructed similarly to cartridge device **300'''**, wherein like components are identified by like-primed reference numbers, except cartridge device **300'''** includes collet **618'** and shuttle **324'''** which may be constructed similarly to shuttle **324** of FIG. **9A**. In this embodiment, cartridge device **300'''** is designed for relatively small sample collection, e.g., less than 10 microliters, preferably 2-5 microliters, of a sample, e.g., from a nasal passageway, from an ear, from blood. Cartridge device **300'''** may be further fitted with a reagent ball(s) intended to identify different target analytes that may be indicative of, for example, inflammation, influenza, testosterone, fertility, HIV, or Vitamin D. FIGS. **18A** and **18B** illustrate the ease of interchangeability between cartridge device **300'''** and cartridge device **300'''**.

[0359] As shown in FIG. **18A**, collet **618'** has one or more protrusions **637** disposed the distal end of collet **618'** configured to contact second end **371'''** of shuttle **324'''** in the pre-mixing position. The one or more protrusions **637** may contact a sealing member, e.g., O-ring, at second end **371'''** of shuttle **324'''** to retain the sealing member in position. The one or more protrusions may have lead-in angles configured to guide the distal portion of the sample collection device into the opening at second end **371'''** of shuttle **324'''**. Similar to that described above, collet **618'** is configured to decouple from shuttle **324'''** responsive to a force applied by the sample collection device on shuttle **324'''**, e.g., via the shoulder of the distal sealing zone on second end **371'''** of shuttle **324'''**, during sample collection device insertion in the input tunnel. In this manner, shuttle **324'''** moves from the pre-mixing position to the mixing position as described above.

[0360] Referring now to FIG. **19**, a process for monitoring temperature during enhanced mixing via the sonication element is described. As is described below, the computerized reader may largely control the operations of the detection system. The reader includes a processor with memory, the memory having instructions stored thereon for implementing various methods needed to successfully detect the presence, absence, and/or quantity of one or more target analytes within a collected sample. For example, the computerized reader may cause the process of FIG. **19** to be performed in an automated manner.

[0361] As is described above, a sonicator element (e.g., sonicator element **327**, **327'**, etc.) of the cartridge may be activated for enhanced mixing of the contents in the sample preparation reservoir when the sample collection device and the cartridge are in the mixing position. At **702**, the process begins. The process may begin responsive to an event. For example, the process may begin when the processor of the reader device receives a signal indicating that a contact switch has been activated in the cartridge device responsive to full insertion of the sample collection device in the cartridge device.

[0362] At **704**, the sonicator element, e.g., a piezoelectric transducer, is activated. Upon activation, the sonicator element emits acoustic waves toward the sample preparation reservoir to move fluid within the reservoir. The sonicator element may emit acoustic waves at a predetermined frequency, e.g., 4 MHz, which may be modified by the reader. As is described above, the acoustic waves may cause fluid within the reservoir to move in a wave pattern to mix the fluid in the reservoir. The

processor of the reader device may activate the sonicator element by causing transmission of electrical signals to the sonicator element, e.g., via one or more electrical leads of the printed circuit board in the cartridge.

[0363] At **706**, a temperature is sampled. The temperature may be indicative of temperature of the fluid within the sample preparation reservoir. For example, the cartridge may include a temperature sensor, e.g., temperature sensor **616** disposed on the circuit board, configured to generate a signal indicative of temperature of the fluid in the sample preparation reservoir. The signal may be transmitted from the cartridge to the reader, e.g., via one or more electrical leads on the printed circuit board of the cartridge, when the cartridge and reader are electrically coupled. The signal may be processed by the processor of the reader device to determine whether the signal indicates a temperature of the fluid in the reservoir outside a threshold.

[0364] At **708**, it is determined whether the sampled temperature is too high. For example, the processor of the reader device may compare the temperature information from the signal generated by the temperature sensor to temperature thresholds stored in memory of the reader device. For example, the memory may store a threshold high temperature, e.g., above 42° C., and/or a look-up table with threshold high temperatures based on cartridge specific parameters such that the processor may compare the sensed temperature to the stored information to determine whether a sensed temperature is too high. If the temperature is determined to be outside a threshold, e.g., too high, too low, not within a range, emission of acoustic waves from the sonicator element may be modified.

[0365] At **710**, if the temperature is determined to be too high, emission of acoustic waves from the sonicator element may be modified by lowering the duty cycle of the sonicator element. For example, the processor of the reader device may modify emission of the acoustic waves from the sonicator element by transmitting electrical signals to the sonicator element to cause the sonicator element to lower the duty cycle. The processor may modify emission of the acoustic waves from the sonicator by deactivating the sonicator if the signal indicates the temperature of the fluid in the reservoir is above the threshold.

[0366] At **712**, it is determined whether the sampled temperature is too low. For example, the processor of the reader device may compare the temperature information from the signal generated by the temperature sensor to temperature thresholds stored in memory of the reader device. For example, the memory may store a threshold low temperature, e.g., below 37° C., and/or a look-up table with threshold low temperatures based on cartridge specific parameters such that the processor may compare the sensed temperature to the stored information to determine whether a sensed temperature is too low.

[0367] At **714**, if the temperature is determined to be too low, emission of acoustic waves from the sonicator element may be modified by raising the duty cycle of the sonicator element. For example, the processor of the reader device may modify emission of the acoustic waves from the sonicator element by transmitting electrical signals to the sonicator element to cause the sonicator element to increase the duty cycle.

[0368] At **716**, it is determined whether the temperature indicative of temperature of the fluid within the sample preparation reservoir is within a threshold range, e.g., below the threshold high temperature and above the threshold low temperature, for a predetermined amount of time (e.g., between 3 minutes to 15 minutes, between 3 minutes to 10 minutes, between 3 minutes to 5 minutes, about 10 minutes), which may be cumulative or consecutive, such that the sonicator element may be timed-out. For example, the memory of the reader may store the predetermined amount of time that the temperature indicative of fluid temperature within the sample preparation reservoir should be within the threshold temperature range. The predetermined amount of time may be the time required for suitable mixing of the fluid within the sample preparation reservoir with the sample and the reagent ball(s) for isothermal amplification for analysis. The acoustic emissions from the sonicator element may both heat fluid within the sample preparation reservoir and mix the

contents of the sample preparation reservoir at the macro and the micro level for isothermal amplification. For example, the sonicator element may pass energy into the sample preparation reservoir sufficient to increase the temperature in the sample preparation reservoir such that an amplification reaction may occur such as isothermal DNA or RNA amplification, thereby generating nucleic acid amplicons for downstream detection. Such downstream detection may be partially based on the binding of nucleic acids or detectable moieties thereon or therein to affinity molecules on magnetic particles (which could be in a combination of antibodies, DNA probes, detectable moieties, and/or enzymes) or it could be based on specific binding to surface bound affinity molecules on one or more working electrodes each with their own population of affinity molecules. Alternatively, or additionally, such a reaction may take place in the sample preparation reservoir and be observed there. If the processor of the reader determines that the sampled temperature has not been within the threshold range for the predetermined amount of time, the process returns to **706**.

[0369] At **718**, the sonicator element is deactivated. For example, the sonicator element may be deactivated if the sampled temperature has been within the threshold range for the predetermined amount of time. The processor of the reader may determine that the sampled temperature has been within the threshold range for the predetermined amount of time.

[0370] At **720**, the process ends after the sonicator element is deactivated.

[0371] Referring now to FIG. **20**, a graph showing temperature in Celsius versus time in seconds is shown. The graph shows the measured temperature of the fluid mixing in the sample preparation reservoir at line **730**, the measured temperature at the thermistor, e.g., temperature sensor **616**, at line **732**, the measured ambient temperature at line **734**, and the measured temperature at the piezoelectric transducer, e.g., sonicator element **327**, **327'**, etc., at line **736**. As will be observed, the duty cycle of the sonicator is reduced at about the 40 second mark on the graph, causing the temperature of the fluid mixed in the sample preparation reservoir to decrease. The duty cycle of the sonicator is again reduced around the 520 second mark on the graph, again causing the temperature of the fluid mixed in the sample preparation reservoir to decrease.

The Reader Device

[0372] The reader device, or reader, of various embodiments is, comprises, or is comprised of, a specialized computer. The computer includes a processor with memory having instructions stored thereon for executing one or more methods for detecting the presence, absence, and/or quantity of one or more target analytes in a sample. In various embodiments, the reader's computer controls the operations of the detection system, controlling when and how various functions of the system occur, such as, for example: mixing of the fluids in the sample preparation reservoir of the cartridge, opening of valves, and/or localization of magnetic particles over the sensors. To control such operations, the computerized reader is configured to receive information from, and send information to, physical components present within the reader or cartridge.

[0373] FIG. **21A** illustrates a perspective view of an exemplary reader device and FIG. **21B** is an exploded view illustrating the internal components of the reader device of FIG. **21A**. Reader device **400** may include opening **401** in cartridge dock **402**, housing **403**, user interface **404**, power supply **405**, first magnetic generator **406**, second magnetic generator **407**, magnetic generator housing **408**, elastic member **409**, light pipe **410**, circuit board **411**, processor **412**, communication circuit **413**, electrical connector **414**, inductive coil **415**, alignment magnets **416**, and/or base **417**.

[0374] Opening **401** of reader device **400** permits the cartridge device to be docked within cartridge dock **402**. The cartridge, when received by reader device **400**, may be disposed on or in, partially or fully, or otherwise coupled to, reader device **400**. Several of the reader components may be strategically positioned in particular locations relative to cartridge dock **402** to achieve desired interactions with the cartridge. For example, electrical connector **414** may be positioned relative to cartridge dock **402** such that the electrical connector of the cartridge device is electrically coupled to electrical connector **414** when the cartridge device is inserted in cartridge dock **402**. In addition,

magnetic field generators **406** and **407** may be positioned relative to cartridge dock **402** such that magnetic field generators **406** and **407** are disposed under the working electrode of the cartridge device when the cartridge device is inserted in cartridge dock **402**.

[0375] Housing **403** is configured to house the internal components of reader device **400** and may cooperate with cartridge dock **402** and base **417** to house the internal components.

[0376] User interface **404** may be used to receive inputs from, and provide outputs to, a user. Illustratively, user interface **404** includes LEDs configured to notify a user when a cartridge device is properly inserted into reader device **400** and/or when a sample collection device is properly inserted in the cartridge device. User interface **404** may be coupled to processor **412**. User interface **404** may include a touchscreen, LED matrix, other LED indicators, or other input/output devices for receiving inputs from, and providing outputs to, a user. In other embodiments, user interface **404** is not present on reader **400**, but is instead provided on a remote computing device communicatively connected to reader **400** via the communication circuit **413**. User interface also may be a combination of elements on the reader and a remote computing device.

[0377] Power supply **405** may be a suitable battery such as a replaceable battery or rechargeable battery and apparatus may include circuitry for charging the rechargeable battery, and a detachable power cord. Power supply **405** may be charged by charger **500** via an inductive coil within the charger and inductive coil **415**. Alternatively, the power supply may be a port to allow reader device **400** to be plugged into a conventional wall socket, e.g., via a cord with an AC to DC power converter, for powering components within the housing.

[0378] Magnetic field generators **406** and **407** may be inductors or other electromagnetic components movably affixed within reader **400**. Magnetic field generators **406** and **407** may be permanent magnets. Magnetic field generators **406** and **407** are positioned such that, when a cartridge is electrically coupled to reader device **400**, the working electrode is disposed directly within a magnetic field created by magnetic field generators **406** and **407**. In various embodiments, the magnetic field(s) are the cause of localization; the magnetic field(s) are what induce magnetic particles and accompanying hybridized molecules to localize within the analysis zone. First and second magnetic generators **406** and **407** may be configured to generate a magnetic field over the length of a single working electrode to promote homogenous distribution of a plurality of magnetic particles over the length of the single working electrode.

[0379] Magnetic field generators **406** and **407** emit a magnetic field sufficiently strong to cause the magnetic particles released into the analysis channel from the sample preparation reservoir to remain localized over magnetic field generators **406** and **407**, and thereby over the working electrode, as the wash solution and/or the fluid carrying chemical substrates flows over the magnetic particles in the analysis channel in the cartridge.

[0380] Magnetic generator housing **408** is configured to house magnetic field generators **406** and **407**. Magnetic generator housing **408** may be coupled to elastic member **409** that permits movement of magnetic field generators **406** and **407**, e.g., upon insertion and removal of a cartridge device from cartridge dock **402**.

[0381] Reader device **400** may include light pipe **410** designed to guide light from LEDs within reader device **400** to user interface **404**.

[0382] Circuit board **411** includes electrical components and permits electrically coupling between processor **412**, communication circuit **413**, and/or electrical connector **414**. One or more electrical components and/or circuits may perform some of or all the roles of the various components described herein. Although described separately, it is to be appreciated that electrical components need not be separate structural elements. For example, processor **412** and communication circuit **413** may be embodied in a single chip. In addition, while processor **412** is described as having memory, a memory chip(s) may be separately provided.

[0383] Processor **412** may be a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other

programmable logic device, discrete gate or transistor logic, discrete hardware components, or any suitable combination thereof designed to perform the functions described herein. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0384] Processor **412** may contain memory and/or be coupled, via one or more buses, to read information from, or write information to, memory. The memory may include processor cache, including a multi-level hierarchical cache in which different levels have different capacities and access speeds. The memory may also include random access memory (RAM), other volatile storage devices, or non-volatile storage devices. The storage devices can include, for example, hard drives, optical discs, flash memory, and Zip drives.

[0385] Processor **412**, in conjunction with firmware/software stored in the memory may execute an operating system, such as, for example, Windows, Mac OS, Unix or Solaris 5.10. Processor **412** also executes software applications stored in the memory. In one non-limiting embodiment, the software comprises, for example, Unix Korn shell scripts. In other embodiments, the software may be programs in any suitable programming language known to those skilled in the art, including, for example, C++, PHP, or Java.

[0386] Communication circuit **413** is configured to transmit information, such as signals indicative of the presence, absence, and/or quantity of one or more target analytes within a sample, locally and/or to a remote location such as a server. Communication circuit **413** is configured for wired and/or wireless communication over a network such as the Internet, a telephone network, a Bluetooth network, and/or a WiFi network using techniques known in the art. Communication circuit **413** may be a communication chip known in the art such as a Bluetooth chip and/or a WiFi chip. Communication circuit **413** may include a receiver and a transmitter, or a transceiver, for wirelessly receiving data from, and transmitting data to a remote computing device. In some such embodiments, the remote computing device may be a mobile computing device that provides the system with a user interface; additionally or alternatively, the remote computing device is a server. In embodiments configured for wireless communication with other devices, communication circuit **413** may prepare data generated by processor **412** for transmission over a communication network according to one or more network standards and/or demodulates data received over a communication network according to one or more network standards.

[0387] Processor **412** is also coupled to electrical connector **414**, which may include an EDGE card or other electrical connector, to send electrical signals to, and receive electrical signals from, the circuit board component of the cartridge (e.g., via electrical connector **312**). Electrical connector **414** may be located on, under, within, or adjacent to cartridge dock **402** and is positioned such that the pins of electrical connector **414** make contact with, and establish electrical connectivity with, the electrical leads of a docked cartridge device. Electrical connector **414** thereby establishes electrical continuity between the sensors on the circuit board of the cartridge and electrochemical circuitry within the reader. Electrical connector **414** of the reader also may establish electrical continuity with one or more heating elements, if present on the circuit board of the cartridge. Reader device **400** may include a portion of an electrochemical circuit, which is completed with the addition of the cartridge based on electrical continuity between electrical connector **414** and the electrical leads of the cartridge. The addition of the cartridge may complete or close the circuit. Coupling the cartridge to reader **400** may activate reader device **400**, causing it to “wake up.” Once awoken, electrical connector **414** may identify signals being received from a portion of the cartridge to identify what type of cartridge is coupled to its dock. Electrical connector **414** may receive signals indicative of information on cartridge type (e.g., inflammation, influenza, testosterone, fertility, Vitamin D), cartridge identification information (e.g., serial number), and/or calibration information from memory within the cartridge and transmit such information to processor **412** for processing.

[0388] Once awoken, reader device **400** also may determine what test protocol to run for the identified cartridge and/or searches for, and connects to, nearby mobile computing devices.

[0389] Reader device **400** may include one or more magnets **416** configured to align with one or more magnets in the charger to facilitate efficient energy transfer between inductive coil **415** and an inductive coil within the charger. Illustratively, four magnets are used. Magnets **416** may be keyed to align with a corresponding keyed magnet in the charger.

[0390] Referring now to FIGS. **22A** through **22D**, insertion of a cartridge device within a reader device is described. As shown in the cutaway view of reader device **400** in FIG. **22A**, first and second magnetic generators **406** and **407** may extend partially into opening **401** through cartridge dock **402**. In FIG. **22A**, first and second magnetic generators **406** and **407** are shown in a raised position. Elastic member **409** may be biased to cause first and second magnetic generators **406** and **407** to rest in the raised position.

[0391] FIG. **22B** shows cartridge device **300**, having sample collection device **200** partially inserted therein, being inserted into reader device **400**. Specifically, cartridge device **300** is inserted in opening **401** at cartridge dock **402**. As cartridge device **300** moves distally into cartridge dock **402**, the top surfaces of first and second magnetic generators **406** and **407** preferably do not contact electrical connector **312** and first contact first ramp portion **313**. First ramp portion **313** causes first and second magnetic generators **406** and **407** to gradually depress during further distal insertion, e.g., by causing a downward force on first and/or second magnetic generators **406**, **407** that depresses elastic member **409**. As cartridge device **300** is inserted past first ramp portion **313**, first and/or second magnetic generators **406**, **407** contact bottom surface **308** of cartridge device **300** and move to a depressed position as shown in FIG. **22B**.

[0392] As cartridge device **300** moves further distally into cartridge dock **402**, first and/or second magnetic generators **406**, **407** contact second ramp portion **314** which ramps up into magnetic generator depression **315**. Second ramp portion **314** gradually guides first and/or second magnetic generators **406**, **407** into magnetic generator depression **315** as shown in FIGS. **22C** and **22D**. FIGS. **22C** and **22D** depict cartridge device **300** inserted in reader device **400** in the analysis position where first and second magnetic generators **406** and **407** are disposed in magnetic generator depression **315** and electrical connector **312** of cartridge device **300** is electrically coupled to electrical connector **414** of reader device **400**. Magnetic generator depression **315** is disposed beneath one or more working electrodes of cartridge device **300** such that first and second magnetic generators **406** and **407** move up into magnetic generator depression **315** into a raised position and are disposed adjacent the one or more working electrodes when cartridge device **300** is fully inserted in the reader. The bias of elastic member **409** may cause first and second magnetic generators **406** and **407** to move up into magnetic generator depression **315**. Second ramp portion **314** also facilitates removal of cartridge device **300** from reader device **400** by gradually depressing first and second magnetic generators **406** and **407** during removal of cartridge device **300**.

[0393] The interaction of the bias of elastic member **409** and magnetic generator depression **315** may allow for first and second magnetic generators **406** and **407** to be positioned as close to the working electrode as possible. The closer first and second magnetic generators **406** and **407** are to the working electrode, the more force the magnet field is able to exert, meaning that smaller magnets or inductors are capable of exerting equivalent magnetic field strengths as larger, more costly magnets or inductors. The use of small magnets or inductors is particularly advantageous in embodiments having multiple magnetic fields and multiple analysis zones (for example, in embodiments configured to detect a plurality of different target analytes), because the smaller the magnet or inductor, the less the magnetic fields overlap. Smaller magnetic fields can limit the amount of cross talk between the magnets or inductors under the different detection sensors.

[0394] Referring to FIGS. **23A** and **23B**, graphs are shown depicting the measured strengths of magnetic fields at varying distances from the analysis channel for designs where one magnetic generator generates a magnetic field for one working electrode (FIG. **23A**) and where two magnetic

generators generates a magnetic field for one working electrode (FIG. 23B). It has been discovered that magnetic fields at or near the absolute peak strength will retain suitable magnetic particles in the analysis channel over the working electrode, but magnetic particles tend to wash off the working electrode as the magnetic field strength decreases from the peaks toward zero, creating dead zones **750** and **751**. Use of two magnetic field generators creates a much more uniform absolute magnetic field over the working electrode as compared to use of one magnetic field generator. Accordingly, dead zone **751** for a dual magnet design is significantly smaller than dead zone **750** for a single magnet design. Accordingly, as described above, first and second magnetic generators **406** and **407** may be used in reader device **400** to generate a magnetic field over the length of a single working electrode to promote homogenous distribution of a plurality of magnetic particles over the length of the single working electrode.

The Computerized Methods of Detection

[0395] The timing of heat delivery and valve opening within the cartridge device may be precisely timed and controlled by the reader device. For example, the reader may control when heat-generating current flows through the heating elements. Current may flow from the reader to the cartridge to cause actuation in the following sequence: (1) valve actuation for sample preparation reservoir, (2) fluidic isolator actuation, if present, (3) valve actuation for wash reservoir, if present, then (4) valve actuation for chemical substrate reservoir. Actuation of each valve may be timed such that: the respective valve fully actuates, the associated reservoir has time to empty its contents into the analysis channel, and at least some of the contents of the reservoir have time to travel to the absorbent pad positioned downstream of the sensors before the contents of the next reservoir is released. In some embodiments, the time between valve actuations is selected to be great enough for the absorbent pad to entirely or substantially absorb fluid present within the analysis channel. Advantageously, in such embodiments, very little mixing occurs between the contents of successive reservoirs. In addition, or alternatively, actuation of each valve may be based on a feedback control system wherein the cartridge and/or reader recognizes when flow has initiated and/or stopped from the respective reservoirs, e.g., using a flow sensor disposed in the analysis channel adjacent to the respective reservoir and electrically coupled to the memory of the cartridge and/or the processor of the reader, such that the valves may be turned on and off with respect to the “state” of the progression of events, e.g., based on signals sensed by the flow sensors disposed adjacent each reservoir.

[0396] As mentioned above, the computerized reader largely controls the operations of the detection system. The reader includes a processor with memory, the memory having instructions stored thereon for implementing various methods needed to successfully detect the presence, absence, and/or quantity of one or more target analytes within a collected sample. For example, an embodiment of one method performed by the computerized reader in an automated manner is provided in FIG. **24**.

[0397] At block **802**, the computerized reader detects the presence of a cartridge loaded into or onto the reader. For example, a cartridge may be coupled to the reader such that electrical leads on the cartridge come into physical contact with electrical pins on the reader, completing a circuit that turns on the reader and signals the reader to the presence of a cartridge.

[0398] At block **804**, the reader detects identification information associated with the cartridge. For example, the cartridge may include cartridge type information stored within its memory, which generates signals unique to the particular type of the cartridge, allowing the reader to distinguish between cartridges and types. The cartridge also may include calibration information stored within its memory.

[0399] The reader's processor receives the cartridge type information signals, and as shown at block **806**, may identify a proper test protocol for the cartridge based on the cartridge type information. The reader's processor may compare cartridge type information signals to a database of protocols based on the cartridge type stored in memory. If the processor does not recognize the

cartridge type information, the processor may communicate with a remote computing device such as a mobile computing device and/or a server to signal that an unidentifiable cartridge has been detected. In some embodiments, the reader downloads updates directly from a server or indirectly with the mobile computing device acting as an intermediary. In some embodiments, when an unknown cartridge type is detected, a user is prompted via the user interface of the mobile computing device, to download updates; in other embodiments, the updates are downloaded automatically. In various embodiments, the updates include newly developed cartridge types and test protocols. Once the new types and test protocols are downloaded, they will be added to the reader's database of supported tests so that future tests with this cartridge type will automatically be recognized and implemented without the need for communicating with remote computing devices. The reader's processor also may receive calibration information signals and may account for the calibration information during the test protocol.

[0400] As shown at block **808**, the computerized reader detects insertion of a sample collection device into the cartridge. For example, the reader may receive electrical signals indicating that the contact switch in the cartridge device has been activated responsive to insertion of the sample collection device into the input tunnel, which may also substantially correspond to the sample entering the sample preparation reservoir. The sample (and reagent ball(s), if present) is then mixed within the fluid of the sample preparation reservoir by introducing the sample (and reagent ball(s), if present) into the sample preparation reservoir.

[0401] At block **810**, the reader's processor sends signals to the sonicator element to instruct it to initiate a sonication protocol to mix a plurality of reagents, affinity molecules, and sample particles within a fluid disposed within the sample preparation reservoir. In various embodiments, the resulting mixture includes magnetic particles bound to: target analytes, target analytes and detector agents, and/or competitive binding agents. As used herein, sandwich complexes refer to magnetic particles bound directly or indirectly to target analytes and detector agents; competitive binding complexes refer to magnetic particles bound to competitive binding agents. Each sandwich complex and competitive binding complex may include a detector agent bound within the complex. In one embodiment described here, the detector agent is an oxidizing enzyme. The sonicator also may pass energy into the sample preparation reservoir sufficient to increase the temperature in the sample preparation reservoir such that an amplification reaction may occur such as isothermal DNA or RNA amplification, thereby generating nucleic acid amplicons for downstream detection. Such downstream detection may be partially based on the binding of nucleic acids or detectable moieties thereon or therein to affinity molecules on magnetic particles (which could be in a combination of antibodies, DNA probes, detectable moieties, and/or enzymes) or it could be based on specific binding to surface bound affinity molecules on one or more working electrodes each with their own population of affinity molecules. Alternatively, or additionally, such a reaction may take place in the sample preparation reservoir and be observed there.

[0402] As shown at block **812**, the reader may monitor temperature of the fluid in the sample preparation reservoir during mixing by the sonicator element and isothermal amplification. For example, a temperature sensor in the cartridge may transmit signals indicative of temperature of the fluid in the sample preparation reservoir, as described above with respect to FIG. **19**.

[0403] As shown at block **814**, the reader may generate a current, which heats or otherwise stimulates a first heating element, thereby causing heat to transfer to a heat-actuated valve sealing the outlet of the sample preparation reservoir within the cartridge. Such heating may cause the valve to melt or undergo another phase change, which allows fluid to flow out of the sample preparation reservoir into an analysis channel via capillary action. As the fluid flows, it transports the mixture with it, and the magnetic particles within the mixture, including magnetic particles within sandwich complexes and/or competitive binding complexes, localize over one or more magnetic fields within the analysis channel, forming one or more localized samples.

[0404] Optionally, at block **816**, the reader generates a current, which heats or otherwise stimulates

a second heating element that causes a fluidic isolator to block the outlet of the sample preparation reservoir from the rest of the analysis channel. In this manner, fluid released from other reservoirs cannot flow into the sample preparation reservoir and/or cause undesirable reactions with leftover reagents.

[0405] Optionally, at block **818**, the reader generates a current, which heats or otherwise stimulates a third heating element such that a second valve within the cartridge undergoes a phase change and a wash solution flows out of a wash reservoir into the analysis channel. In various embodiments, the wash solution removes, from the one or more localized samples, oxidizing enzymes (or other detector agents) that are not indirectly bound to magnetic particles.

[0406] At block **820**, the reader generates a current, which heats or otherwise stimulates a fourth heating element such that a third valve within the cartridge undergoes a phase change and a solution of substrates flows out of a substrate reservoir into the analysis channel. In various embodiments, when the detector agent is an oxidizing enzyme, the oxidizing enzymes within the sandwich complexes and/or competitive binding complexes of each localized sample oxidize the substrate molecules present in the aqueous media used to transport said substrate molecules. In embodiments in which sandwich complexes are present, oxidation occurs at an electrochemical cell formed by an electrochemical sensor and the volume of fluid substantially over it and electrons flow from the working electrode of the electrochemical sensor to the volume substantially above the sensor in a quantity proportional to a quantity of target analyte present within the localized sample (e.g., magnetic particle bound complexes and/or surface bound complexes). In embodiments in which competitive binding complexes are present, oxidation occurs at an electrochemical cell formed by an electrochemical sensor and the volume of fluid substantially over the sensor and electrons flow from working electrode of the electrochemical sensor in a quantity inversely proportional to a quantity of target analyte present within the localized sample.

[0407] At block **822**, the reader's processor receives from the reader's electric connector a first signal detected at the positive control working electrode within the analysis channel in the cartridge. In various embodiments, the signal is a voltage or current or resistivity signal. At least a portion of the signal may be caused by the oxidation of the substrate bound to surface bound antibodies at the positive control working electrode. At block **824**, the reader's processor receives from the reader's electric connector a second signal detected at the working electrode within the analysis channel in the cartridge. In various embodiments, the signal is a voltage or current or resistivity signal. At least a portion of the signal is caused by the oxidation of the substrate over the working electrode, e.g., substrates magnetically bound to magnetic particles magnetically held over the working electrode. At block **826**, the reader's processor receives from the reader's electric connector a third signal detected by a negative control working electrode within the analysis channel in the cartridge. At block **828**, the reader's processor processes and analyzes the signal from the working electrode (and, optionally, the signal from the positive control working electrode and/or the signal from the negative control working electrode) to identify the presence and/or quantity of one or more target analytes. The reader's processor may determine whether the parameter(s), e.g., current, voltage, of the first signal are within a predetermined range stored in the reader's memory, e.g., in a lookup table. Alternatively, the predetermined range may be stored in the memory of the cartridge, stored in the memory for the device running the software application, or stored in a server in the system's network. The first signal may be used for error detection and diagnostic of faulty cartridges. The third signal may be indicative of noise present within the system. The reader's processor may subtract or apply another algorithm to remove the third signal from the second signal to account for and/or eliminate noise that may be present within the system. Alternatively, the third signal may be used for error detection and diagnostic of faulty cartridges. Optionally, as shown at block **830**, the reader may transmit signals indicative of a test result to a mobile computing device for further processing, storage, transmission to a server, and/or display of results to a user.

The Charger

[0408] FIG. 25A illustrates a perspective view of an exemplary charger and FIG. 25B is an exploded view illustrating the internal components of the charger of FIG. 25A. Charger 500 is an optional feature that may be used to charge reader device 400. Charger 500 may include alignment magnets 501, inductive coil 502, and batteries 503. Charger 500 may include one or more alignment magnets 501 configured to align with one or more alignment magnets in the reader device to facilitate efficient energy transfer between inductive coil 502 and the inductive coil within the reader. Illustratively, four magnets are used. Magnets 501 may be keyed to align with a corresponding keyed magnets in the reader. Charger 500 may be plugged into a conventional socket, e.g., via a cord or a cord with an AC to DC power converter, for charging components within charger 500 such as batteries 503 to permit charging of reader 400.

The Reactants and the Reactions

[0409] Various devices, systems, kits, and methods disclosed herein are intended to isolate, tag, and detect one or more target analytes within a sample taken from a specimen. Chemical reactions may be employed to enable such detection. Chemical reactions may take place in a reservoir such as the sample preparation reservoir described above. For example, the sample preparation reservoir may hold a fluid such as water, saline solution, water/saline solution mixed with one or more of magnetic particles, affinity molecules, connection molecules, signaling agents, competitor binding molecules, competitor molecules, labels, and/or signaling agents. One or more reagent balls also may hold one or more of magnetic particles, affinity molecules, connection molecules, signaling agents, competitor binding molecules, competitor molecules, labels, and/or signaling agents outside of the fluid in the sample preparation reservoir. A reaction may begin when a sample potentially having one or more target analytes (and, optionally, one or more reagent balls) is (are) mixed with fluid in the sample preparation reservoir, e.g., by introducing a sample collected on a distal portion of a sample collection device into the sample preparation reservoir when the sample collection device is fully inserted into an input tunnel of a cartridge. Exemplary chemical reactions are discussed below and depicted in FIGS. 26A-28H.

[0410] Referring to FIGS. 26A and 26C, target analyte 910a, 910b is added to a solution of sample preparation reagents, e.g., in the sample preparation reservoir. Target analyte 910a, 910b may be any molecule such as a nucleic acid, protein, small molecule, or heavy metal or in the case of a biologic or large molecule, the target analyte is a fragment thereof associated with a particular condition or possible contamination or presence of a particular cell type, e.g., to detect a target biomarker present on the surface of cell. Non-limiting examples include specific pathogens, e.g. bacteria, viruses, parasites; toxins; hormones; immune regulatory molecules; or detectable fragments thereof.

[0411] Target analyte 910a, 910b may be detected using the systems and methods described herein to for studying a condition such as molecular levels indicative of inflammation, influenza, testosterone, fertility, and/or Vitamin D. The sample preparation reagents may include magnetic microbeads or nanoparticles 920a, 920b (referred to herein as “magnetic particles”). Magnetic particles 920a, 920b are magnetically responsive such that they will be attracted to a magnetic field emitted from one or more magnetic generators. In this manner, magnetic particles 920a, 920b released from the sample preparation reservoir travel downstream in the analysis channel until they are localized over the one or more working electrodes of the sensor within the analysis channel of the cartridge responsive to magnetic fields generated by one or more magnetic generators of the reader. Each magnetic particle 920a, 920b may have affinity molecule 930a, 930b bound to its surface. The magnetic particles may be different sizes and may have a diameter between 50 nanometers to 5000 nanometers, between 100 nanometers to 4000 nanometers, between 100 nanometers to 3000 nanometers, between 100 nanometers to 2000 nanometers, between 100 nanometers to 1000 nanometers, between 500 nanometers to 4000 nanometers, between 1000 nanometers to 4000 nanometers, or between 1500 nanometers to 3000 nanometers.

[0412] The affinity molecule may be any suitable molecule or moiety that can bind to or capture a target molecule. Non-limiting examples of affinity molecules include antibodies (including single chain, multi-chain antibodies, diabodies, humanized antibodies, etc.), antibody fragments with affinity, ligands, polypeptide or protein molecules and moieties with binding affinity for substrates, nucleic acid molecules (e.g., aptamers), other molecules with binding affinity, and the like. The affinity molecules include chemically modified naturally occurring molecules. Affinity molecules for a particular target analyte may be selected according to generally understood methods. For example, methods of generating antibodies and fragments thereof are well known in the literature and are exemplified by *Antibodies: A Laboratory Manual* (1988) Eds. Harlow and Lane, Cold Spring Harbor Laboratories Press, and U.S. Pat. Nos. 4,381,292, 4,451,570, and 4,618,577. Further, affinity molecules are commercially available for specific target analytes. A listing of such sources may be found in *Linscott's Directory of Immunological and Biological Reagents*.

[0413] As used herein the terms “hybridize” and “hybridization” intend the specific interaction between two entities, such as an antigen and antibody or two complementary nucleic acids, such that specific binding (covalent or non-covalent) can occur.

[0414] FIGS. 26A and 26B depict antibody **930a** and FIGS. 26C and 26D depict nucleic acid probe **930b**, although any suitable affinity molecule could be used, including a nucleic acid aptamer or other binding protein or molecule. The sample preparation reagents also may include detector agent **940a**, **940b**, such as, for example, an antibody **960a** conjugated to signaling agent **950a** (FIG. 26A) or labeled nucleic acid probe **960b** bound to signaling agent **950b** (FIG. 26C). Detector agents **940** may each include signaling agent **950**, such as, for example, an oxidizing enzyme or other signaling enzyme, alkaline phosphatase (AP), methylene blue or other electrochemically responsive tag, or a fluorescent tag such as ethidium bromide, fluorescein, green fluorescent protein, or other fluorophore.

[0415] In embodiments that include detector agents **940**, the various reagents listed above may hybridize together to form sandwich complexes. Exemplary sandwich complexes **900a**, **900b** are illustrated in FIGS. 26B and 26D. Each sandwich complex may be formed of: (1) magnetic particle **920a**, **920b** having surface-bound affinity molecule **930a**, **930b**, (2) target analyte **910a**, **910b**, and (3) detector agent **940a**, **940b**. As such, sandwich complex **900a**, **900b** held over the working electrode of the sensor within the analysis channel of the cartridge due to the magnetic attraction between magnetic particle **920a**, **920b** and the magnetic field generators of the reader will include target analyte **910a**, **910b** and detector agent **940a**, **940b**. The exemplary sandwich complex **900a** of FIG. 26B uses antibodies as affinity molecules, and the target analyte is a protein or small molecule of interest. The exemplary sandwich complex **900b** of FIG. 26D uses nucleic acid probes designed to capture a particular sequence of nucleic acids. Also, competitor molecules may be used where the competitor molecules are each pre-bound to HRP, such as testosterone-HRP.

[0416] In various embodiments, signaling agent **950a**, **950b** is an oxidizing enzyme such as, for example, horseradish peroxidase (HRP) or soybean peroxidase (SBP). In such embodiments, the enzyme induces an oxidation reaction to occur at an electrochemical cell when in the presence of a particular chemical substrate, e.g., released from the substrate reservoir, such as TMB and/or OPD which may be in a substrate solution including acceptor molecules such as hydrogen peroxide. Thus, if the particular substrate flows over, or otherwise encounters, the oxidizing enzyme bound to a target analyte and magnetic particle at an electrochemical cell, an oxidation reaction occurs. In such embodiments, electrons are accordingly released from a working electrode of the electrochemical cell to replenish electrons stripped from the substrate by the oxidizing enzyme in a quantity proportional to the amount of target analyte present. The release or flow of electrons results in a current, which is detectable by the sensor in the analysis channel (e.g., at the working electrode), for example, as a change in current or a change in voltage. Advantageously, signaling agents that are not bound to magnetic particles are not magnetically held over the working electrode and instead wash further downstream (e.g., caused by release of fluid from the sample

preparation, wash, and/or substrate reservoirs) so as to not interfere with sensor readings. Accordingly, a signal indicative of the presence, absence, and/or quantity of one or more target analytes within a sample may be generated at the sensor in the cartridge and transmitted to the reader for further processing. In addition, or alternatively, surface bound target analytes with HRP may also produce signal that may be sensed by the sensor and transmitted to the reader for further processing.

[0417] Referring now to FIGS. 27A and 27B, the sample preparation reagents may include a population of magnetic particles **1020**, each having affinity molecule **1030** bound to its surface. Competitive binding agent **1040** and a sample containing target analyte **1010** may be added to the sample preparation reagents, e.g., by mixing the collected sample (and, optionally, one or more reagent balls) with fluid in the sample preparation reservoir. Competitive binding agent **1040** may include pre-bound target analyte **1070**, which comes pre-bound to signaling agent **1050**, for example, any of the signaling agents described above. The pre-bound target analyte **1070** may be indirectly bound to the signaling agent **1050**, for example, via an antibody, a nucleic acid probe, a nucleic acid aptamer, or other affinity molecule **1060**. Unbound target analyte **1010** from a sample and competitive binding agent **1040** may compete with each other to bind to affinity molecules **1030** on magnetic particles **1020**. The amount of competitive binding agent **1040** and signaling agent **1050** that successfully binds to magnetic particles **1020** is inversely proportional to the amount of target analyte **1010** present in a sample. In embodiments where signaling agent **1050** of competitive binding agent **1040** is an oxidizing enzyme, an oxidation reaction occurs if a particular substrate (e.g., from the substrate reservoir) flows over, or otherwise encounters, magnetic particles **1020** bound to competitive binding agents **1040** at the sensor within the analysis channel of the cartridge. Electrons are accordingly released from a working electrode of the sensor to replenish electrons stripped from the substrate by the oxidizing enzyme in a quantity inversely proportional to the amount of target analyte present in the sample. The release or flow of electrons results in a current, which is detectable by an electrode coupled to electrical circuitry with a current-to-voltage topology. Accordingly, a signal indicative of the presence, absence, and/or quantity of one or more target analytes within a sample may be generated at the sensor in the cartridge and transmitted to the reader for further processing.

[0418] Referring now to FIGS. 28A through 28H, a process for detecting the presence, absence, and/or quantity of a target analyte(s) within a sample in a cartridge is described. As shown in FIG. 28A, a sample having a plurality of sample target analytes **1110** may be collected with sample collection device **1100**. Sample collection device **1100** may be constructed in the manner described above with respect to any of the sample collection devices. Each sample target analyte **1110** may be bound to sample binding molecule **1115**. Sample target analyte **1110** may be any type of analyte described herein such as 25-hydroxy vitamin D3 or 25-hydroxy vitamin D2 or 1 α ,25-dihydroxyvitamin D2 or 1 α ,25-dihydroxyvitamin D. Sample binding molecule **1115** of the plurality of sample binding molecules may be any type of binding molecule such as the naturally occurring vitamin D binding protein also known as gc-component (group-specific component).

[0419] One or more reagent balls may be provided for mixing with the collected sample. The reagent ball(s) may be stored within the cartridge, e.g., within a shuttle stored in the cartridge as described above. Referring to FIG. 28B, reagent ball **1116** may include a plurality of competitor molecules **1170**, a plurality of labels **1175**, and a plurality of competitor binding molecules **1180**. Each competitor molecule **1170** may be pre-bound to competitor binding molecule **1180** of the plurality of competitor binding molecules. Each competitor molecule **1170** may possess label **1175** of the plurality of labels. Competitor molecule **1170** may be any type of competitor molecule such as a 25-hydroxy vitamin D2 or 25-hydroxy vitamin D3 possessing a label. Label **1175** may be any type of label such as biotin, and competitor binding molecule **1180** may be any type of competitor binding molecule such as vitamin D binding protein. Each label **1175** is configured to bind to signaling agent **1150**, e.g., via attachment or affinity molecule **1160**. In addition, or alternatively,

label **1175** acts as signaling agent **1150**. Reagent ball **1117** may include a plurality of solid particles **1120**, a plurality of affinity molecules **1130**, a plurality of detector agents **1140** which may each have signaling agent **1150** and other affinity molecule **1160**, and a plurality of de-binding agents **1190**. Reagent balls **1116**, **1117** may be constructed in the manner, and used in a cartridge as, described above with respect to any reagent ball described herein including reagent balls **375**, **375'**, **375''**, **375'''**.

[0420] The plurality of solid particles **1120** may comprise magnetically responsive material such as the magnetic particles, as described herein, or non-magnetically responsive material such as gold nanoparticles. Preferably, each solid particle **1120** is bound to affinity molecule **1130**, as described herein. Affinity molecule **1130** may be any affinity molecule described herein and preferably has an affinity to bind to sample target analyte **1110** and/or competitor molecule **1170**. Detector agent **1140** having signaling agent **1150** and affinity molecule **1160** may be similar to the respective agent/molecule described herein. For example, signaling agent **1150** may be HRP and affinity molecule **1160** may be streptavidin. Each de-binding agent **1190** of the plurality of depending agents is configured to de-bind competitor molecule **1170** from competitor binding molecule **1180** and, in some embodiments, to then bind to competitor binding molecule **1180**, and/or to de-bind sample target analyte **1110** from sample binding molecule **1115** and, in some embodiments, to then bind the sample binding molecule **1115**.

[0421] The molecules may be lyophilized in multiple reagent balls configured to be used with a single cartridge or one reagent ball configured to be used with a single cartridge. The types of molecules may be distributed amongst the reagent balls in a desired manner, for example as shown in FIG. **28B**, or randomly. In addition, certain types of molecules may be stored in the fluid within the sample preparation reservoir while other types are stored within a reagent ball(s).

[0422] As shown in FIG. **28C**, all the molecules of reagent ball **1116** and reagent ball **1117** of FIG. **28B** may be lyophilized in a single reagent ball. Reagent ball **1118** may include a plurality of solid particles **1120'**, a plurality of affinity molecules **1130'**, a plurality of detector agents **1140'** (having a plurality of signaling agents **1150'** and a plurality of affinity molecules **1160'**), a plurality of competitor molecules **1170'**, a plurality of labels **1175'**, a plurality of competitor binding molecules **1180'**, and/or a plurality of de-binding agents **1190'**. Reagent ball **1118** may be constructed in the manner, and used in a cartridge as, described above with respect to any reagent ball described herein including reagent balls **375**, **375'**, **375''**, **375'''**.

[0423] Referring now to FIG. **28D**, a plurality of solid particles, a plurality of affinity molecules, a plurality of detector agents (having a plurality of signaling agents and a plurality of affinity molecules), a plurality of competitor molecules, a plurality of labels, a plurality of competitor binding molecules, and/or a plurality of de-binding agents may be mixed in fluid held in sample preparation reservoir **1119** with a sample collected with sample collection device **1100**. The sample may be introduced into sample preparation reservoir **1119** while on the distal portion of sample collection device **1100** or after being released from the distal portion of sample collection device **1100**, both of which are described above. The sample may have a plurality of sample target analytes that may each be pre-bound to sample binding molecules. The plurality of solid particles, plurality of affinity molecules, plurality of detector agents (having the plurality of signaling agents and the plurality of affinity molecules), plurality of competitor molecules, plurality of labels, plurality of competitor binding molecules, and/or plurality of de-binding agents may be pre-stored in the fluid within sample preparation reservoir **1119** or some or all of those molecules may be introduced into the fluid from a reagent ball(s), such as reagent balls **1116**, **1117**, **1118**. Sample preparation reservoir **1119** may be constructed in the same manner as any sample preparation reservoir described herein including sample preparation reservoirs **317**, **317'**, **317''**, **317'''**.

[0424] Referring now to FIG. **28E**, sample target analyte **1110** having sample binding molecule **1115** pre-bound thereto may further mix within sample preparation reservoir **1119**. While not necessary, mixing within sample preparation reservoir **1119** may be enhanced by actuation of a

sonicator element disposed adjacent sample preparation reservoir **1119**, as described above.

[0425] As shown in FIG. **28F**, de-binding agent **1190** may de-bind competitor molecule **1170** from pre-bound competitor binding molecule **1180** and then de-binding agent **1190** may bind to competitor binding molecule **1180** leaving competitor molecule **1170** having label **1175** unbound. Another de-binding agent **1190** may de-bind sample target analyte **1110** from sample binding molecule **1115** and then the other de-binding agent **1190** may bind to sample binding molecule **1190** leaving sample target analyte **1110** unbound.

[0426] Referring to FIG. **28G**, label **1175** of de-bound competitor molecule **1170** may be configured to bind to signaling agent **1150** (e.g., via bonding with affinity molecule **1160**).

[0427] De-bound competitor molecule **1170** may be configured to bind to affinity molecule **1130** of the plurality of affinity molecules which may be pre-bound to solid particle **1120** as shown in FIG. **28H**. In this manner, solid particle **1120** may be indirectly bound to competitor molecule **1170** indirectly bound to signaling agent **1150**. A sandwich complex may be formed of solid particle **1120**, affinity molecule **1130**, competitor molecule **1170**, label **1175**, affinity molecule **1160**, and/or signaling agent **1150**. Illustratively, solid particle **1120** is bound to affinity molecule **1130** which is bound to competitor molecule **1170** which is bound to label **1175** which is bound to affinity molecule **1160** which is bound to signaling agent **1150**.

[0428] Sample target analyte **1110** and competitor molecule **1170** may compete with each other to bind to affinity molecules **1130** on solid particles **1120**. The amount of competitor molecule **1170** and signaling agent **1150** that successfully binds to solid particles **1120** is inversely proportional to the amount of unbound target analyte **1110** present in a sample. In embodiments where signaling agent **1150** of competitive binding agent **1140** is an oxidizing enzyme, an oxidation reaction occurs if a particular substrate (e.g., from the substrate reservoir) flows over, or otherwise encounters, solid particles **1120** bound to competitor molecules **1140** at the sensor within the analysis channel of the cartridge. Electrons are accordingly released from a working electrode of the sensor to replenish electrons stripped from the substrate by the oxidizing enzyme in a quantity inversely proportional to the amount of target analyte present in the sample. For example, as shown in the graph of FIG. **29A** comparing electrochemical reading (microamps) versus concentration (ng/ml), the higher the electrochemical reading, the lower the concentration of sample target analytes. The release or flow of electrons results in a current, which is detectable by an electrode coupled to circuitry, for example, as a change in current or a change in voltage. Accordingly, a signal indicative of the presence, absence, and/or quantity of one or more target analytes within a sample may be generated at the sensor in the cartridge and transmitted to the reader for further processing. FIG. **29B** shows a graph comparing electrochemical reading (microamps) versus concentration (ng/mL) when competitor binding molecule is not pre-bound for vitamin D quantification analysis. As FIG. **29B** illustrates, the electrochemical reading does not show a relationship with the concentration of vitamin D.

[0429] As will be readily apparent to one skilled in the art, while one type of molecule may be shown as present in an illustrated reaction, for example, one sample target analyte **1110** in FIGS. **28D** through **28H**, a plurality of the types of molecules may be present. In addition, all the types of molecules shown in FIGS. **28A** through **28H**, wherein like reactions are identified by like-primed reference numbers, need not be included in a reaction. For example, referring to FIGS. **30A** through **30H**, the reactions are similar to FIGS. **28A** through **28H** except affinity molecule **1130"** is not bound to a solid particle and the plurality of solid particles are not needed.

[0430] Referring now to FIGS. **31A** through **31J**, a process for detecting the presence, absence, and/or quantity of a target analyte(s) within a sample in a cartridge is described. As shown in FIG. **31A**, a sample having a plurality of sample target analytes **1210** may be collected with sample collection device **1200**. Sample collection device **1200** may be constructed in the manner described above with respect to any of the sample collection devices. Sample target analyte **1210** may be any type of analyte described herein.

[0431] One or more reagent balls may be provided for mixing with the collected sample. The reagent ball(s) may be stored within the cartridge, e.g., within a shuttle stored in the cartridge as described above. Referring to FIG. 31B, reagent ball **1215** may include a plurality of solid particles **1120**, a plurality of affinity molecules **1230**, a plurality of detector agents **1240** which may each include signaling agent **1250** and affinity molecule **1260**, a plurality of control targets **1270**, and/or a plurality of control detector agents **1275** which may each include control signaling agent **1285** and control affinity molecule **1280**. Reagent ball **1215** may be constructed in the manner, and used in a cartridge as, described above with respect to any reagent ball described herein including reagent balls **375**, **375'**, **375''**, **375'''**.

[0432] The plurality of solid particles **1220** may comprise magnetically responsive material such as the magnetic particles, as described herein, or non-magnetically responsive material such as gold nanoparticles. The plurality of solid particles **1220** may be similar to the magnetic particles described above. Preferably, each solid particle **1220** is pre-bound to affinity molecule **1230**, as described herein. Affinity molecule **1230** may be any affinity molecule described herein and preferably has an affinity to bind to sample target analyte **1210**. Detector agent **1240** having signaling agent **1250** and affinity molecule **1260** may be similar to the respective agent/molecule described herein. For example, signaling agent **1250** may be HRP and affinity molecule **1260** may be streptavidin. The plurality of control targets **1270** may be configured to bind to affinity molecules pre-bound to the surface of the sensor of the cartridge, for example, affinity molecules pre-bound to the positive control working electrode of the sensor (e.g., see FIGS. 7D, 7E, 7F) and/or the working electrode of the sensor (e.g., see FIG. 7D). Control targets **1270** may be a protein. Control affinity molecule **1280** may be any affinity molecule described herein and preferably has an affinity to bind to control target **1270**. Control signaling agent **1285** may be similar to the signaling agents described herein, including signal agent **1250**.

[0433] The molecules may be lyophilized in multiple reagent balls configured to be used with a single cartridge or one reagent ball configured to be used with a single cartridge. The types of molecules may be distributed amongst the reagent balls in a desired manner or randomly. In addition, certain types of molecules may be stored in the fluid within the sample preparation reservoir while other types are stored within a reagent ball(s).

[0434] Methods encapsulate materials are known in the art but have not, to the best of Applicants' knowledge, been utilized to encapsulate reaction reagents for use in a device, cartridge or system as described herein. As noted above, the reagent ball(s) may include one or more of magnetic particles, affinity molecules, connection molecules, signaling agents, competitor binding molecules, competitor molecules, labels, signaling agents, primers, nucleic acid probes, and/or polymerases, and other enzymes or components as described in further detail herein and the components, encapsulation material and dimensions may be the same or different from each other.

[0435] Referring now to FIG. 31C, a plurality of solid particles, a plurality of affinity molecules, a plurality of detector agents (having a plurality of signaling agents and a plurality of affinity molecules), a plurality of control targets, and/or a plurality of control detector agents (having a plurality of control signaling agents and a plurality of control affinity molecules) may be mixed in fluid held in sample preparation reservoir **1290** with a sample collected with sample collection device **1200**. The sample may be introduced into sample preparation reservoir **1290** while on the distal portion of sample collection device **1200** or after being released from the distal portion of sample collection device **1200**, both of which are described above. The sample may have a plurality of sample target analytes. The plurality of solid particles, the plurality of affinity molecules, the plurality of detector agents (having the plurality of signaling agents and the plurality of affinity molecules), the plurality of control targets, and/or the plurality of control detector agents (having the plurality of control signaling agents and the plurality of control affinity molecules) may be pre-stored in the fluid within sample preparation reservoir **1290** or some or all of those molecules may be introduced into the fluid from a reagent ball(s), such as reagent ball **1215**.

Sample preparation reservoir **1290** may be constructed in the same manner as any sample preparation reservoir described herein including sample preparation reservoirs **317**, **317'**, **317''**, **317'''**.

[0436] Referring now to FIG. **31D**, sample target analyte **1210** may further mix within sample preparation reservoir **1290**. While not necessary, mixing within sample preparation reservoir **1290** may be enhanced by actuation of a sonicator element disposed adjacent sample preparation reservoir **1290**, as described above. Affinity molecule **1230** (pre-bound to solid particle **1220**) may bind with sample target analyte **1210** and detector agent **1240** may bind to sample target analyte **1210**. For example, affinity molecule **1260** (pre-bound to signaling agent **1250**) may bind to sample target analyte **1210**. A sandwich complex may be formed of solid particle **1220**, affinity molecule **1230**, target analyte **1210**, affinity molecule **1260**, and/or signaling agent **1250**. Illustratively, solid particle **1220** is bound to affinity molecule **1230** which is bound to target analyte **1210** which is bound to affinity molecule **1260** which is bound to signaling agent **1250**. Control detector agent **1275** may bind to control target **1270**. For example, control affinity molecule **1280** (pre-bound to signaling agent **1285**) may bind to control target **1270**. A partial sandwich complex may be formed of target control **1270**, control affinity molecule **1280**, and/or control signaling agent **1285**. Illustratively, control target **1270** is bound to control affinity molecule **1280** which is bound to control signaling agent **1285**.

[0437] Referring now to FIG. **31E**, a surface of a sensor for use in a cartridge device described herein is shown. Sensor surface **1292** may include a plurality of affinity molecules **1294** pre-bound to sensor surface **1292** within the cartridge. Surface affinity molecule **1294** may be any affinity molecule described herein and preferably has an affinity to bind to control target **1270**. Sensor surface **1292** is preferably positioned within the analysis channel of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. **31E** shows sensor surface **1292** prior to exposure to fluid from the reservoir(s). Sensor surface **1292** may be used on any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''**. Sensor surface **1292** may be used for a working electrode of the sensor such as working electrode **340''** and/or may be used for a positive control working electrode of the sensor such as positive control working electrodes **376**, **376'**, **376''**.

[0438] Referring now to FIG. **31F**, another surface of a sensor for use in a cartridge device described herein is shown. Sensor surface **1296** may have a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol for added stability. Additionally or alternatively, sensor surface **1296** may be backfilled with mercaptoundecanoic acid, mercaptohexanol, or other suitable backfiller. Sensor surface **1296** is preferably positioned within the analysis channel of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. **31F** shows sensor surface **1296** prior to exposure of fluid from the reservoir(s). Sensor surface **1296** may be used on any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''**. Sensor surface **1296** may be used for a working electrode of the sensor such as working electrodes **340**, **340'**, **340''**, **340'''**. Sensor surface **1296** is configured to be exposed to magnetic fields from magnetic field generator **1298**, e.g., when the cartridge is inserted in the reader. For example, magnetic field generator **1298** may be similar to first magnetic generator **406** and second magnetic generator **407** of reader **400** described above.

[0439] Referring now to FIG. **31G**, a surface of a sensor for use in a cartridge device described herein is shown. Sensor surface **1293** has a conductive base preferably made of metal such as gold. Sensor surface **1293** has been modified with self-assembled monolayer **1295** for added stability. Illustratively, self-assembled monolayer **1295** is made from hexaethylene glycol dithiol (HEGT) laid on a gold sensor base, although other self-assembled monolayer molecules may be used such as thiolated ethylene glycol and/or another dithiol. Sensor surface **1293** has been further modified as self-assembled monolayer **1295** has been backfilled with backfiller **1297**. Illustratively,

backfiller **1297** is mercaptoundecanoic acid (MUDA) although other suitable backfillers may be used such as mercaptohexanol. Sensor surface **1293** has been further modified with a plurality of affinity molecules pre-bound to self-assembled monolayer **1295**. Illustratively, affinity molecule **1299** is pre-bound to backfiller **1297** which is pre-bound to self-assembled monolayer **1295** which is pre-bound to the metallic base of sensor surface **1293**. Affinity molecule **1299** may be any affinity molecule described herein and preferably has an affinity to bind to a control target(s) (e.g., control target **1270**) or a target analyte(s) (e.g., target analyte **1210**). Sensor surface **1293** is preferably positioned within the analysis channel of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. **31G** shows sensor surface **1293** prior to exposure to fluid from the reservoir(s). Sensor surface **1293** may be used on any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''**. Sensor surface **1293** may be used for a reference electrode of the sensor(s) such as reference electrodes **339**, **339'**, **339''**, **339'''**, **339''''**. Additionally or alternatively, sensor surface **1293** may be used for a working electrode of the sensor(s) such as working electrodes **340**, **340'**, **340''**, **340'''**, **340''''**. Additionally or alternatively, sensor surface **1293** may be used for a counter electrode of the sensor(s) such as counter electrodes **341**, **341'**, **341''**, **341'''**, **341''''**. Additionally or alternatively, sensor surface **1293** may be used for a negative control working electrode (also referred to herein as a background working electrode) of the sensor(s) such as negative control working electrodes **342**, **342'**, **342''**, **342'''**, **342''''**. Additionally or alternatively, sensor surface **1293** may be used for a positive control working electrode of the sensor(s) such as positive control working electrodes **376**, **376'**, **376''**.

[0440] Referring now to FIG. **31H**, sensor surface **1292** is shown after exposure to reagents, e.g., from fluid flowing into the analysis channel from the sample preparation reservoir. As shown in FIG. **31H**, partial sandwich complexes of control molecules may bind to surface affinity molecules **1294** to complete the sandwich complexes. For example, surface affinity molecule **1294** may bind to control target **1270** which is bound to control affinity molecule **1280** which is bound to control signaling agent **1285**. A chemical reaction may occur when the sandwich complexes are exposed to a substrate, e.g., from substrate reservoir, such that the sensor may detect electrical signals resulting from chemical reactions over the sensor. For example, the mixed fluid from the sample preparation reservoir may be introduced into the analysis channel such that control signaling agents **1285** directly or indirectly bound to control targets **1270** from the mixed fluid from the sample preparation reservoir localize over sensor surface **1292** by binding with pre-bound surface affinity molecules **1294**. The chemical reactions may occur when fluid from the substrate reservoir reacts with particles from the mixed fluid from the sample preparation reservoir localized over the sensor. For example, substrate solution having a substrate may be introduced from the substrate reservoir and the sensor having sensor surface **1292** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS, other electron mediator) and the signaling agents (e.g., HRP, SBP, redox dye) localized over the sensor. The reactions may cause electrons to be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules of the substrate solution) thereby generating electrical signals detectable by the sensor. If sensor surface **1292** is on a working electrode, such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. If sensor surface **1292** is on a positive control working electrode, such detected electrical signals may be used for error detection. For example, if a parameter(s), e.g., voltage, current, of the detected electrical signals is not within a predetermined range(s), there may be an error and the test may be rejected. If the parameter(s) is within the predetermined range, electrical signals detected by the working electrode of the sensor may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signals from the positive control working electrode and/or the working electrode may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300**

and reader device **400**.

[0441] Referring now to FIG. **31I**, sensor surface **1296** is shown after exposure to reagents, e.g., from fluid flowing into the analysis channel from the sample preparation reservoir. It should be understood that sensor surface **1296** and sensor surface **1292** may be exposed to the sample preparation reservoir fluid at substantially the same time, e.g., when sensor surface **1296** corresponds to working electrode **340'** or working electrode **340''** and sensor surface **1292** corresponds to positive control working electrode **376'** of sensor **338'''** or positive control working electrode **376''** of sensor **338''''**. Similarly, sensor surface **1296** and sensor surface **1292** may be exposed to the substrate reservoir fluid at substantially the same time, thereby causing reactions to occur between the substrate and reagents (e.g., signaling agents) at substantially the same time. A chemical reaction may occur when the sandwich complexes are exposed to a substrate, e.g., from substrate reservoir, such that the sensor may detect electrical signals resulting from chemical reactions over the sensor. For example, the mixed fluid from the sample preparation reservoir may be introduced into the analysis channel such that signaling agents directly or indirectly bound to target analytes from the mixed fluid from the sample preparation reservoir localize over sensor surface **1296** responsive to magnetic fields from magnetic field generator **1298** holding solid particles **1220** directly or indirectly bound to signaling agents **1250**. The chemical reactions may occur when fluid from the substrate reservoir reacts with particles from the mixed fluid from the sample preparation reservoir localized over the sensor. For example, a substrate solution having a substrate may be introduced from the substrate reservoir and the sensor having sensor surface **1296** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS) and the signaling agents (e.g., HRP, SBP, redox dye) localized over the sensor. The reactions may cause electrons to be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules in the substrate solution) thereby generating electrical signals detectable by the sensor. If sensor surface **1296** is on a working electrode, such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signal from the working electrode may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300** and reader device **400**.

[0442] Referring now to FIG. **31J**, sensor surface **1293** is shown after exposure to reagents, e.g., from fluid flowing into the analysis channel from the sample preparation reservoir. As shown in FIG. **31J**, partial sandwich complexes of control or target molecules may bind to affinity molecules **1299** to complete the sandwich complexes. For example, affinity molecule **1299** may bind to target **1212** (which may correspond to target analyte **1210** or control target **1270**) which is bound to affinity molecule **1214** (which may correspond to affinity molecule **1260** or control affinity molecule **1280**) which is bound to signaling agent **1216** (which may correspond to signaling agent **1250** or control signaling agent **1285**). A chemical reaction may occur when the sandwich complexes are exposed to a substrate, e.g., from substrate reservoir, such that the sensor may detect electrical signals resulting from chemical reactions over the sensor. For example, the mixed fluid from the sample preparation reservoir may be introduced into the analysis channel such that signaling agents **1216** directly or indirectly bound to targets **1212** from the mixed fluid from the sample preparation reservoir localize over sensor surface **1293** by binding with pre-bound affinity molecules **1299**. The chemical reactions may occur when fluid from the substrate reservoir reacts with particles from the mixed fluid from the sample preparation reservoir localized over the sensor. For example, substrate solution having a substrate may be introduced from the substrate reservoir and the sensor having sensor surface **1293** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS) and the signaling agents (e.g., HRP, SBP, redox dye) localized over the sensor. The reactions may cause electrons to be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules of the substrate solution) thereby generating electrical signals detectable by the sensor. If sensor surface

1293 is on a working electrode, such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. If sensor surface **1293** is on a positive control working electrode, such detected electrical signals may be used for error detection. For example, if a parameter(s), e.g., voltage, current, of the detected electrical signals is not within a predetermined range(s), there may be an error and the test may be rejected. If the parameter(s) is within the predetermined range, electrical signals detected by the working electrode of the sensor may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signals from the positive control working electrode and/or the working electrode may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300** and reader device **400**.

[0443] As will be readily apparent to one skilled in the art, while one type of molecule may be shown as present in an illustrated reaction, for example, one sample target analyte **1210** in FIGS. **31C** and **31D**, a plurality of the types of molecules may be present. In addition, all the types of molecules shown in FIGS. **31A** through **31J** need not be included in a reaction.

[0444] Referring now to FIGS. **32A** through **32N**, a process for detecting the presence, absence, and/or quantity of a target analyte(s) within a sample in a cartridge is generally described. The process may involve amplification such as isothermal amplification as shown in general in FIG. **32D**. As used herein, the term “isothermal amplification” refers to a method of amplifying a nucleic acid, e.g. DNA or RNA, wherein the temperature remains constant. In one aspect, the reaction system is in contact with an outside source having a temperature differential but the temperature change, if any, occurs slowly enough to allow the system to continuously adjust the temperature. Non-limiting exemplary of isothermal amplification include rolling circle amplification (RCA), loop-mediated isothermal amplification (LAMP), strand displacement amplification (SDA), recombinase polymerase amplification (RPA), helicase dependent amplification (HDA), polymerase spiral reaction (PSR), and nicking enzyme amplification reaction (NEAR). These and further non-limiting exemplary methods are disclosed in Zhao et al. (2015) Chem. Rev. 115 (22): 12491-12545; and Dong et al. (2015) Scientific Reports 5:12723; Yan et al. (2014) Mol. BioSyst. 10:970-1003.

[0445] Depending on the isothermal amplification method selected, the reagent ball(s) may include one or more amplification enzymes, e.g., reverse transcriptases, endonucleases (e.g. a nicking endonuclease, restriction endonuclease, flap endonuclease, or endonuclease III), polymerases (e.g. a strand displacing polymerase), method specific primers or probes (e.g. a padlock probe, linked probe, or LAMP primers), helicases, recombinases, and/or single-stranded DNA binding proteins. The requisite combination of reagents required to carry out each method of isothermal amplification is appreciated by an ordinary skilled artisan. Non-limiting exemplary combinations of reagents required to perform the various methods of isothermal amplification are described in Chinese Patent Application No. 104232622; U.S. Pat. Nos. 5,223,414; 6,410,278; 5,455,166; 5,470,723; 5,714,320; 6,235,502; and 7,282,328.

[0446] As used herein, the term “dNTPs” intend the nucleotides that are the building blocks for DNA. They principally include (d)ATP, (d)GTP, (d)CTP, (d)TTP and (d)UTP and are necessary for nucleic acid amplification reactions.

[0447] RT specific primers intends a primer, typically unlabelled, that is designed to hybridize with an RNA molecule containing the target region and allowing for a reverse transcriptase to polymerize a cDNA strand from 3' end of the primer. The cDNA strand will is then used as a template for isothermal amplification. The RT specific primer may have design characteristics that differ from the primers that are optimized for amplification of DNA within the isothermal amplification reaction that proceeds subsequently to the reverse transcription of the target RNA. Applicant has determined that use of RT specific primers will increase amplification efficiency.

[0448] As shown in FIG. **32A**, a sample having a plurality of sample target analytes **1310** and/or **1312** may be collected with sample collection device **1300**. Sample collection device **1300** may be

constructed in the manner described above with respect to any of the sample collection devices. Sample target analyte **1310** may be any type of analyte described herein and illustratively is RNA. Sample target analyte **1312** may be any type of analyte described herein and illustratively is DNA. It should be understood that sample target analyte **1310** and/or **1312** may contain an upstream and downstream sequence region of the target region, but is understood to at least contain the target region to be amplified and detected. Sample target analyte **1310** and/or **1312** may be contained with a cell, microbe, or virus or could be cell-free. The sample preparation reservoir and/or the reagent ball(s) may include lysis agents to free the target analyte from the cell, microbe, or virus.

[0449] One or more reagent balls may be provided for mixing with the collected sample. The reagent ball(s) may be stored within the cartridge, e.g., within a shuttle stored in the cartridge as described above. Referring to FIG. 32B, reagent ball **1315** may include a plurality of solid particles **1320**, a plurality of affinity molecules **1322**, a plurality of detector agents **1324** which may each include signaling agent **1326** and affinity molecule **1328**, a plurality of control targets **1330**, a plurality of control detector agents **1332** which may each include control signaling agent **1334** and control affinity molecule **1336**, a plurality of solid particles **1338** pre-conjugated to primers **1340** (e.g., via affinity molecule **1342** pre-bound to capture element **1344** pre-bound to spacer **1346** pre-bound to primer **1340** or through a direct covalent bond), a plurality of detector agents **1348** which may each include signaling agent **1350** and affinity molecule **1352**, a plurality of internal control reverse primers **1354** which may each be pre-bound to signaling agent **1356**, e.g., each optionally via spacer **1358**, a plurality of reverse primers **1360** which may each be pre-bound to label **1362**, e.g., via spacer **1364**, a plurality of internal control forward primers **1366** which may each be pre-bound to capture element **1368**, e.g., via spacer **1370**, a plurality of polymerase **1372**, a plurality of reverse transcriptases (RT) **1374**, a plurality of dNTPs **1376**, a plurality of RT specific primers **1378**, a plurality of RNA template controls **1380**, a plurality of DNA template controls **1382**, a plurality of reverse primers **1384** which may each be pre-bound to signaling agent **1386**, e.g., via spacer **1388**, a plurality of forward primers **1390** which may each be pre-bound to capture element **1392**, e.g., via spacer **1394**, and/or a plurality of internal control reverse primers **1396** which may each be pre-bound to internal control label **1394**, e.g., via internal control spacer **1400**. Reagent ball **1315** may be constructed in the manner, and used in a cartridge as, described above with respect to any reagent ball described herein including reagent balls **375**, **375'**, **375''**, **375'''**.

[0450] The plurality of solid particles **1320** may comprise magnetically responsive material such as the magnetic particles, as described herein, or non-magnetically responsive material such as gold nanoparticles. The plurality of solid particles **1320** may be similar to the magnetic particles described above. Preferably, each solid particle **1320** is pre-bound to affinity molecule **1322**, as described herein. Affinity molecule **1322** may be any affinity molecule described herein and preferably has an affinity to bind to a sample target analyte(s). Detector agent **1324** having signaling agent **1326** and affinity molecule **1328** may be similar to the respective agent/molecule described herein. For example, signaling agent **1326** may be HRP and affinity molecule **1328** may be streptavidin. The plurality of control targets **1330** may be configured to bind to affinity molecules pre-bound to the surface of the sensor of the cartridge, for example, affinity molecules pre-bound to the positive control working electrode of the sensor (e.g., see FIGS. 7D, 7E, 7F) and/or the working electrode of the sensor (e.g., see FIG. 7D). Control targets **1330** may be a protein. Control affinity molecule **1336** may be any affinity molecule described herein and preferably has an affinity to bind to control target **1330**. Control signaling agent **1334** may be similar to the signaling agents described herein, including signal agent **1326**. The plurality of solid particles **1338** may comprise magnetically responsive material such as the magnetic particles, as described herein, or non-magnetically responsive material such as gold nanoparticles. The plurality of solid particles **1338** may be similar to the magnetic particles described above. Preferably, each solid particle **1338** is pre-bound to affinity molecule **1342**, as described herein. Affinity molecule **1342** may be any affinity molecule described herein and preferably has an affinity to bind to

capture element **1344**. For example, affinity molecule **1342** may be pre-bound to capture element **1344**. Capture element **1344** contains a spacer **1346** and the primer **1340**. Detector agent **1348** having signaling agent **1350** and affinity molecule **1352** may be similar to the respective agent/molecule described herein. For example, signaling agent **1350** may be HRP and affinity molecule **1352** may be streptavidin. Affinity molecule **1352** may have an affinity to bind to a label such as label **1362** and/or internal control label **1398**. Internal control reverse primers **1354** may be configured to bind to affinity molecules pre-bound to the surface of the sensor of the cartridge, for example, affinity molecules pre-bound to the positive control working electrode of the sensor (e.g., see FIGS. 7D, 7E, 7F) and/or the working electrode of the sensor (e.g., see FIG. 7D). Internal control reverse primers **1354** may bind to affinity molecules pre-bound to the surface of the sensor via capture element **1368** bound to internal control forward primer **1366** (e.g., via spacer **1370**). Signaling agent **1356** may be any signaling agent described herein such as HRP. Reverse primers **1360** may be configured to bind to the target analyte(s) and to bind directly or indirectly to a solid particle to localize at the surface of the sensor of the cartridge, for example, via magnetic fields at the working electrode of the sensor. Label **1362** may be any type of label such as biotin and is configured to bind with an affinity molecule, e.g., affinity molecule **1352** bound to signaling agent **1350**. Internal control forward primers **1366** may be configured to bind to affinity molecules pre-bound to the surface of the sensor of the cartridge, for example, affinity molecules pre-bound to the positive control working electrode of the sensor (e.g., see FIGS. 7D, 7E, 7F) and/or the working electrode of the sensor (e.g., see FIG. 7D) via, for example, capture element **1368**. The plurality of polymerase **1372**, the plurality of reverse transcriptases (RT) **1374**, the plurality of dNTPs **1376**, the plurality of RT specific primers **1378**, the plurality of RNA template controls **1380**, and/or the plurality of DNA template controls **1382** may be used to amplify the nucleic acid, e.g., RNA and/or DNA, via, for example, isothermal amplification. The plurality of RNA template controls **1380**, and/or the plurality of DNA template controls **1382** may be contained with a cell, microbe, or virus or could be free floating. The sample preparation reservoir and/or the reagent ball(s) may include lysis agents to free the template control(s) from the cell, microbe, or virus. Reverse primers **1384** may be configured to bind to a target analyte(s) and to bind directly or indirectly to a solid particle to localize at the surface of the sensor of the cartridge, for example, via magnetic fields at the working electrode of the sensor. Signaling agent **1386** may be any signaling agent described herein such as HRP. Forward primers **1390** may be configured to bind to a target analyte and to bind directly or indirectly to a solid particle to localize at the surface of the sensor of the cartridge, for example, via magnetic fields at the working electrode of the sensor via, for example, capture element **1392**. Internal control reverse primers **1396** may be configured to bind to affinity molecules pre-bound to the surface of the sensor of the cartridge, for example, affinity molecules pre-bound to the positive control working electrode of the sensor (e.g., see FIGS. 7D, 7E, 7F) and/or the working electrode of the sensor (e.g., see FIG. 7D). Internal control reverse primers **1396** may bind to affinity molecules pre-bound to the surface of the sensor via capture element **1368** bound to internal control forward primer **1366** (e.g., via spacer **1370**). Label **1398** may be any type of label such as biotin and may be configured to bind with an affinity molecule, e.g., affinity molecule **1352** bound to signaling agent **1350**.

[0451] The molecules may be lyophilized in multiple reagent balls configured to be used with a single cartridge or one reagent ball configured to be used with a single cartridge. The types of molecules may be distributed amongst the reagent balls in a desired manner or randomly. In addition, certain types of molecules may be stored in the fluid within the sample preparation reservoir while other types are stored within a reagent ball(s).

[0452] Referring now to FIG. 32C, any combination of the types of particles shown in reagent ball **1315** may be mixed in fluid held in sample preparation reservoir **1410** with a sample collected with sample collection device **1300**. The sample may be introduced into sample preparation reservoir **1410** while on the distal portion of sample collection device **1300** or after being released from the

distal portion of sample collection device **1300**, both of which are described above. The sample may have a plurality of sample target analytes. The types of particles shown in reagent ball **1315** may be pre-stored in the fluid within sample preparation reservoir **1410** or some or all of those molecules may be introduced into the fluid from a reagent ball(s), such as reagent ball **1315**. Sample preparation reservoir **1410** may be constructed in the same manner as any sample preparation reservoir described herein including sample preparation reservoirs **317**, **317'**, **317''**, **317'''**.

[0453] Sample target analyte **1310** and/or **1312** may further mix within sample preparation reservoir **1410**. While not necessary, mixing within sample preparation reservoir **1410** may be enhanced by actuation of a sonicator element disposed adjacent sample preparation reservoir **1410**, as described above. The mixing within sample preparation reservoir **1410** may amplify nucleic acids, e.g., DNA and/or RNA, within the mixture of fluid within sample preparation reservoir which includes the sample and optionally reagent ball(s). The sample target analyte may be a target nucleic acid, e.g., a RNA for the detection and/or diagnosis of an influenza or HIV infection, and DNA for DNA viruses or bacteria or RNA for the detection of 16s rRNA for bacteria. The amplification proceeds by action of a polymerase beginning at 3' end of a bound primer (forward or reverse) wherein the polymerase moves along the template strands comprising the target nucleic acid and incorporates nucleotides (dNTP) to synthesize a complementary strand. Since the primer may have a labelled end, these labelled ends are incorporated into the resulting amplicon. In the case of a capture element labelled end, this allows the amplicon to bind to an affinity molecule bound to a solid particle or to a pre-bound surface affinity molecule on a sensor surface while the signaling agent on the other end of the amplicon can be conjugated directly or indirectly to the signaling agent. As used herein, "conjugated" intends a covalent or non-covalently bond, e.g., a biotin-Streptavidin binding complex. And these binding events can happen simultaneously with the amplification. In a further aspect, a separate amplification reaction is simultaneously occurring and lysis of viral particles, bacteria, other microbes, and cells may be occurring freeing the cellular contents, e.g., nucleic acid. In addition to the labeled primers, unlabeled primers designed to amplify at least the same target region but perhaps some sequence flanking the target sequence may be provided to facilitate the amplification reaction simultaneously with the labelled primers. The presence of unlabeled primers can make amplification more efficient since labelled primers may be more sterically hindered as they bind to other elements e.g. solid particles or signaling agents.

[0454] This polymerization of the complement strand in the amplification can be facilitated by the presence of single stranded binding proteins known to those of ordinary skill in the art such as but not limited to SSB from *E. coli*. Strand displacing polymerases can be used to allow for the polymerization of the complement strand to occur without thermally denaturing the double strand prior to synthesis of complementary strands. Since the primer may have a labelled end, these labelled ends are incorporated into the resulting amplicon. In the case of a capture element labelled end, this allow the amplicon to bind to an affinity molecule bound to a solid particle or to a pre-bound surface affinity molecule on a sensor surface while the signaling agent on the other end of the amplicon can be conjugated directly or indirectly to the signaling agent. As used herein, "conjugated" intends a covalent or non-covalently bond, e.g., a biotin-Streptavidin binding complex. And these binding events can happen simultaneously with the amplification.

[0455] The amplification may be an isothermal reaction such as isothermal amplification. FIG. **32D** shows an exemplary process for isothermal amplification of nucleic acids. The amplification of nucleic acids within sample preparation reservoir **1410** generates amplicons as shown in FIG. **32E**. Actuation of the sonication element to mix the contents of the sample preparation reservoir may promote the amplification. The amplicons may form sandwich complexes which are labeled amplicon complexes **1422** comprising a capture label and a signaling label, or labeled amplicon complexes **1434**. For example, a forward primer and reverse primer amplicon may be bound to a signaling agent and/or bound to a solid particle as a result of the mixing. Amplicon complexes **1422**

and **1434** may each include a solid particle configured to be magnetically held over a working electrode in a sensor such that a signaling agent of the complex can react with a substrate from a substrate solution. The amplicons may form partial sandwich complexes which may be labeled partial amplicon complexes **1420**, labeled partial amplicon complexes **1423**, labeled partial amplicon complexes **1426**, and/or labeled partial amplicon complexes **1435**. For example, an internal control forward primer and an internal control reverse primer amplicon may be bound to a signaling agent as a result of the mixing. Partial amplicon complexes **1420**, **1423**, **1426**, and **1435** may be configured to bind to affinity molecules directly pre-bound to a metallic sensor surface or in-directly pre-bound to a metallic sensor surface (e.g., via a self-assembled monolayer which may be backfilled) of an electrode (e.g., working electrode and/or positive control working electrode) of a sensor.

[0456] Pre-conjugated primers (either covalently or through a linkage such as biotin-streptavidin) may be used wherein either the forward or the reverse is conjugated to a particle and the other side (forward if reverse was conjugated to particle and reverse if forward was conjugated to particle) is conjugated to a signaling agent either directly or indirectly and there may also be a population of unlabelled and unconjugated primers that facilitates the reaction to occur more efficiently.

[0457] Referring now to FIG. **32F**, a surface of a sensor for use in a cartridge device described herein is shown. Sensor surface **1412** may include a plurality of affinity molecules **1414** pre-bound to sensor surface **1412** within the cartridge. Surface affinity molecule **1414** may be any affinity molecule described herein and preferably has an affinity to bind to a partial amplicon complex. For example, surface affinity molecule **1414** may have an affinity to bind to an internal control capture element bound to an internal control forward primer and an internal control reverse primer amplicon bound to a signaling agent. The internal control reverse primer amplicon may be bound to a signaling agent, e.g., via a spacer, or may be bound to a label, e.g., via a spacer, which is bound to an affinity molecule bound to a signaling agent. Sensor surface **1412** is preferably positioned within the analysis channel of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. **32F** shows sensor surface **1412** prior to exposure to fluid from the reservoir(s). Sensor surface **1412** may be used on any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''**. Sensor surface **1412** may be used for a working electrode of the sensor such as working electrode **340''** and/or may be used for a positive control working electrode of the sensor such as positive control working electrodes **376**, **376'**, **376''**.

[0458] Referring now to FIG. **32G**, another surface of a sensor for use in a cartridge device described herein is shown. Sensor surface **1416** may have a self-assembled monolayer such as thiolated ethylene glycol and/or a dithiol such as hexaethylene glycol dithiol for added stability. Additionally or alternatively, sensor surface **1416** may be backfilled with mercaptoundecanoic acid, mercaptohexanol, or other suitable backfiller. Sensor surface **1416** is preferably positioned within the analysis channel of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. **32G** shows sensor surface **1416** prior to exposure of fluid from the reservoir(s). Sensor surface **1416** may be used on any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''**. Sensor surface **1416** may be used for a working electrode of the sensor such as working electrodes **340**, **340'**, **340''**, **340'''**. Sensor surface **1416** is configured to be exposed to magnetic fields from magnetic field generator **1418**, e.g., when the cartridge is inserted in the reader. For example, magnetic field generator **1418** may be similar to first magnetic generator **406** and second magnetic generator **407** of reader **400** described above.

[0459] Referring now to FIG. **32H**, a surface of a sensor for use in a cartridge device described herein is shown. Sensor surface **1413** has a conductive base preferably made of metal such as gold. Sensor surface **1413** has been modified with self-assembled monolayer **1415** for added stability. Illustratively, self-assembled monolayer **1415** is made from hexaethylene glycol dithiol (HEGT)

laid on a gold sensor base, although other self-assembled monolayer molecules may be used such as thiolated ethylene glycol and/or another dithiol. Sensor surface **1413** has been further modified as self-assembled monolayer **1415** has been backfilled with backfiller **1417**. Illustratively, backfiller **1417** is mercaptoundecanoic acid (MUDA) although other suitable backfillers may be used such as mercaptohexanol. Sensor surface **1413** has been further modified with a plurality of affinity molecules pre-bound to self-assembled monolayer **1415**. Illustratively, affinity molecule **1419** is pre-bound to backfiller **1417** which is pre-bound to self-assembled monolayer **1415** which is pre-bound to the metallic base of sensor surface **1413**. Affinity molecule **1419** may be any affinity molecule described herein and preferably has an affinity to bind to a partial target amplicon complex(es) or partial internal control amplicon complex(es). For example, affinity molecule **1419** may have an affinity to bind to an internal control capture element bound to an internal control forward primer and an internal control reverse primer amplicon bound to a signaling agent. The internal control reverse primer amplicon may be bound to a signaling agent, e.g., via a spacer, or may be bound to a label, e.g., via a spacer, which is bound to an affinity molecule bound to a signaling agent. As another example, affinity molecule **1419** may have an affinity to bind to a target capture element bound to a target forward primer and a target reverse primer amplicon bound to a signaling agent. The target reverse primer amplicon may be bound to a signaling agent, e.g., via a spacer, or may be bound to a label, e.g., via a spacer, which is bound to an affinity molecule bound to a signaling agent. Sensor surface **1413** is preferably positioned within the analysis channel of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. 32H shows sensor surface **1413** prior to exposure to fluid from the reservoir(s). Sensor surface **1413** may be used on any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''**. Sensor surface **1413** may be used for a reference electrode of the sensor(s) such as reference electrodes **339**, **339'**, **339''**, **339'''**, **339''''**. Additionally or alternatively, sensor surface **1413** may be used for a working electrode of the sensor(s) such as working electrodes **340**, **340'**, **340''**, **340'''**, **340''''**. Additionally or alternatively, sensor surface **1413** may be used for a counter electrode of the sensor(s) such as counter electrodes **341**, **341'**, **341''**, **341'''**, **341''''**. Additionally or alternatively, sensor surface **1413** may be used for a negative control working electrode (also referred to herein as a background working electrode) of the sensor(s) such as negative control working electrodes **342**, **342'**, **342''**, **342'''**, **342''''**. Additionally or alternatively, sensor surface **1413** may be used for a positive control working electrode of the sensor(s) such as positive control working electrodes **376**, **376'**, **376''**.

[0460] Referring now to FIG. 32I, sensor surface **1414** is shown after exposure to reagents, e.g., from fluid flowing into the analysis channel from the sample preparation reservoir. As shown in FIG. 32I, partial amplicon complexes **1420** of control amplicon molecules may bind to surface affinity molecules **1414** to complete the amplicon complexes. For example, surface affinity molecule **1414** may bind to an internal control capture element bound to an internal control forward primer amplicon and an internal control reverse primer amplicon bound to a signaling agent as shown in FIG. 32I. A chemical reaction may occur when the amplicon complexes are exposed to a substrate, e.g., from substrate reservoir, such that the sensor may detect electrical signals resulting from chemical reactions over the sensor. For example, the mixed fluid from the sample preparation reservoir may be introduced into the analysis channel such that the signaling agents directly or indirectly bound to the amplicon primers from the mixed fluid from the sample preparation reservoir localize over sensor surface **1412** by binding with pre-bound surface affinity molecules **1414**, e.g., via capture elements. The chemical reactions may occur when fluid from the substrate reservoir reacts with particles from the mixed fluid from the sample preparation reservoir localized over the sensor. For example, a substrate solution having a substrate may be introduced from the substrate reservoir and the sensor having sensor surface **1412** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS) and the signaling agents (e.g., HRP, SBP, redox dye) localized over the sensor. The reactions may cause electrons to

be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules from the substrate solution) thereby generating electrical signals detectable by the sensor. If sensor surface **1412** is on a working electrode, such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. If sensor surface **1412** is on a positive control working electrode, such detected electrical signals may be used for error detection. For example, if a parameter(s), e.g., voltage, current, of the detected electrical signals is not within a predetermined range(s), there may be an error and the test may be rejected. If the parameter(s) is within the predetermined range, electrical signals detected by the working electrode of the sensor may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signals from the positive control working electrode and/or the working electrode may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300** and reader device **400**.

[0461] Referring now to FIG. **32J**, sensor surface **1416** is shown after exposure to reagents, e.g., from fluid flowing into the analysis channel from the sample preparation reservoir. It should be understood that sensor surface **1416** and sensor surface **1412** or sensor surface **1424** may be exposed to the sample preparation reservoir fluid at substantially the same time, e.g., when sensor surface **1416** corresponds to working electrode **340'''** or working electrode **340''''** and sensor surface **1412** corresponds to positive control working electrode **376'** of sensor **338'** or positive control working electrode **376''** of sensor **338''''**. Similarly, sensor surface **1416** and sensor surface **1412** or sensor surface **1424** may be exposed to the substrate reservoir fluid at substantially the same time, thereby causing reactions to occur between the substrate and reagents (e.g., signaling agents) at substantially the same time.

[0462] As shown in FIG. **32J**, amplicon complexes **1422** of target amplicon molecules may localize over sensor surface **1416**. Target amplicon complexes **1422** may be formed of a solid particle bound to target amplicon primers bound to a signal agent. For example, the solid particle may be bound to an affinity molecule bound to a capture element bound to a target forward primer amplicon and a target reverse primer amplicon bound to the signaling agent as shown in FIG. **32J**. A chemical reaction may occur when the target amplicon complexes are exposed to a substrate, e.g., from substrate reservoir, such that the sensor may detect electrical signals resulting from chemical reactions over the sensor. For example, the mixed fluid from the sample preparation reservoir may be introduced into the analysis channel such that signaling agents directly or indirectly bound to target amplicons from the mixed fluid from the sample preparation reservoir localize over sensor surface **1416** responsive to magnetic fields from magnetic field generator **1418** holding the solid particles directly or indirectly bound to the signaling agents. The chemical reactions may occur when fluid from the substrate reservoir reacts with particles from the mixed fluid from the sample preparation reservoir localized over the sensor. For example, a substrate solution having a substrate may be introduced from the substrate reservoir and the sensor having sensor surface **1416** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD ferricyanide, PBS) and the signaling agents (e.g., HRP, SBP, redox dye) localized over the sensor. The reactions may cause electrons to be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules from the substrate solution) thereby generating electrical signals detectable by the sensor. If sensor surface **1416** is on a working electrode, such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signal from the working electrode may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300** and reader device **400**.

[0463] Referring now to FIG. **32K**, sensor surface **1413** is shown after exposure to reagents, e.g., from fluid flowing into the analysis channel from the sample preparation reservoir. As shown in FIG. **32K**, partial amplicon complexes **1423** (which may be similar to partial amplicon complexes

1420 as illustrated) of control amplicon molecules or target amplicon complexes may bind to affinity molecules **1419** to complete the amplicon complexes. For example, affinity molecule **1419** may bind to an internal control capture element bound to an internal control forward primer amplicon and an internal control reverse primer amplicon bound to a signaling agent. As another example, affinity molecule **1419** may bind to a capture element bound to a target forward primer amplicon and a target reverse primer amplicon bound to the signaling agent. A chemical reaction may occur when the amplicon complexes are exposed to a substrate, e.g., from substrate reservoir, such that the sensor may detect electrical signals resulting from chemical reactions over the sensor. For example, the mixed fluid from the sample preparation reservoir may be introduced into the analysis channel such that the signaling agents directly or indirectly bound to the amplicon primers from the mixed fluid from the sample preparation reservoir localize over sensor surface **1413** by binding with pre-bound affinity molecules **1419**, e.g., via capture elements. The chemical reactions may occur when molecules in the fluid from the substrate reservoir reacts with particles from the mixed fluid from the sample preparation reservoir localized over the sensor. For example, a substrate solution having a substrate or multiple substrates may be introduced from the substrate reservoir and the sensor having sensor surface **1413** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS) and the substrates specific signaling agent or signaling agents (e.g., HRP, SBP, redox dye) localized over the sensor. The reactions may cause electrons to be shifted to or from the substrate by the signaling agents thereby generating electrical signals detectable by the sensor. If sensor surface **1413** is on a working electrode, such detected electrical signals may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. If sensor surface **1413** is on a positive control working electrode, such detected electrical signals may be used for error detection. For example, if a parameter(s), e.g., voltage, current, of the detected electrical signals is not within a predetermined range(s), there may be an error and the test may be rejected. If the parameter(s) is within the predetermined range, electrical signals detected by the working electrode of the sensor may be used to generate the signal indicative of the presence, absence, and/or quantity of one or more analytes within the sample. The signals from the positive control working electrode and/or the working electrode may be transmitted to reader device **400**, e.g., via respective electrical connectors of cartridge device **300** and reader device **400**.

[0464] Referring now to FIG. **32L**, sensor surface **1424** is similar to sensor surface **1412** of FIG. **32I** and the control amplicon complexes are similar to the control amplicon complexes of FIG. **32I** except control amplicon complexes **1426** include a label bound to the internal control reverse primer amplicon which is bound to an affinity molecule bound to a signaling agent. For example, surface affinity molecule **1428** may bind to an internal control capture element bound to an internal control forward primer amplicon and an internal control reverse primer amplicon bound to a label bound to an affinity molecule bound to a signaling agent as shown in FIG. **32L**. The reaction and signal processing may occur as described above with respect to FIG. **32I**.

[0465] Referring now to FIG. **32M**, sensor surface **1430** is similar to sensor surface **1416** and magnetic field generator **1432** is similar to magnetic field generator **1418** of FIG. **32J** and the target amplicon complexes are similar to the target amplicon complexes of FIG. **32J** except target amplicon complexes **1434** include a label bound to the target reverse primer amplicon which is bound to an affinity molecule bound to a signaling agent. For example, a solid particle may be bound to an affinity molecule bound to a capture element bound to a target forward primer amplicon and a target reverse primer amplicon bound to a label bound to an affinity molecule bound to a signaling agent as shown in FIG. **32M**. The reaction and signal processing may occur as described above with respect to FIG. **32J**.

[0466] Referring now to FIG. **32N**, sensor surface **1436** is similar to sensor surface **1413**, self-assembled monolayer **1438** is similar to self-assembled monolayer **1415**, backfiller **1440** is similar to backfiller **1417**, and affinity molecule **1142** is similar to affinity molecule **1419** of FIG. **32K** and

the amplicon complexes are similar to the amplicon complexes of FIG. 32K except amplicon complexes **1435** (which may be control amplicon complexes or target amplicon complexes) include a label bound to the reverse primer amplicon which is bound to an affinity molecule bound to a signaling agent. For example, affinity molecule **1442** may bind to a capture element bound to a forward primer amplicon and a reverse primer amplicon bound to a label bound to an affinity molecule bound to a signaling agent as shown in FIG. 32N. The reaction and signal processing may occur as described above with respect to FIG. 32K.

[0467] As will be readily apparent to one skilled in the art, while one type of molecule may be shown as present in an illustrated reaction, for example, one solid particle **1320** in FIGS. 32B, a plurality of the types of molecules may be present. In addition, all the types of molecules shown in FIGS. 32A through 32N need not be included in a reaction.

[0468] As a further embodiment for multiplexing analytes, the reagents from the sample preparation reservoir include different types of signaling agents with different detectable potential for different types of target analytes. For example, the signaling agents can be a redox dye of differing redox potential for each of the target analytes. A capture moiety is retained on the other primer such that amplicons captured on surface-bound affinity molecules (e.g., at sensor surface **1412**, **1413**, **1424**, or **1436**) or capture via magnetic forces on magnetic particle (e.g., at sensor surface **1416** or **1430**). Signals are sensed by the sensor(s) in response to the substrate reacting with the signaling agents localized over the sensor as described above, however, the reactions will differ based on the potential of the signaling agents for the particular type of target analyte. A scanning technique such as cyclic voltammetry (CV), square wave voltammetry, or amperometry may be used to probe the localized sample over the sensor(s) for the presence and/or quantity of a target analyte generated in response to a target analyte (e.g., nucleic acid targets) being introduced in the sample and or any internal control reactions. Such techniques could be used to detect multiple target analyte types including multiple amplicon types per sensor to allow for multiplexing on each sensor or different sensors. For example, if there exist three different redox dyes that can be detected at different redox potentials during CV scanning or square wave voltammetry or linear sweep over the known potential range (e.g., 0.6 V to 0.5 V), then there can be three analytes measured per sensor. If there are three sensors, then nine targets could be analyzed within the analysis channel having three sensors. Instructions for executing the scanning technique may be stored in memory coupled to the processor of the reader such that the reader processes the electrical signals sensed by the sensor(s) of the cartridge that are transmitted to the reader, e.g., via respective electrical connectors. For example, the processor of the reader may direct one or more electrodes of the one or more sensors in the cartridge to apply a voltage sweep. The different types of signaling agents may respond to different voltages during the voltage sweep. The response for each type of signaling agent may be used to determine the presence, absence, and/or quantity of the corresponding target analyte.

[0469] Amplification and immunoassays can be proceed simultaneously such that targets of the nucleic acid variety (RNA, DNA) and target small molecules, proteins of interest, cell surface antigens or any other analyte of interest can be measured by combining various affinity and signal molecules and a combination of the above methods.

[0470] In another embodiment, assays using affinity agent-based binding to targets are utilized (immunoassays). A detection antibody labelled with a redox dye such as methylene blue, ferrocene, Nile blue, anthraquinone such that the analyte itself is labelled through the affinity agent with a redox dye specific to that analyte (such specificity conferred by the affinity agent). By labeling a set of differing target analytes with redox dyes, quantitative detection of multiple analytes simultaneously is possible through CV square wave voltammetry, or amperometry scanning at the sensor. Such signal detection may happen through directly measuring the electrical response of the redox dye at a certain potential on the working electrode, such as during a voltage sweep across the potential known to be the redox or oxidation potential of the dye itself, or by measuring the

electrical response of a substrate such as ferricyanide that interacts with the redox dye. Moreover, the substrate solution from the substrate reservoir may have multiple types of substrate molecules such as TMB for interacting with HRP, ferricyanide for interacting with methylene blue, such that multiple substrate molecules interact specifically with a set of signaling agents specific to the presence, absence or quantity of a target analyte. The target analyte can be captured by particles of a single size, multiple sizes or magnetic contents, and/or captured onto a sensor through use of an affinity agent such as an antibody, as described herein. Through a method of capture or methods of capture, multiple analytes can be localized over a sensor or multiple sensors, scanned at different redox potentials, and then the target analyte concentrations can be determined using the processor of the reader, as described herein.

[0471] Conjugation techniques to apply redox dyes to affinity agents such as antibodies or primers are known such as by adding carboxylic acid groups to the redox dyes and doing NHS/EDC reactions to covalently bond the dye with the affinity agent or primer (primers having groups such as amine groups).

[0472] Referring now to FIGS. 33A through 33H, a multiplexing process for detecting the presence, absence, and/or quantity of multiple target analytes within a sample in a cartridge is described. First, a sample is collected. Illustratively, the sample in FIG. 33A is blood having monocytes, eosinophils, basophils, lymphocytes, neutrophils, hemoglobin, and platelets, although any sample described herein may be collected. The sample may be collected with any of the sample collection devices described herein.

[0473] Reagents for mixing with the collected sample are provided. For example, one or more reagent balls may be stored within the cartridge, e.g., within a shuttle stored in the cartridge as described above. Referring to FIG. 33B, the reagents (shown below the molecules in the blood from FIG. 33A) may include affinity molecules configured to bind with different types of target analytes, solid particles, and multiple types of signaling agents (illustratively, Label 1 and Label 2). As an alternative to multiple types of signaling agents, or in addition to multiple types of signaling agents, multiple types of substrates in the substrate solution may be used for a multiplexing scheme. The solid particles may comprise magnetically responsive material such as the magnetic particles, as illustrated and described herein, or non-magnetically responsive material such as gold nanoparticles. Preferably, each solid particle is pre-bound to at least one affinity molecule, as described herein. Illustratively, the solid particle is pre-bound to three affinity molecules. The affinity molecules may be any affinity molecule described herein and each preferably have an affinity to bind to a type of target analyte in the collected sample. The reagents also include a plurality of detector agents each having a signaling agent (referred to in FIG. 33B as a Label) pre-bound to an affinity molecule. For example, the signaling agent may be HRP and the affinity molecule may be streptavidin. As another example, the signaling agent may be redox dye and preferably has a detectable potential for a type of target analyte. For example, the signaling agent called Label 1 in FIG. 33B preferably has a first detectable potential for a first type of target analyte and the signaling agent called Label 2 has a second detectable potential, different from the first detectable potential, for a second type of target analyte. As an alternative to multiple types of signaling agents, or in addition to multiple types of signaling agents, a first type of substrate having another detectable potential may be used for the second type of target analyte or for a third type of target analyte, and/or a second type of substrate having yet another detectable potential may be used for the third type of target analyte or for a fourth type of target analyte. The reagents may be configured to localize over electrodes of one or more sensors in the cartridge such as a working electrode and/or a positive control working electrode. The reagents for the multiplexing process also or alternatively may include the reagents described above with respect to FIGS. 31A-31J and/or FIGS. 32A-32N.

[0474] Referring now to FIG. 33C, the reagents and the sample are mixed in fluid held in sample preparation reservoir 1450. The sample may be introduced into sample preparation reservoir 1450

while on the distal portion of a sample collection device or after being released from the distal portion of a sample collection device, both of which are described above. The sample may have a plurality of sample target analytes. The reagents may be pre-stored in the fluid within sample preparation reservoir **1450** or some or all of those molecules may be introduced into the fluid from a reagent ball(s). Sample preparation reservoir **1450** may be constructed in the same manner as any sample preparation reservoir described herein including sample preparation reservoirs **317**, **317'**, **317''**, **317'''**. Mixing within sample preparation reservoir **1450** may be enhanced by actuation of a sonicator element disposed adjacent sample preparation reservoir **1450**, as described above. Preferably, affinity molecules bind to target analytes in the sample during mixing to form sandwich complexes as illustrated in FIG. **33C**.

[0475] Referring now to FIG. **33D**, first sensor **1452** and second sensor **1454** on circuit board **1456** in analysis channel **1458** are shown. First sensor **1452** and second sensor **1454** may be constructed similarly to any of the sensors described above including sensors **338**, **338'**, **338''**, **338'''**, **338''''** and the sensors described in FIGS. **31A-32N**. In the multiplexing scheme, each sensor may be individually addressable. For example, first sensor **1452** and second sensor **1454** are individual addressable. Illustratively, first sensor **1452** does not have pre-bound surface affinity molecules and the sandwich complexes are magnetically held over first sensor using magnet **1460**. First sensor **1452** may have a self-assembled monolayer that may be backfilled as described above. In FIG. **33D**, second sensor **1454** has two types of pre-bound surface affinity molecules that bind to two different types of sandwich complexes. Circuit board **1456** may be constructed similarly to any of the circuit boards described above including circuit board **331**, **331'**, **331''**, and **331'''**. Analysis channel **1458** may be constructed similarly to any of the analysis channels described above including analysis channel **330**, **330'**, **330''**, and **330'''**. First sensor **1452** and/or second sensor **1454** is configured to be exposed to magnetic fields from magnet **1460**, e.g., when the cartridge is inserted in the reader. For example, magnet **1460** may be similar to first magnetic generator **406** and second magnetic generator **407** of reader **400** described above.

[0476] First sensor **1452** and second sensor **1454** are preferably positioned within analysis channel **1458** of the cartridge for exposure to fluid released from the sample preparation reservoir and/or fluid released from the substrate reservoir. FIG. **33D** shows first sensor **1452** and second sensor **1454** after exposure to reagents, e.g., from fluid flowing into analysis channel **1458** from sample preparation reservoir **1450**. As shown in FIG. **33D**, partial sandwich complexes may bind to affinity molecules pre-bound to second sensor **1454** to complete the sandwich complexes. In addition, sandwich complexes with magnetic particles are localized over first sensor **1452** using magnet **1460** as shown in FIG. **33D**. A chemical reaction may occur when the sandwich complexes are exposed to a substrate, e.g., from substrate reservoir, such that first sensor **1452** may detect electrical signals resulting from chemical reactions over first sensor **1452** and second sensor **1454** may detect electrical signals resulting from chemical reactions over second sensor **1454**. For example, the mixed fluid from sample preparation reservoir **1450** may be introduced into analysis channel **1458** such different types of signaling agents directly or indirectly bound to different types of targets from the mixed fluid from sample preparation reservoir **1450** localize over first sensor **1452** by magnetic attraction and localize over second sensor **1454** by binding with pre-bound affinity molecules. The chemical reactions may occur when fluid from the substrate reservoir reacts with particles from the mixed fluid from sample preparation reservoir **1450** localized over first sensor **1452** and when fluid from the substrate reservoir reacts with particles from the mixed fluid from sample preparation reservoir **1450** localized over second sensor **1454**. For example, substrate solution having a substrate may be introduced from the substrate reservoir and first sensor **1452** or second sensor **1454** may detect electrical signals resulting from the reactions between the substrate (e.g., TMB, OPD, ferricyanide, PBS) and the signaling agents (e.g., HRP, SBP, redox dye) localized over the respective sensor. The reactions may cause electrons to be stripped from the substrate by the signaling agents (which electrons may be donated to acceptor molecules of the

substrate solution) thereby generating electrical signals detectable by the respective sensor. First sensor **1452** may include a working electrode and the detected electrical signals may be used to generate a first signal indicative of the presence, absence, and/or quantity of a first type of target analyte within the sample and, in a multiplexing scheme, a second signal indicative of the presence, absence, and/or quantity of a second type of target analyte within the sample. Additional types of target analytes within the sample may be detected by first sensor **1452** by using additional types of signaling agents configured to localize over first sensor **1452**. Second sensor **1454** may include a working electrode and the detected electrical signals may be used to generate a third signal indicative of the presence, absence, and/or quantity of a third type of target analyte within the sample and, in a multiplexing scheme, a fourth signal indicative of the presence, absence, and/or quantity of a fourth type of target analyte within the sample. Additional types of target analytes within the sample may be detected by second sensor **1454** by using additional types of signaling agents configured to localize over second sensor **1454**. For further multiplexing, more than two sensors may be used as well. First and second sensors **1452** and **1454** may include other types of electrodes described herein such as a positive control working electrode. First and second sensors **1452** and **1454** may each run a voltage sweep for the multiplexing scheme to generate respective signals from the sensors for transmission for processing, e.g., transmitted to reader device **400** via respective electrical connectors of cartridge device **300** and reader device **400**.

[0477] Instructions for executing a scanning technique (e.g., cyclic voltammetry) may be stored in memory coupled to the processor of the reader such that the reader processes the electrical signal sensed by first sensor **1452** and the electrical signal sensed by second sensor **1454** of the cartridge that are transmitted to the reader, e.g., via respective electrical connectors. For example, the processor of the reader may direct one or more electrodes that are individually addressable of the first sensor **1452** or second sensor **1454** in the cartridge to apply a voltage sweep. The different types of signaling agents may respond to different voltages during the voltage sweep. The response for each type of signaling agent may be used to determine the presence, absence, and/or quantity of the corresponding target analyte.

[0478] Referring now to FIGS. **33E** and **33F**, the graphs show how peak currents are generated for each type of signaling agent at each sensor in the multiplexing scheme. As shown in FIG. **33E** current is measured in nA across applied voltages during a sweep. FIG. **33F** illustrates that the first type of signaling agent (referred to as Label 1) reaches a peak current at the first sensor in response to a first voltage during the voltage sweep while the second type of signaling agent (referred to as Label 2) reaches a peak current at the first sensor in response to a second voltage (different from the first) during the voltage sweep. The first peak corresponds to the number of a first type of target analyte cells directly or indirectly bound to the first type of signaling agent localized over the first sensor. The second peak corresponds to the number of a second type of target analyte cells directly or indirectly bound to the second type of signaling agent localized over the first sensor.

[0479] FIG. **33F** further illustrates that the first type of signaling agent (referred to as Label 1) reaches a peak current at the second sensor in response to the same first voltage during the voltage sweep while the second type of signaling agent (referred to as Label 2) reaches a peak current at the second sensor in response to the same second voltage during the voltage sweep. The first peak corresponds to the number of a third type of target analyte cells directly or indirectly bound to the first type of signaling agent localized over the second sensor. The second peak corresponds to the number of a fourth type of target analyte cells directly or indirectly bound to the second type of signaling agent localized over the second sensor. Advantageously, in this multiplexing scheme, the number of sensors used in the cartridge may be a fraction of the number of different types of target analytes that are detected.

[0480] Referring now to FIGS. **33G** and **33H**, the graphs show how the signal generated at the respective sensor may be used to determine the number of cells for a type of target analyte detected at the sensor in the multiplexing scheme. The curves in the graphs may be generated from lot-

specific data stored in memory of the cartridge (e.g., memory 332, 332', 332'', 332'''). For example, one or more cartridge devices of a select group of cartridges (e.g., common lot of production cartridges) may be tested using a known quantity of target analytes to determine electrical properties associated with a type of target analyte. Data indicative of multiplexing curves like those in FIGS. 33G and 33H based on the test results may be stored in the cartridge memory in the select group of cartridges to precisely and consistently determine the presence, absence, and/or quantity of that type of target analytes using the electrical signals generated by the sensor of the cartridge. The cartridge memory may also store such data for multiple types of target analytes for the detection scheme. As is shown in FIG. 33H, the signal sensed by the first sensor is compared to the data indicative of multiplexing curves to determine the number of cells of the particular type(s) of target analyte localized at the first sensor. Also shown in FIG. 33H, the signal sensed by the second sensor is compared to the data indicative of multiplexing curves to determine the number of cells of the particular type(s) of target analyte localized at the second sensor.

[0481] The sample reagents may include only one population of magnetic particles and one population of detector agents or competitive binding agents. Such embodiments may be tailored for detection of a single target analyte of interest.

[0482] In other embodiments, multiple populations of magnetic particles and detector agents and/or competitive binding agents and/or multiple populations of distinct affinity molecules are provided, each population constructed to have its own affinity. For example, each population of magnetic particles has a unique affinity molecule bound to its surface, and each population of magnetic particles is thereby designed to bind with a different target analyte. Similarly, each population of detector agents includes a unique affinity molecule and is thereby designed to bind with a different target analyte. A multiplexing scheme may be used such that a first size of magnetic particles is more magnetically responsive to a first working electrode, and a second size of magnetic particles is more magnetically responsive to a second working electrode, etc. (third, fourth, fifth, etc. sizes and working electrodes may be used). Alternatively or additionally, different surface binding schemes may be used for different working electrodes such that a first set of affinity molecules directed to a first target analyte population is immobilized at the surface of a first working electrode and a second set of affinity molecules directed to a second target analyte population is immobilized at the surface of a second working electrode of the electrochemical cell, etc. (third, fourth, fifth, etc. affinities and working electrodes may be used). In this manner, the respective target analyte population binds to the respective affinity molecules (and signaling agent) and binds to the surface bound affinity molecules on the surface of the respective working electrode. Accordingly, multiple sets of affinity molecules and addressable working electrodes may be used to execute a multiplexing scheme. One or more populations of magnetic particles having unique magnetic properties and/or unique affinity molecules bound thereto, as described above, may be multiplexed with one or more different surface binding schemes. For example, the process of FIGS. 31A-31J may occur at substantially the same time as the process of FIGS. 32A-32N and/or at substantially the same time as the process of FIGS. 33A-33H within the same cartridge using the same or different sample preparation reservoirs and the same or different sensors. In embodiments employing the competitive binding approach, each population of competitive binding agents may include a different pre-bound target analyte and is thereby designed to compete with a different target analyte. Such embodiments allow for the detection of a plurality of target analytes, including detection of multiple foodborne pathogens, multiple contaminants, and multiple ailments.

[0483] Those skilled in the art will appreciate that the possibilities for forming the magnetic particle-bound complexes are numerous and all such possibilities are contemplated herein. For example, the sample preparation reagents may include a biotin-labelled antibody, which binds to a portion of the target analyte. In some embodiments, antibodies and/or nucleic acids present among the sample preparation reagents may be pre-biotinylated such that a streptavidin conjugated signaling enzyme can bind with the biotinylated detector to form a complex. One such streptavidin

conjugated signaling enzyme is HRP. The tagging combination is not limited to biotin-streptavidin. Any suitable tagging scheme will work. In another example, multiple HRP enzymes are conjugated together into a molecule commonly known as a Poly-HRP molecule in order to enhance the signal generating capability of the resultant sandwich complex.

[0484] In addition to the components that form the magnetic particle-bound complexes, the sample preparation reagents of various embodiments may include one or more of: (a) agents that facilitate formation of the magnetic particle-bound complexes, such as salts; (b) agents that facilitate access and specificity to target analytes, such as detergents and enzymes for lysis of bacteria or viruses or cutting of large molecules or nucleotides; (c) blocker proteins to decrease nonspecific binding; and (d) stabilizers such as, for example, trehalose, which can improve the shelf life of the sample preparation reagents.

[0485] For the sample preparation reagents, salts may be necessary to enhance the likelihood of binding. For example, phosphate buffered saline (PBS) may be the fluid held in the sample preparation reservoir. Any salt which does not interfere with electrochemical detection may be provided within the reagents.

[0486] Blocker proteins, such as the well-known Bovine Serum Albumin, casein, fibrinogen, or other blocker protein may be provided to help stabilize the antibodies, enzymes, and/or other proteins present among the sample preparation reagents. Such blocker proteins may also help prevent non-specific binding of signaling enzymes to the magnetic particles and to the walls of the systems and devices described elsewhere herein.

[0487] Additionally, for embodiments that require lysis to access the molecules or nucleic acids of interest, detergents may be employed. In various embodiments, nonionic detergents, rather than ionic detergents, are provided to prevent denaturation of the signaling enzyme and/or antibodies. Detergents may enhance lysis of bacteria, but are also useful for gently lysing various viruses, such as the influenza virus. Such lysing may be desirable to improve access to target analytes such as nucleoproteins internal to a virus. Additionally, the sample preparation reagents may include enzymes that enhance lysis and reduce viscosity during lysis; such reagents may be necessary for the preparation of some samples, for example, samples containing bacteria such as *E. coli*. The enzymes that enhance and facilitate lysis may include lysozymes and DNAses that chop up released genomic DNA without disrupting nucleic acid probes on the surface of the magnetic particles.

[0488] Enzymes such as RNAses or DNAses, which selectively chop larger nucleotide sequences into smaller sequences, may be useful for generating smaller fragments having favorable binding kinetics. Such enzymes may be present in the sample preparation reagents. Other components also may be included within the sample preparation reagents. For example, a stabilizer agent such as trehalose, may be present; such stabilizer agents help protect proteins from oxidation and thereby increase the shelf-life of the reagents, especially at room temperature.

[0489] Referring now to FIG. 34, detection system **100** is shown in use where sample collection device **200** is inserted in cartridge device **300** which is inserted in reader device **400**. Reader device **400** may transmit signals indicative of the presence, absence, and/or quantity of one or more target analytes to a mobile device running software-based detection interface system **600** such that a user may review and interact with the analyzed results on the presence, absence, and/or quantity of one or more target analytes. Such results may be transmitted to server **1500** via network **1510**.

Software-based detection interface system **600** may be downloaded onto the mobile device.

Software-based detection interface system **600** may be a dedicated application or “app” and may be downloaded from an online store such as iTunes™ (Apple, Inc., Cupertino, CA), the App Store (Apple, Inc.), Google™ Play (Google, Inc., Mountain View, CA), the Android™ Marketplace (Google, Inc.), Windows™ Phone Store (Microsoft Corp., Redmond, WA), or BlackBerry™ World (BlackBerry, Waterloo, Ontario, Canada). Preferably, software-based detection interface system **600** need only be downloaded once—although updates may be downloaded.

[0490] Sample collection device **200** may be disposable and configured for one-time use. It may come within removable sterile packaging. Once inserted into the input tunnel of cartridge device **300**, sample collection device **200** may be locked into a permanent fixed engagement and cannot be used again. Cartridge device **300** also may be disposable and configured for one-time use. Once sample collection device **200** locks into place within the input tunnel of cartridge **300**, cartridge **300** cannot be used again. Cartridge **300**, may, however, be removed from the reader **400**. Cartridge **300** and reader **400** may be configured to be separably coupled, and cartridge **300** may be inserted and removed from the dock of reader **400** at least before and after implementation of a detection protocol. Reader **400** may include a locking mechanism for temporarily locking cartridge **300** into place, and limiting removal, during the duration of a detection test cycle. Reader **400** of various embodiments is reusable.

[0491] Reader **400**, and the entire detection system **100**, may be configured for non-clinical, consumer-directed use. Accordingly, system **100** is easy to use and generates results quickly. Results of a target analyte detection protocol may be generated in 30 minutes or less from the time a sample from sample collection device **200** is inserted into the system's cartridge **300**. Results may be generated in less than 20 minutes, less than 10 minutes, and/or less than 5 minutes. Additionally, the consumer-directed system may be small for an unobtrusive presence within a home, school, office, or other place of employment. Cartridge **300**, sample collection device **200**, and reader **400** together may be approximately the size of a smartphone or other mobile computing device. System **100** may be sized and configured to be portable. In such embodiments, in addition to a compact, hand-held design, all fluids within the sample are properly sealed and separated such that no leaking or premature oxidation reactions will occur due to jostling of the system components while on the go.

[0492] To promote use by lay people in non-clinical settings, system **100** may be designed to be “dummy proof” by including a self-activating and self-run detection protocol. For example, FIG. **34** depicts an example in which cartridge **300** has been placed into the dock of reader **400** and sample collection device **200** has been inserted into the input tunnel of cartridge **300**. In the depicted embodiment, loading cartridge **300** into reader **400** may establish an electrical connection between the pins of cartridge **300** and reader **400**, thereby completing a circuit within reader **400**, which automatically activates reader **400**. Upon being activated, reader **400** may determine if sample collection device **200** is properly inserted in cartridge **300**, e.g., upon receipt of an electrical signal indicating that the contact switch in cartridge **300** has been activated. Upon detection, reader **400** may initiate a detection protocol automatically without any further human intervention. The automated start ensures that mixing of reagents and sample within the sample preparation reservoir occurs consistently at a fixed time following insertion of sample collection device **200**, leading to consistent test results. Alternatively, the testing protocol may initiate when a user presses a “go”, “run”, “start”, or other similar button or icon on reader **400** or computing device **601** running software **600**.

[0493] In various embodiments, computing device **601** may be included within the system: to provide for more computing power and/or more memory; to provide a wireless transceiver for pulling data from, and transmitting data to, a remote server; and/or to provide a display screen and user interface. Computing device **600** is not needed within every embodiment.

[0494] One skilled in the art will appreciate that the embodiment in FIG. **34** is illustrative in nature only and various components may be added, deleted, or substituted and various different hierarchies and modes of communication between the devices may be employed. Communication network **1510** through which some or all of the various devices communicate with one another. The network may be a local area network (LAN) or a wide area network (WAN). Network **1510** may be a wireless communication network, such as, for example, a mobile WiMAX network, LTE network, Wi-Fi network, or other wireless network. The communication between computing device **601** having a user interface software **600** and server **1500** may occur over the internet via a wired

network, such as a DSL cable connection.

[0495] Communication between reader **400** and computing device **601** may occur, wirelessly, for example, using Bluetooth®, Wi-Fi, near-field communications, or other radiofrequency technology. Alternatively, transmission of signals between reader **400** and computing device **601** may occur over a cord, cable, or other wired or direct connection. In various embodiments, computing device **601** or other device having a user interface software **600** includes a software application for a front-end, graphical user interface for presenting test results to a user.

[0496] Reader **400** may be configured to control the tests and processes needed to detect and/or quantify one or more target analytes within a sample. To do so, a significant amount of information may be stored within the memory of reader **400**. Alternatively, some or all of the information may be stored within computing device **601** and/or server **1500** and accessible by reader **400** via the communication network **1510**. Such information may include, for example, a database of cartridge keys, which identifies each cartridge type by the signal generated by the cartridge's unique identifier stored in the cartridge's memory. The information also may include test protocols associated with each type of cartridge. The test protocols may specify details such as how long to mix sample preparation reagents through sonication, the frequency of the sonication, when to heat the various heat-sensitive valves, etc. The information may also include correlation tables for each cartridge type, which correlate detected sensor signals to the absence, presence, and/or a specific quantity of a target analyte. Additionally, the information stored by reader **400** and/or server **1500** may include one or more past results. Reader **400** may store test results at least until reader **400** comes into communication with a remote computing device; at such time, the results may be transmitted to the remote computing device (e.g., computing device **601**, server **1500**) for display and/or long-term storage.

[0497] Server **1500** also may store user profiles, which may include biographical information entered into the system by a user through computing device **601** having user interface software **600**. A log of test results for each user may also be stored by server **1500** and accessible for viewing by the user through transmission of such data to computing device **601** with user interface software **600**.

Definitions

[0498] Unless otherwise defined, each technical or scientific term used herein has the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. In accordance with the claims that follow and the disclosure provided herein, the following terms are defined with the following meanings, unless explicitly stated otherwise.

[0499] The term “about” or “approximately,” when used before a numerical designation or range (e.g., pressure or dimensions), indicates approximations which may vary by (+) or (−) 5%, 1% or 0.1%.

[0500] As used in the specification and claims, the singular form “a”, “an” and “the” include both singular and plural references unless the context clearly dictates otherwise. For example, the term “a molecule” may include, and is contemplated to include, a plurality of molecules. At times, the claims and disclosure may include terms such as “a plurality,” “one or more,” or “at least one;” however, the absence of such terms is not intended to mean, and should not be interpreted to mean, that a plurality is not conceived.

[0501] As used in the specification and claims, “at least one of” means including, but not limited to, one or more of any combination of the following. For example, “at least one of A, B, and C” or “at least one of A, B, or C” means including, but not limited to, A(s) or B(s) or C(s) or A(s) and B(s) or A(s) and C(s) or B(s) and C(s) or A(s) and B(s) and C(s); none of which excludes other elements such as D(s), E(s), etc.

[0502] As used herein, the term “comprising” or “comprises” is intended to mean that the devices, systems, and methods include the recited elements, and may additionally include any other elements. “Consisting essentially of” shall mean that the devices, systems, and methods include the

recited elements and exclude other elements of essential significance to the combination for the stated purpose. Thus, a device or method consisting essentially of the elements as defined herein would not exclude other materials or steps that do not materially affect the basic and novel characteristic(s) of the claimed invention. "Consisting of" shall mean that the devices, systems, and methods include the recited elements and exclude anything more than a trivial or inconsequential element or step. Embodiments defined by each of these transitional terms are within the scope of this disclosure.

[0503] Although the foregoing has included detailed descriptions of some embodiments by way of illustration and example, it will be readily apparent to those of ordinary skill in the art in light of the teachings of these embodiments that numerous changes and modifications may be made without departing from the spirit or scope of the appended claims.

Claims

1-27. (canceled)

28. A kit for detecting at least one of a presence, absence, or quantity of a first and second types of target analytes in a sample comprising: a sample collection device; a sample analysis cartridge; and a sample analysis reader configured to receive the sample analysis cartridge; wherein the sample analysis cartridge comprises: a sample preparation reservoir holding a first fluid comprising a first signaling agent to detect the first type of target analyte of the sample and a second signaling agent to detect the second type of target analyte of the sample; and a reagent ball comprising nucleic acid amplification reagents; wherein the first type of target analyte directly or indirectly binds to the first signaling agent in the first fluid in the sample preparation reservoir, the first signaling agent having a first detectable potential for the first type of target analyte; and the second type of target analyte directly or indirectly binds to the second signaling agent in the first fluid in the sample preparation reservoir; the second signaling agent having a second detectable potential, different from the first detectable potential, for the second type of target analyte; and a substrate reservoir configured to hold a second fluid comprising a substrate that reacts with the first signaling agent and the second signaling agent to form a substrate-first signaling agent complex and a substrate-second signaling complex; and an analysis channel configured to receive the second fluid from the substrate reservoir and comprising a sensor for detecting both the first and second types of target analytes based upon a first signal and a second signal respectively generated by the substrate-first signaling agent complex and the substrate-second signaling agent complex respectively following electrical stimulation of the second fluid; and wherein the signal for each substrate-signaling agent complex is based on the respective potential of the signaling agent for the analyte.

29. The kit of claim 28, wherein the sample collection device comprises a distal portion adapted to be exposed to the sample, optionally wherein the sample collection device is a swab.

30. The kit of claim 28, wherein the sensor is an electrochemical sensor.

31. The kit of claim 28, wherein the sensor comprises: a positive control working electrode configured to generate the first signal for detecting the first type of target analytes; and a working electrode configured to generate the second signal for detecting the second type of target analytes; wherein the positive control working electrode or the working electrode comprises a self-assembled monolayer pre-bound to an affinity molecule configured to directly or indirectly bind to the first signaling agent or the second signaling agent.

32. The kit of claim 31, wherein the self-assembled monolayer comprises a thiolated ethylene glycol and/or a dithiol.

33. The kit of claim 32, wherein the self-assembled monolayer is backfilled with a backfiller that binds directly to the affinity molecule.

34. The kit of claim 33, wherein the backfiller is a carboxylic acid terminated thiol.

35. The kit of claim 34, wherein the carboxylic acid terminated thiol is a mercaptoundecanoic acid

(MUDA) or a mercaptohexanol.

36. The kit of claim 32, wherein the dithiol is an hexaethylene glycol dithiol (HEGT).

37. The kit of claim 35, wherein the self-assembled monolayer comprises an hexaethylene glycol dithiol (HEGT) and the backfiller comprises the mercaptoundecanoic acid (MUDA).

38. The kit of claim 31, wherein the positive control working electrode comprises the self-assembled monolayer pre-bound to affinity molecules and the working electrode comprises a self-assembled monolayer.

39. The kit of claim 31, further comprising a negative control working electrode configured to generate a third signal based on a reaction between the first and second signaling agents localized at the negative control working electrode and the substrate.

40. The kit of claim 39, wherein the negative control working electrode comprises a self-assembled monolayer.

41. The kit of claim 40, wherein the self-assembled monolayer of the negative control working electrode is pre-bound to affinity molecules.

42. The kit of claim 31, wherein the second signal is indicative of at least one of the presence, absence, or quantity of the first and second types of analytes within the sample.

43. The kit of claim 31, wherein the first signal is indicative of reliability of a test.

44. The kit of claim 43, wherein the test is determined to be reliable if the first signal indicates a quantity of the first and second types of analytes within a predetermined range.

45. The kit of claim 39, wherein the third signal is indicative of reliability of a test.

46. The kit of claim 45, wherein the test is determined to be reliable if the third signal indicates a quantity of the reaction that is below a threshold.

47. The kit of claim 31 further comprising a processor configured to process the second signal to generate information indicative of at least one of the presence, absence, or quantity of one or more analytes within a sample.

48. The kit of claim 28, wherein the reagent ball comprises a plurality of forward and reverse primers, a plurality of dNTPs, a polymerase, a reverse transcriptase, a recombinase, and a single stranded binding protein.

49. The kit of claim 48, wherein the reagent ball further comprises a recombinase loading factor.

50. The kit of claim 28, wherein the reagent ball comprises nucleic acid amplification reagents for an isothermal amplification reaction.
